The Public-Stall Allocation Problem

A Mathematical Theory of Position, Privacy, Hygiene, and Collective Choice

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Abstract

When several public toilet stalls are free, which one is chosen, and what distribution of use emerges after many people make that choice? The hygiene version of the question adds a feedback loop: a stall may be attractive because it has been used less, but choosing it changes the fact on which that judgment was based.

This monograph constructs the distribution from the sequence of individual decisions. Three kinds of information enter one arrival's comparison. Position describes the stall even when the room is empty. Current occupancy determines which stalls are available and where immediate neighbors would be. Past use, decay, and cleaning determine a remembered hygiene state. The opening chapter introduces these ingredients one at a time before combining them into a choice rule.

Repeated arrivals and departures turn that rule into a population process. The analysis gives an exact finite-state solution when hygiene is held fixed, derives population equilibria when historical use feeds back into choice, and finds the precise linear threshold at which delayed hygiene information makes a steady allocation unstable. Position can favor a near, interior, or far stall; current privacy can favor boundaries; availability redistributes demand as a room fills; and historical-use avoidance can weaken a persistent positional advantage or make the preferred label vary over time.

The empirical target is kept explicit throughout. A visit count, an occupancy snapshot, and a depleted paper roll are different observations unless visit durations and consumption marks are homogeneous. The direct restroom evidence is therefore used at the level it supports, while exact calculations, simulations, and sensitivity analyses show what stronger event-level data could identify. The result is a conditional mathematical answer: given geometry, traffic, holding times, cleaning, information, and behavioral sensitivity, the model produces the distribution of stall use and states which mechanism created it.

The thought experiment

Imagine approaching a row of closed stall doors. Before introducing any symbol, change the scene three times.

First, suppose the room is empty and newly cleaned. The doors may look identical, but their positions are not identical. One is nearest the entrance, one may be screened by a turn, one has an end wall, and one requires the longest walk. Any systematic choice in this scene must come from position or stall-specific amenities; neither current privacy nor differential history is available to explain it.

Second, keep the same room but occupy one stall. The remaining options now differ for two new reasons. One door is unavailable, and the free doors place the entrant at different distances from the current user. This is the instantaneous privacy problem. Its input is a snapshot of the room now.

Third, empty the room again, but reveal that the stalls have had different numbers and timings of previous visits since cleaning. Current privacy has disappeared, yet history can still matter. This is the hygiene problem. Its input is not merely a count: recent visits can matter more than old ones, a cleaning event can remove part of the accumulated state, and users may see only a delayed or noisy signal of it.

The observed usage distribution is generated by repeating these scenes in one continuous process. A choice changes occupancy immediately, occupancy changes the options facing the next arrival, a departure restores availability, and use changes the historical state. The central mathematical task is therefore not to name a universally preferred stall. It is to turn this feedback loop into a probability law and determine how that law changes with geometry, traffic, and memory.

How to read this monograph

The argument is written for a reader who has not studied probability, game theory, or differential equations. Every symbol is introduced where it first appears, and the mathematical tools are developed before they are used. A reader seeking the conclusion first should read Chapter 1 and then Chapters 9 and 11. A reader interested in the derivation should continue in order. Proofs that carry the main insight remain in the chapters; longer algebra and auxiliary results are collected in the appendices.

The document makes a strict distinction among three kinds of statement:

- *empirical results*, which are observations reported in prior work;
- *theorems*, which follow from explicit assumptions;
- *predictions*, which connect the model to measurements not yet made.

The distinction lets each claim be checked against the evidence appropriate to it.

Notation at a glance

Symbol	Meaning
n	number of stalls in the row
i, j	stall indices, normally counted from the entrance
x_i	physical position or walking distance of stall i
$s_i(t)$	occupancy indicator: 1 when stall i is occupied, 0 otherwise
$S(t)$	set of occupied stalls at time t
g_i	baseline positional cost of stall i
K_{ij}	spatial privacy kernel linking stalls i and j
J	sensitivity to instantaneous privacy loss
$m_i(t)$	accumulated, perceived hygiene load of stall i
H	sensitivity to hygiene load
β	choice precision; larger values mean more deterministic choice
Λ	arrival rate to the stall system
μ_i	departure rate from stall i; $1/\mu_i$ is mean occupied time
ρ	offered load Λ/μ when service rates are common
p_i	long-run entry share or population choice probability of stall i
q_i	long-run fraction of time stall i is occupied
κ	hygiene decay or cleaning rate
τ	delay between hygiene state and the information used for choice
h	reduced-form own-load coefficient induced by use and cleaning
Φ	population potential or free-energy functional

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Part I

The question and the evidence

1

The question, stated precisely

1.1 A single arrival

Suppose a man enters a public restroom containing a row of n stalls. Number the stalls $1, 2, \ldots, n$ from the entrance toward the far end. At the instant of arrival, some doors are free and some are occupied. Write $F(t)$ for the set of free stall numbers at time t. For example, if a four-stall row has only stalls 1 and 3 free, then $F(t) = \{1, 3\}$. The entrant compares only the members of this set.

The model will assign a *cost* to each free stall. Cost here is a common numerical scale for anything that makes a stall less attractive. It need not be money, and it need not be consciously calculated. A lower-cost stall is more likely to be chosen. We now build that cost one piece at a time.

1.1.1 Position in an empty room

Begin with an empty, newly cleaned room. Stall locations can still differ: the first may be exposed to the entrance, the last may require a longer walk, and an end wall or an accessibility feature may change the experience. Before writing an equation, collect all such fixed features of stall i into one number called g_i. Smaller g_i means that the stall is more attractive when current occupancy and use history are held equal. The first model is therefore simply

$$C_i^{\text{position}} = g_i. \tag{1.1}$$

Nothing probabilistic has happened yet. Equation (1.1) is only a ledger for position and amenities. Chapter 4 later replaces this ledger entry by physical components such as decaying door exposure and increasing walking effort.

1.1.2 Current occupancy and immediate privacy

Next, keep the same row but occupy one stall. For each stall j, define $s_j(t)$ to be 1 when j is occupied at time t and 0 when it is free. This zero–one switch tells the model which current users are present.

Presence alone is not enough: an occupant in the adjacent stall usually matters more than one several partitions away. Let K_{ij} measure how strongly occupancy of stall j affects privacy in stall i. Nearby, poorly screened pairs have larger K_{ij}; well-isolated or distant pairs have smaller values. Set $K_{ii} = 0$, because an occupied stall is unavailable rather than its own neighbor. Finally, let J measure how strongly the entrant values this form of immediate privacy.

With every object now defined, the current-privacy cost of choosing stall i is

$$C_i^{\text{privacy}}(t) = J \sum_{j=1}^{n} K_{ij} s_j(t). \tag{1.2}$$

The summation sign means “add the effects of the currently occupied stalls.” If stall 3 is the only occupied

stall, every term except $j = 3$ vanishes and

$$C_i^{\text{privacy}}(t) = JK_{i3}.$$

Thus the formula reduces to the ordinary statement that the privacy cost is larger when the sole occupant is more consequential for stall i.

1.1.3 Accumulated use and hygiene history

Now empty the room again but allow the stalls to have different histories since cleaning. Let $m_i(t)$ be the hygiene load associated with stall i at time t. The word *load* denotes a state variable: visits can increase it, natural decay and cleaning can decrease it, and recent use can be weighted more heavily than old use. A larger value means more unresolved evidence of past use on the interpretation being modeled.

Let H be the cost assigned to one unit of this load. The history-dependent cost is then

$$C_i^{\text{hygiene}}(t) = Hm_i(t). \tag{1.3}$$

This expression says only that perceived hygiene cost rises in proportion to the modeled load. Chapter 7 derives $m_i(t)$ from visit events, occupied time, decay, and cleaning rather than treating it as a mysterious score.

1.1.4 The combined perceived cost

We have introduced three quantities with three different meanings: g_i describes the stall itself, $C_i^{\text{privacy}}(t)$ describes privacy now, and $C_i^{\text{hygiene}}(t)$ describes use history. Their combined cost is

$$\begin{aligned} C_i(t) &= g_i + C_i^{\text{privacy}}(t) + C_i^{\text{hygiene}}(t) \\ &= g_i + J\sum_{j=1}^{n} K_{ij}s_j(t) + Hm_i(t). \end{aligned} \tag{1.4}$$

The second line is not a new assumption; it substitutes (1.2) and (1.3) into the first. Every term is measured on the same latent cost scale, and the entrant compares this cost only across $i \in F(t)$.

1.1.5 Choice probabilities

Different people can make different choices in the same state, and the same person need not act identically on repeated occasions. We therefore need a probability rule rather than a command to choose the smallest cost.

Introduce one last number, $\beta \geq 0$, called *choice precision*. For two free stalls i and k, the model says that their odds obey

$$\frac{\mathbb{P}\{I(t) = i\}}{\mathbb{P}\{I(t) = k\}} = \exp[-\beta\{C_i(t) - C_k(t)\}], \qquad i, k \in F(t). \tag{1.5}$$

Here $I(t)$ means "the stall chosen at time t." Equation (1.5) has an immediate reading. Equal costs give equal odds. If stall i costs more than stall k, its odds are smaller. When $\beta = 0$, cost differences are ignored; as β grows, even a small cost difference has a strong effect.

Probabilities must add to one over the free stalls. Normalizing the odds gives

$$\mathbb{P}\{I(t) = i \mid F(t), \boldsymbol{m}(t)\} = \frac{\exp[-\beta C_i(t)]}{\sum_{k \in F(t)} \exp[-\beta C_k(t)]}, \qquad i \in F(t), \tag{1.6}$$

and an occupied stall has probability zero. This is the logit rule. It is introduced here through its odds interpretation; Chapter 3 derives it from random utility and explains its invariances.

1.1.6 From a single choice to a usage distribution

Equation (1.6) describes one arrival in one state. The requested distribution of usage is produced by repeating the event:

1. an entrant observes the free set and the available information;
2. one free stall is selected;
3. that stall becomes unavailable for the duration of the visit;
4. the visit changes the stall's historical load;
5. a departure makes the stall available again; and
6. the next entrant faces the new state.

Availability therefore redirects traffic even if preferences never change. Historical-use avoidance adds a second feedback: choosing a stall alters one reason for choosing it next time. Delayed information can make that response overshoot. The long-run distribution must be calculated from this entire loop, not read from the empty-room probabilities.

The central conclusion

The model does not begin by assuming that a middle, end, or entrance stall is best. Position determines the ranking in an empty room; current occupants change the ranking from one arrival to the next; availability converts those choices into traffic; and remembered use feeds traffic back into later choice. Consequently, the most-used stall can be near, interior, or far, and the ranking can be steady or time varying. A recommendation to use a historically less-used stall is part of the system it predicts: adoption changes that stall's subsequent history.

1.2 What "distribution of usage" means

The phrase *stall usage* hides at least three different random quantities. For stall i over an observation window $[0, T]$, define

$$N_i^{\text{in}}(T) = \text{number of visits that begin in stall } i, \tag{1.7}$$

$$O_i(T) = \int_0^T s_i(t)\ \mathrm{d}t, \tag{1.8}$$

$$D_i(T) = \sum_{r=1}^{N_i^{\text{in}}(T)} X_{ir}, \tag{1.9}$$

where X_{ir} is a consumptive mark, such as sheets of toilet paper used on visit r. The corresponding long-run rates, when they exist, are

$$a_i = \lim_{T\to\infty} \frac{\mathbb{E}N_i^{\text{in}}(T)}{T}, \qquad q_i = \lim_{T\to\infty} \frac{\mathbb{E}O_i(T)}{T}, \qquad d_i = \lim_{T\to\infty} \frac{\mathbb{E}D_i(T)}{T}. \tag{1.10}$$

The normalized entry share is $p_i = a_i / \sum_j a_j$.

These quantities coincide only under restrictive conditions. If visits in stall i have mean duration $1/\mu_i$, stationarity gives the flow identity

$$q_i = \frac{a_i}{\mu_i}. \tag{1.11}$$

If the mean consumptive mark is $\xi_i = \mathbb{E}[X_{ir} \mid I_r = i]$, then

$$d_i = a_i \xi_i. \tag{1.12}$$

Consequently a paper-roll study identifies entry shares only when ξ_i is effectively constant across stalls; an occupancy sensor identifies entry shares only when mean durations are effectively constant. Long visits sorting toward more private stalls break the occupancy equivalence in a predictable direction. They also change the depletion-to-entry mapping when the consumptive mark ξ_i covaries with duration or otherwise differs by stall.

This separation resolves an apparent contradiction in ordinary speech. "Frequently chosen" means large a_i; "usually occupied" means large q_i; "uses the most paper" means large d_i. A serious model must say which quantity it predicts before it compares a prediction with data.

1.3 The four clocks

The complete process contains four characteristic times.

1. The **decision time** is the few seconds in which an entrant sees the available stalls and chooses.
2. The **occupancy time**, with mean roughly $1/\mu$, is the duration of a visit.
3. The **hygiene-memory time**, roughly $1/\kappa$, is how long a use continues to affect the perceived or physical state of a stall.
4. The **information delay** τ is the age of the evidence on which users act: memory, signage, reputation, or a published recommendation.

Privacy lives mainly on the first two clocks. Hygiene lives on the last three. This is the precise sense in which the two forces are dual. Both read the same occupancy field $s_j(u)$, but privacy reads across space near the present while hygiene reads backward through time at one location. A single space–time kernel makes the statement literal:

$$C_i(t) = g_i + \sum_{j=1}^{n} \int_{-\infty}^{t} \mathcal{K}_{ij}(t-u) s_j(u)\, \mathrm{d}u, \tag{1.13}$$

with

$$\mathcal{K}_{ij}(v) = JK_{ij}\delta_0(v) + H\mathbf{1}\{i=j\}\varphi(v), \qquad v \geq 0. \tag{1.14}$$

The Dirac mass δ_0 represents an instantaneous response, while the nonnegative memory kernel φ distributes the effect of past use over future time. With H measured in cost per unit hygiene load, φ has units of load per unit time; an occupancy-dose example is $\varphi(v) = \sigma e^{-\kappa v}$.

1.4 Limits of single-mechanism explanations

1.4.1 "People choose the middle"

Christenfeld's field study is the central empirical anchor: in a four-stall public restroom, the two middle stalls accounted for about 60 % of depleted rolls rather than the 50 % expected under equal use (Christenfeld 1995).

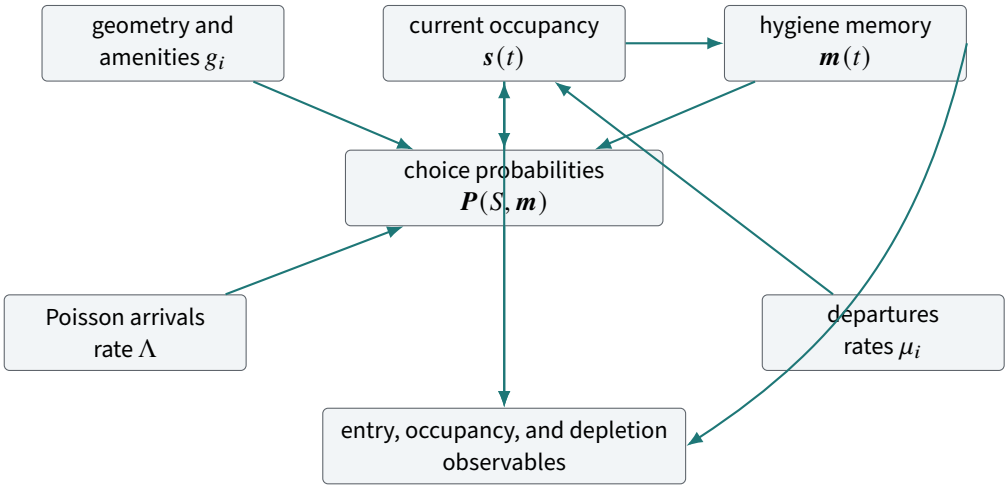


Figure 1.1: The causal layers of the model. Position changes choice directly; occupancy changes both availability and immediate privacy; the history of occupancy changes hygiene; arrivals and departures convert choices into a population process.

The same paper found middle selection in other arrays of nominally identical objects. That observation establishes a position effect in the measured setting. It does not determine whether the effect came from an abstract preference for centrality, distance from the entrance, walking cost, wall effects, stall-specific visit durations, or consumption per visit.

The model therefore does not insert "center bias" as an unexplained constant. It begins with physical position. If exposure to the doorway decays with distance x while walking cost rises with x, a parsimonious baseline is

$$g(x) = ae^{-x/\lambda} + wx. \tag{1.15}$$

The first term falls; the second rises. On an installed interval $[0, L]$, their sum has an interior minimum exactly when $\lambda w < a < \lambda w e^{L/\lambda}$. The central preference is then a geometric compromise, not a primitive psychological label. Chapter 4 derives the exact discrete condition and shows how it can be estimated.

1.4.2 "People maximize distance"

Distance matters, but a current-distance rule sees only the occupied snapshot. It cannot determine first choice in an empty room because every stall has the same privacy cost at that instant. Nor does it account for a user who chooses an end stall to reduce the chance of a future neighbor. The urinal literature formalizes several versions of that forward-looking problem (Kranakis and Krizanc 2010); the field experiment of Middlemist et al. (1976) establishes that interpersonal distance in a men's lavatory can alter measurable behavior and physiology. Neither result by itself supplies a dynamic stall-use distribution.

1.4.3 "The least-used stall is cleanest"

Use is an input to hygiene, not a synonym for it. A hygiene stock must specify what a visit deposits, how fast the stock decays, what cleaning removes, and how well users observe it. Even if all of those links are monotone, a least-used-stall recommendation changes the process it describes. As more people act on the recommendation, use shifts toward that stall and its historical advantage shrinks. This is congestion feedback.

There is also no contradiction between a positional middle preference and a hygiene preference for low use. They are terms in the same cost. The equilibrium balances them. Increasing hygiene sensitivity flattens

the entry shares; it need not reverse their order immediately, and it cannot make a permanently popular stall remain the least used.

1.4.4 The queueing story

The number of busy stalls can indeed have the classical Erlang loss distribution. That statement is exact when arrivals are Poisson, holding times are independent exponential random variables with a common rate, and every arrival takes some stall whenever one is free. Yet the busy-count distribution does not say which stalls are busy. Two systems can have identical Erlang blocking probabilities and radically different per-stall use. If entrants balk because the only free stall is too exposed, too close to an occupant, or visibly unclean, even the Erlang count law changes.

1.5 Seven model results

The analysis ahead can be summarized as seven claims, each proved or computed under explicit conditions.

1. **An interior baseline optimum is mechanistic.** Exponentially decaying entrance exposure plus linearly increasing walking cost has an interior minimizer exactly when entrance exposure is strong enough to move the optimum away from the near boundary but not so strong that the optimum lies beyond the far boundary.

2. **Choice and occupancy are different levels.** A logit rule over free stalls defines birth rates in a Markov chain. The stationary distribution, not the empty-room choice vector, determines long-run use.

3. **Normalization changes the physics.** Always choosing one of the free stalls generally violates detailed balance. A local accept-or-balk rule has an exact Gibbs measure. The two behavioral protocols cannot be interchanged merely because both contain an exponential.

4. **Privacy is a boundary force at weak interaction.** With no positional differences, a symmetric repulsive kernel gives edge stalls lower exposure to the rest of a finite row. The first correction away from uniform use is therefore edge-favoring.

5. **Hygiene is quadratic congestion at equilibrium.** Linear deposition and exponential or Poisson cleaning make steady hygiene load proportional to own use. Integrating that marginal cost gives the potential term $(h/2)\sum_i p_i^2$.

6. **Advice destroys its own advantage.** In the weak-heterogeneity regime, positional dispersion shrinks in inverse proportion to $n + \beta h$. A public belief that one stall is less used raises its use until the benefit is reduced.

7. **Delayed hygiene response has an exact linear instability threshold.** In the primitive event-dose model, let q be the admitted visit-start rate and let η be hygiene cost per unit stock. If each visit contributes dose δ_m, then $A = \delta_m q \beta \eta / n$; symmetric use is stable for every delay when $A \le \kappa$. When $A > \kappa$, it first loses stability at

$$\tau_c = \frac{\arccos(-\kappa/A)}{\sqrt{A^2 - \kappa^2}}.$$

   Beyond that point, linear nonuniform hygiene perturbations grow oscillatory; the nonlinear choice rule determines the bounded post-threshold pattern.

2

What has been measured, and what it means

2.1 A fragmented empirical record

The literatures reviewed here address the components separately. Four streams contain pieces of the answer:

1. studies of positional choice among nominally identical options;
2. studies of interpersonal distance and privacy in lavatories;
3. mathematical analyses of sequential spacing, especially urinals and unfriendly seating;
4. environmental microbiology of restroom surfaces and cleaning.

Their outcome variables differ. Position studies count selections or depletion. Privacy studies measure behavior conditional on interpersonal distance. Sequential-spacing papers specify rules and derive consequences without estimating how real users combine the rules. Microbiological studies measure surface communities or viable organisms, not choice. The contribution of the present theory is to give these measurements distinct places in one generative model.

The broader positional-choice record confirms that central selection is not unique to restrooms. Similar-option experiments found central choices among objects and chairs (Shaw et al. 2000); eye-tracking studies of consumer arrays showed a central-gaze cascade (Atalay et al. 2012); and Bar-Hillel (2015) reviewed the competing mechanisms that can generate position effects. These studies motivate comparison mechanisms. They do not turn one geometry's restroom result into a universal coefficient.

2.2 The positional result

Christenfeld (1995) reported six studies of choices from arrays of identical or equivalent options. The second study used a public restroom with four side-by-side stalls. Each stall contained four paper-roll positions. During a ten-week observation period, exhausted rolls were recorded by stall and by within-stall position. Eighty-six rolls were depleted. Fifty-two came from the two middle stalls and thirty-four from the two ends: shares of 60.5 % and 39.5 %, respectively. Under an equal-depletion null model the corresponding shares are 50 % and 50 %. The four individual stall shares were approximately 22 %, 31 %, 29 %, and 17 % from entrance end to far end (Bar-Hillel 2015; Christenfeld 1995).

The result is valuable because it is behavioral and unobtrusive. It is also a coarse aggregate from one four-stall men's restroom, measured through paper depletion rather than event-level choice. Its transfer to other facility types, user populations, visit purposes, and accessibility needs is an empirical question for the multi-site design below. Let N_i be the unobserved number of visits and X_{ir} the paper used on visit r. The study observes a thresholded version of

$$D_i = \sum_{r=1}^{N_i} X_{ir},$$

Table 2.1: Empirical and theoretical anchors, organized by what they identify.

Source or literature	Object observed	Contribution to this model	What it does not identify
Christenfeld (1995)	depleted toilet-paper rolls in a four-stall setting; choices from other arrays	evidence of a middle-versus-end positional inequality	visit counts, holding times, state at choice, hygiene, or a causal mechanism
Middlemist et al. (1976)	urination onset and persistence under manipulated interpersonal distance	direct evidence that lavatory proximity changes behavior and physiological response	enclosed-stall choice probabilities or a long-run use distribution
Kranakis and Krizanc (2010)	formal sequential urinal-choice models	forward-looking privacy and sensitivity to behavioral protocol	estimated parameters or validation for toilet stalls
Friedman and Rothman (1964), Georgiou et al. (2009), and MacKenzie (1962)	random sequential exclusion on lines and grids	hard-privacy packing and jamming benchmarks	behavior when adjacent use is permitted or holding times are finite
Flores et al. (2011) and Gibbons et al. (2015)	microbial communities on restroom surfaces and their re-establishment	evidence that restroom surface state has a temporal ecology affected by use and cleaning	a monotone map from stall visit count to personal infection risk

because a roll is counted when cumulative consumption crosses its capacity. Even with equal roll capacities, D_i is proportional to N_i only if the conditional distribution of X_{ir} does not vary by stall or over the observation schedule. The study's middle share therefore establishes a depletion inequality. Interpreting it as a visit-count inequality adds a homogeneous-consumption assumption.

The same distinction prevents an overconfident hygiene conclusion. A stall responsible for fewer depleted rolls need not have fewer touch events, less occupied time, lower viable microbial burden, or more recent cleaning. The positional result is evidence for a model term g_i. It is not a direct assay of m_i.

2.2.1 A small occupied-time snapshot

Heston (2023) reported 37 opportunistic snapshots of occupancy in one three-stall bank. Across those snapshots there were 46 occupied-stall instances: 23 in the wall-adjacent end stall, 11 in the middle stall, and 12 in the end stall beside the open side of the room. The asymmetry between the two ends is useful: "end stall" is not a complete geometric description, because a wall, an open side, the entrance, and an accessibility designation need not coincide.

These observations estimate neither an arrival-choice distribution nor a population preference. A snapshot samples occupied time, so a stall with longer visits is more likely to appear; it is conditioned on an undocumented sequence of free sets; and 37 observations from one site leave large sampling uncertainty. Write $K_{\text{occ}} = \sum_j s_j$ for the number occupied. Because only snapshots with $K_{\text{occ}} > 0$ were recorded, $23/37$, $11/37$, and $12/37$ are the observed conditional sample proportions corresponding to the marginal events $\{s_i = 1 \mid K_{\text{occ}} > 0\}$; without a probability sampling frame they do not estimate population probabilities. Normalizing the 46 occupied-stall instances instead gives the observed ratio

$$\frac{\sum_{r=1}^{37} s_{ir}}{\sum_{r=1}^{37} K_{\text{occ},r}}.$$

Under a specified probability sampling model, the corresponding population functional would be $\mathbb{E}[s_i \mid K_{\text{occ}} > 0]/\mathbb{E}[K_{\text{occ}} \mid K_{\text{occ}} > 0]$. Both are occupied-time observables that must be distinguished from the entry share p_i; the distinction becomes exact in Chapter 5.

2.2.2 A useful binomial reconstruction

At the middle-versus-end level, let $Y = 52$ of $N = 86$ depleted rolls denote the middle count. Under the equal-depletion null, $Y \sim \text{Binomial}(86, 1/2)$. The standardized difference is

$$Z = \frac{Y - 43}{\sqrt{86/4}}. \tag{2.1}$$

This gives $Z \approx 1.94$; the exact two-sided binomial tail is approximately 0.066, while the directional upper tail is approximately 0.033. The paper reports the predicted middle/end comparison as $p < .05$. This reconstruction is not a replacement for its analysis or week-level structure. It shows what the headline aggregate can and cannot carry by itself: the effect is substantively visible, while precise stall-specific parameters require the individual counts and exposure schedule.

2.3 The privacy result

Middlemist et al. (1976) used a men's lavatory as a natural setting in which interpersonal distance could be manipulated while compensatory movement was limited. In the main field experiment, sixty users were assigned to close, moderate, or control conditions in a three-urinal room. Mean delay before urination rose from 4.9 s in the control condition to 6.2 s at the moderate distance and 8.4 s at the close distance. Mean persistence after onset fell from 24.8 s to 23.4 s and 17.4 s, respectively. The multivariate condition effect was reported as statistically significant.

For the present model, the decisive fact is not the historical interpretation of arousal. It is that spatial proximity in a lavatory altered measurable behavior. This motivates testing a proximity effect—especially in the holding-time or departure model—but it does not establish a nonzero stall-choice coefficient J. Partitions, door orientation, acoustic isolation, visit purpose, and the self-selected population using stalls all change the mapping from physical distance to experienced intrusion.

The study also demonstrates why duration cannot automatically be treated as independent of privacy. If proximity changes the onset or persistence of a visit, then the service rate μ_i can depend on the occupancy state. In that case a stall's occupied-time share is not simply a rescaled entry share, and the classical Erlang reduction becomes an approximation rather than an identity.

2.4 Stated bathroom values and the interaction order

A complementary literature asks people what makes a public bathroom acceptable and observes how they manage a socially delicate setting. Corradi et al. (2020) developed a 14-item public-bathroom perception scale from a Spanish convenience sample of 654 respondents. Its factor structure separated privacy, ease of use, and cleanliness. These are stated evaluations, not revealed stall choices, but they justify keeping privacy and cleanliness as separate modeled costs rather than collapsing both into a single aversion.

Ethnographic work describes public bathrooms as places with a distinctive interaction order: users coordinate gaze, speech, bodily exposure, and territorial boundaries (Cahill et al. 1985; Haslam 2012). That record explains why a theory-of-mind component is scientifically natural. It does not imply that every user performs the same strategic calculation. The population model therefore permits random utility, heterogeneous coefficients, and bounded levels of anticipation.

2.5 The urinal and sequential-spacing literature

The urinal problem asks where an initial user should stand to reduce the probability that a subsequent user occupies an adjacent position. Kranakis and Krizanc (2010) analyzed several rules for how later users behave and showed that the answer depends on the rule. The point carries over unchanged to stalls: "maximize privacy" is not a complete behavioral specification.

Three decisions must be separated.

1. Does an entrant consider every free position or encounter stalls in a fixed walking order?
2. Is an adjacent position merely costly, or is it unacceptable?
3. Does the current user value only present separation, or expected separation over the remainder of his visit?

Hard exclusion connects the problem to random sequential adsorption and the unfriendly-seating problem. On a long one-dimensional row, sequentially placing users uniformly among positions whose two neighbors are empty produces a maximal independent set with limiting occupied fraction

$$\theta_{\text{jam}} = \frac{1 - e^{-2}}{2} \approx 0.43233 \tag{2.2}$$

(Friedman and Rothman 1964; MacKenzie 1962). The capacity optimum is $1/2$, attained by occupying alternating positions. Random selfish placement therefore achieves

$$\frac{\theta_{\text{jam}}}{1/2} = 1 - e^{-2} \approx 0.86466 \tag{2.3}$$

of the best hard-exclusion capacity, leaving a relative gap of about 13.5 %.

This is a precise and illuminating benchmark, but it is not the ordinary stationary restroom. The hard-exclusion process has no departures and stops when every remaining vacancy touches an occupant. A functioning stall system has departures, and an entrant may accept adjacency rather than leave. The benchmark describes a privacy-limited filling episode or a policy of balking, not the complete use distribution.

2.6 What environmental microbiology adds

Surface sampling has established that public restrooms carry spatially structured human-associated microbial communities. Flores et al. (2011) sampled multiple surface types in public restrooms and linked community composition to likely sources such as skin, gut, and soil. Gibbons et al. (2015) followed restroom surfaces after decontamination and found rapid ecological succession toward a stable community, while also distinguishing DNA-based detection from viability. Flush-generated droplets and aerosols provide another transport pathway. Schreck et al. (2021) measured aerosol generation by flushing toilets and urinals in a public-restroom setting, while Abney et al. (2021) reviews the larger evidence base and its unresolved measurement questions. These results establish sources, transport, and temporal change; they do not establish that a stall's cumulative visit count is a sufficient statistic for an entrant's infection risk.

Those results motivate a dynamical state variable. They do not justify treating "number of previous users" as a measured pathogen dose. A general linear surface-load model is

$$\mathrm{d}m_i(t) = -\kappa_i m_i(t)\ \mathrm{d}t + \delta_{m,i}\ \mathrm{d}N_i^{\text{in}}(t) + \sigma_i s_i(t)\ \mathrm{d}t - m_i(t^-)\ \mathrm{d}N_i^{\text{cl}}(t), \tag{2.4}$$

where $\delta_{m,i}$ is load deposited per entry, σ_i is load accumulated per occupied unit time, and $N_i^{\text{cl}}(t)$ is a

cleaning-counting process whose jump resets the modeled load. The symbol m_i may represent visible soil, odor, perceived recent use, a selected organism, or a composite index. Its interpretation and measurement unit must be declared before H can be estimated.

This disciplined definition makes the hygiene extension stronger. It permits separate tests of the statements "use raises the selected load," "cleaning reduces it," and "users avoid it." No single microbiological proxy is forced to carry all three claims.

2.7 A generative measurement model

Let visits be indexed by $r = 1, \dots, R$. For visit r, record the arrival time t_r, free set F_r, chosen stall I_r, duration L_r, a visit-type covariate Z_r, and a consumptive mark X_r. A complete observation model factors as

$$\begin{aligned} &f\big(I_r, L_r, X_r \mid F_r, \boldsymbol{s}(t_r^-), \boldsymbol{m}(t_r^-), Z_r\big) \\ &\quad = P_{I_r}\big(F_r, \boldsymbol{s}(t_r^-), \boldsymbol{m}(t_r^-), Z_r\big) f_L\big(L_r \mid I_r, \boldsymbol{s}(t_r^-), Z_r\big) f_X\big(X_r \mid I_r, L_r, Z_r\big). \end{aligned} \tag{2.5}$$

The first factor estimates choice; the second estimates service and any privacy effect on duration; the third connects visits to depletion. The occupancy and hygiene processes update between visits.

If only choices and free sets are required, the conditional log-likelihood is

$$\ell_{\text{choice}}(\vartheta) = \sum_{r=1}^{R} \left[-\beta C_{I_r}(t_r; \vartheta) - \log \sum_{k \in F_r} e^{-\beta C_k(t_r; \vartheta)} \right], \tag{2.6}$$

where ϑ collects the estimable parameters. This likelihood uses each arrival as a risk set, exactly as a conditional logit model does. Observing the free set is crucial: a person cannot reveal a preference for an occupied stall.

2.7.1 What aggregate data lose

Suppose only total entries $(A_1, \dots, A_n)$ are observed. Then the data discard which alternatives were available and which neighbors were occupied. Baseline cost g_i remains partly identifiable, but instantaneous privacy J is not: many pairs (g, J) induce the same aggregate shares after averaging over states. If only depletion $(D_1, \dots, D_n)$ is observed, the nuisance means $(\xi_1, \dots, \xi_n)$ introduce an additional nonidentifiability. This is not an abstract statistical warning; it is the reason a middle-heavy roll count cannot separate centrality, door exposure, duration sorting, and consumption sorting.

2.8 A field design that identifies the model

A decisive field study does not require cameras, audio, or personally identifying data. The minimal design uses anonymous stall-door and occupancy sensors, an entrance counter, and time-stamped cleaning logs. It records:

- arrival and departure times to the nearest second;
- the free set and occupancy configuration immediately before each choice;
- stall identity, without recording user identity;
- cleaning start, end, and stall coverage;
- fixed geometry: distance from entrance, walls, partitions, dimensions, and accessibility features;

- optional non-imaging hygiene measurements collected on a declared schedule.

Three planned perturbations make the causal pieces separable.

1. **Availability perturbation.** Rotate brief, preannounced maintenance closures across stalls. This changes the choice set without changing geometry.
2. **Cleaning perturbation.** Randomize the order or interval of otherwise identical cleaning across comparable time blocks. This changes m_i independently of stable position.
3. **Information perturbation.** Display accurate, delayed, or no recent-use indicators in randomized blocks. This changes perceived hygiene and delay without altering physical load at the instant of assignment.

The resulting panel identifies the positional coefficients from empty and balanced states, J and the spatial kernel from within-free-set neighbor variation, holding-time effects from departures, deposition and decay from hygiene samples and cleaning, and delay sensitivity from the information perturbation. Chapter 10 develops the likelihood and power calculations.

2.9 Scientific status of the conclusions

What is established and what is derived

Established in prior observations: a middle-versus-end depletion inequality occurred in the four-stall setting studied by Christenfeld (1995); interpersonal distance affected measured urination behavior in Middlemist et al. (1976); one small snapshot study found unequal snapshot counts across three stalls (Heston 2023); survey respondents distinguished privacy and cleanliness in stated bathroom evaluations (Corradi et al. 2020); and restroom surface communities respond to human use and re-establish after decontamination (Flores et al. 2011; Gibbons et al. 2015).
Derived in this monograph: mathematical consequences of combining a positional cost, a spatial privacy kernel, stochastic occupancy, an own-use hygiene stock, and delayed response under the declared exact, reduced, and delayed models.
Predicted for new data: state-conditional edge advantage from privacy, hygiene-induced flattening, advice attenuation, and delay-driven autocorrelation or oscillation. These are falsifiable properties of a future event-level data set.

The evidence is therefore neither empty nor complete. It is sufficient to motivate the components but insufficient to estimate the full model without new measurements. The next chapters supply the mathematics that maps those measurements to the distribution of use.

Part II

From individual choice to collective use

3

Mathematical foundations from first principles

3.1 States, events, and distributions

With n stalls, write

$$\mathcal{I} = \{1, 2, \ldots, n\}.$$

Stall 1 is nearest the entrance unless a different indexing convention is declared. At time t, the binary variable

$$s_i(t) = \begin{cases} 1, & \text{stall } i \text{ is occupied,} \\ 0, & \text{stall } i \text{ is free} \end{cases}$$

records occupancy. The vector

$$\boldsymbol{s}(t) = (s_1(t), \ldots, s_n(t)) \in \{0, 1\}^n$$

is a complete occupancy state, and

$$S(t) = \{i : s_i(t) = 1\}$$

is the equivalent set notation. There are 2^n states because every stall has two possibilities. The number busy is

$$K(t) = |S(t)| = \sum_{i=1}^{n} s_i(t).$$

A **probability distribution** assigns a nonnegative number $\pi(S)$ to each state, with

$$\sum_{S \subseteq \mathcal{I}} \pi(S) = 1.$$

The marginal occupancy probability of stall i is

$$q_i = \mathbb{P}\{i \in S\} = \sum_{S \ni i} \pi(S). \tag{3.1}$$

Two stalls can have the same marginal q_i while participating in different joint patterns. For example, alternating configurations can make neighboring occupancies strongly negatively correlated even if all marginals equal $1/2$. The state distribution contains that dependence; the vector of marginals does not.

3.2 Poisson arrivals

Let $N(t)$ be the number of potential arrivals during an interval of length t. A homogeneous Poisson process with rate $\Lambda > 0$ satisfies

$$\mathbb{P}\{N(t) = k\} = e^{-\Lambda t}\frac{(\Lambda t)^k}{k!}, \qquad k = 0, 1, 2, \ldots. \tag{3.2}$$

Its mean and variance are both Λt. More important for the dynamics is its small-time behavior:

$$\mathbb{P}\{\text{one arrival in } [t, t + \mathrm{d}t]\} = \Lambda\, \mathrm{d}t + o(\mathrm{d}t), \tag{3.3}$$

$$\mathbb{P}\{\text{two or more}\} = o(\mathrm{d}t). \tag{3.4}$$

The notation $o(\mathrm{d}t)$ means a quantity whose ratio to $\mathrm{d}t$ tends to zero as $\mathrm{d}t \downarrow 0$. Thus, to first order, at most one event occurs in a very short interval.

The time T_A between Poisson arrivals is exponentially distributed:

$$\mathbb{P}\{T_A > t\} = e^{-\Lambda t}, \qquad f_{T_A}(t) = \Lambda e^{-\Lambda t}. \tag{3.5}$$

The exponential distribution is memoryless:

$$\mathbb{P}\{T_A > u + v \mid T_A > u\} = \mathbb{P}\{T_A > v\}.$$

It is a useful baseline for public traffic because independent contributions from many unscheduled visitors often approximate Poisson input. Time-of-day variation is handled by replacing Λ with $\Lambda(t)$; bursts caused by events or queues require a richer arrival process.

3.3 Holding times and hazards

Suppose a visit to stall i lasts for a random time L_i. An exponential holding time with departure rate $\mu_i > 0$ has density

$$f_i(\ell) = \mu_i e^{-\mu_i \ell}, \qquad \ell \geq 0,$$

mean $\mathbb{E}L_i = 1/\mu_i$, and constant hazard

$$\lim_{\mathrm{d}t \downarrow 0} \frac{\mathbb{P}\{\ell < L_i \leq \ell + \mathrm{d}t \mid L_i > \ell\}}{\mathrm{d}t} = \mu_i. \tag{3.6}$$

Constant hazard means that the probability of ending in the next instant does not depend on elapsed duration. Real restroom visits are heterogeneous and need not be exponential. The exponential model matters because it makes the occupancy vector Markov: the present state then contains all information needed for the next-event distribution. Phase-type or visit-class mixtures preserve Markov structure after augmenting the state.

3.4 Random utility and the logit rule

3.4.1 From cost plus noise to choice probabilities

Let the perceived cost of free stall i be

$$\widetilde{C}_i = C_i - \epsilon_i^U.$$

The entrant selects the smallest perceived cost. Suppose the ϵ_i^U are independent maximum-type Gumbel variables at scale $1/\beta$. Equivalently, an additive cost error $\epsilon_i^C = -\epsilon_i^U$ has the corresponding minimum-type

extreme-value distribution, and utility is $U_i = -C_i + \epsilon_i^U$. Then the probability that i has the largest utility is

$$P_i = \frac{e^{-\beta C_i}}{\sum_{k \in F} e^{-\beta C_k}}, \qquad i \in F, \tag{3.7}$$

where F is the free set. This is the multinomial or conditional logit model (Luce 1959; McFadden 1974; Yellott 1977).

For completeness, let the Gumbel cumulative distribution at scale $1/\beta$ be

$$G(x) = \exp[-e^{-\beta x}], \qquad g(x) = \beta e^{-\beta x} \exp[-e^{-\beta x}].$$

The event that option i wins conditional on $\epsilon_i^U = z$ is

$$\epsilon_k^U \le z + U_i^0 - U_k^0 \quad \text{for every } k \ne i,$$

where $U_i^0 = -C_i$. Integrating over z gives

$$\begin{aligned} P_i &= \int_{-\infty}^{\infty} g(z) \prod_{k \ne i} G(z + U_i^0 - U_k^0)\, \mathrm{d}z \\ &= \frac{e^{\beta U_i^0}}{\sum_k e^{\beta U_k^0}} = \frac{e^{-\beta C_i}}{\sum_k e^{-\beta C_k}}. \end{aligned} \tag{3.8}$$

3.4.2 Interpretation of choice precision

Only products such as βC_i affect the probabilities, so the absolute scale of cost is not separately identified from β. One coefficient or the error scale must be normalized. The limiting cases are instructive:

$$\beta \downarrow 0 \quad \Longrightarrow \quad P_i \to \frac{1}{|F|}, \tag{3.9}$$

$$\beta \uparrow \infty \quad \Longrightarrow \quad P_i \to 0 \text{ off the cost-minimizing set.} \tag{3.10}$$

At finite β, every free stall with finite cost has positive choice probability. A model with truly unacceptable stalls therefore needs either an outside option or infinite cost; a large but finite coefficient is not logical exclusion.

3.4.3 The independence restriction

The logit odds of two free stalls satisfy

$$\frac{P_i}{P_j} = e^{-\beta(C_i - C_j)},$$

independent of the other alternatives. This independence-of-irrelevant- alternatives property is convenient and sometimes too strong. Nested logit, mixed logit, or random coefficients can represent groups such as accessible stalls, wall stalls, or user types. The present work uses logit because it isolates the dynamic consequences of position, privacy, and hygiene; the state process remains definable under any smooth choice map (McFadden and K. Train 2000; K. E. Train 2009).

3.5 Continuous-time Markov chains

3.5.1 Transition rates

A continuous-time Markov chain on the 2^n occupancy states is specified by a generator Q. For distinct states S and T, $Q_{S,T} \geq 0$ is the instantaneous transition rate from S to T, and

$$Q_{S,S} = -\sum_{T \neq S} Q_{S,T}.$$

Only one-stall changes are possible under the basic arrival-departure model:

$$Q_{S,S\cup\{i\}} = \lambda_i(S), \qquad i \notin S, \tag{3.11}$$

$$Q_{S,S\setminus\{i\}} = \mu_i(S), \qquad i \in S. \tag{3.12}$$

If an arrival stream has rate Λ and every nonfull arrival selects according to $P_i(S)$, then

$$\lambda_i(S) = \Lambda P_i(S), \qquad \sum_{i\notin S} \lambda_i(S) = \Lambda \quad \text{when } |S| < n. \tag{3.13}$$

3.5.2 Stationarity

Let π_t be the row vector of state probabilities at time t. It obeys the Kolmogorov forward equation

$$\frac{\mathrm{d}}{\mathrm{d}t}\pi_t = \pi_t Q. \tag{3.14}$$

A stationary distribution π satisfies

$$\pi Q = \mathbf{0}, \qquad \sum_S \pi(S) = 1. \tag{3.15}$$

For a finite irreducible chain, this solution exists, is unique, and equals the long-run fraction of time spent in each state. The per-stall occupancy probabilities follow from (3.1); entry rates follow from

$$a_i = \sum_{S: i\notin S} \pi(S)\lambda_i(S). \tag{3.16}$$

3.6 Detailed balance and Gibbs distributions

A stationary chain is **reversible** if, in stationarity, every directed transition is balanced by its reverse:

$$\pi(S)Q_{S,T} = \pi(T)Q_{T,S} \quad \text{for all } S, T. \tag{3.17}$$

Summing (3.17) over T gives $\pi Q = 0$, so detailed balance is sufficient for stationarity. It is stronger than stationarity: a nonreversible chain can have circulating probability currents whose net divergence is zero.

For birth and death of one occupant, detailed balance would require

$$\frac{\pi(S\cup\{i\})}{\pi(S)} = \frac{\lambda_i(S)}{\mu_i(S\cup\{i\})}. \tag{3.18}$$

If the right side is an exponential of the energy increment, the state law can be reconstructed by multiplying increments. Let

$$E(S) = \sum_{i\in S} g_i + J \sum_{\substack{i<j \\ i,j\in S}} K_{ij}. \tag{3.19}$$

Then

$$E(S \cup \{i\}) - E(S) = g_i + J \sum_{j \in S} K_{ij}.$$

If

$$\frac{\lambda_i(S)}{\mu} = \rho_0 \exp\{-\beta[E(S \cup \{i\}) - E(S)]\}, \tag{3.20}$$

the stationary distribution is the lattice-gas law

$$\pi_{\mathrm{G}}(S) = \frac{1}{Z} \rho_0^{|S|} e^{-\beta E(S)}, \qquad Z = \sum_{T \subseteq \mathcal{I}} \rho_0^{|T|} e^{-\beta E(T)}. \tag{3.21}$$

The normalizer Z is called the partition function.

3.6.1 The cycle test

Reversibility can be checked without knowing π. Kolmogorov's cycle criterion requires the product of transition rates around every closed cycle to equal the product in the reverse direction. Consider the square

$$S \to S \cup \{i\} \to S \cup \{i, j\} \to S \cup \{j\} \to S.$$

With common departure rates, the condition reduces to

$$\lambda_i(S) \lambda_j(S \cup \{i\}) = \lambda_j(S) \lambda_i(S \cup \{j\}). \tag{3.22}$$

This single equation will expose why normalizing choice over all free stalls usually destroys the Gibbs representation.

3.7 Potentials, entropy, and equilibrium

At a population scale, let

$$\Delta_n = \left\{ \boldsymbol{p} \in \mathbb{R}^n : p_i \geq 0, \ \sum_{i=1}^{n} p_i = 1 \right\}$$

be the probability simplex. A potential $\Phi(\boldsymbol{p})$ is a scalar function whose partial derivative gives the marginal cost faced by choosing stall i, up to a common constant. The central potential in this monograph is

$$\Phi(\boldsymbol{p}) = \boldsymbol{g}^{\top} \boldsymbol{p} + \frac{\alpha}{2} \boldsymbol{p}^{\top} K \boldsymbol{p} + \frac{h}{2} \|\boldsymbol{p}\|_2^2 + \frac{1}{\beta} \sum_{i=1}^{n} p_i \log p_i. \tag{3.23}$$

Its four terms represent baseline position, privacy cross-load, hygiene own-load, and dispersed choice.

The function $p \log p$ is defined continuously at zero by $0 \log 0 = 0$. For $p > 0$,

$$\frac{\mathrm{d}}{\mathrm{d}p}(p \log p) = 1 + \log p, \qquad \frac{\mathrm{d}^2}{\mathrm{d}p^2}(p \log p) = \frac{1}{p}.$$

The first derivative tends to $-\infty$ at zero. Therefore a finite-β minimizer of (3.23) with finite costs lies in the interior of the simplex. The first-order conditions give

$$p_i = \frac{\exp\{-\beta[g_i + \alpha(K\boldsymbol{p})_i + h p_i]\}}{\sum_k \exp\{-\beta[g_k + \alpha(K\boldsymbol{p})_k + h p_k]\}}, \tag{3.24}$$

which is a logit quantal-response fixed point (Blume 1993; McKelvey and Palfrey 1995; Monderer and Shapley 1996; Sandholm 2010).

The Hessian is

$$\nabla^2\Phi(\boldsymbol{p}) = \alpha K + hI + \frac{1}{\beta}\operatorname{diag}(1/p_i). \tag{3.25}$$

Positive definiteness on directions whose coordinates sum to zero throughout the simplex interior implies global strict convexity and hence a unique equilibrium. Repulsive privacy does not automatically make the potential convex: a symmetric matrix with positive off-diagonal entries can have negative eigenvalues. Hygiene and entropy add positive diagonal curvature; sufficiently strong privacy can instead produce several anti-coordinated local minima.

3.8 Linearization and spatial modes

Let $\boldsymbol{p}^*$ be an equilibrium and perturb it by a small vector $\boldsymbol{\epsilon}$ with $\mathbf{1}^\mathsf{T}\boldsymbol{\epsilon} = 0$. A differentiable map $F(\boldsymbol{p})$ has the first-order expansion

$$F(\boldsymbol{p}^* + \boldsymbol{\epsilon}) = F(\boldsymbol{p}^*) + DF(\boldsymbol{p}^*)\boldsymbol{\epsilon} + O(\|\boldsymbol{\epsilon}\|^2).$$

The matrix DF is the Jacobian. Linearization turns a nonlinear local problem into matrix algebra.

If a symmetric spatial matrix K has orthonormal eigenvectors $\boldsymbol{v}_r$ and eigenvalues λ_r, then every perturbation can be written

$$\boldsymbol{\epsilon} = \sum_r c_r \boldsymbol{v}_r.$$

The modes evolve independently whenever the remaining linearized operators are functions of K. On a line, the low-frequency modes describe broad left-versus-right or center-versus-edge differences; high-frequency modes describe alternating popularity. This decomposition is the cleanest way to understand how privacy can amplify one spatial pattern while hygiene damps all nonuniform patterns.

3.9 Delay differential equations

An ordinary differential equation uses the current state:

$$\dot{x}(t) = f(x(t)).$$

A delay differential equation also uses a past state:

$$\dot{x}(t) = f(x(t), x(t-\tau)).$$

To determine the future, one must specify an initial history on $[-\tau, 0]$, not just $x(0)$.

The linear scalar equation relevant to hygiene is

$$\dot{x}(t) = -\kappa x(t) - Ax(t-\tau). \tag{3.26}$$

Trying a mode $x(t) = e^{zt}$ gives the characteristic equation

$$z + \kappa + Ae^{-z\tau} = 0. \tag{3.27}$$

Unlike an ordinary differential equation, it has infinitely many complex roots. Stability requires every root to have negative real part. A conjugate-pair crossing of the imaginary axis is an oscillatory loss of linear stability.

Calling the nonlinear transition a Hopf bifurcation additionally requires transversality, nondegeneracy, and a Lyapunov-coefficient calculation; a stable cycle does not follow from the linear crossing alone. Setting $z = i\omega$ separates (3.27) into two real equations:

$$\kappa + A\cos(\omega\tau) = 0, \qquad \omega - A\sin(\omega\tau) = 0. \tag{3.28}$$

Chapter 8 solves these equations and verifies the crossing direction.

3.10 The role of computation

For $n \leq 12$ or so, all 2^n occupancy states can be enumerated and the stationary equations (3.15) solved directly. This gives an exact numerical answer up to floating-point error. Larger systems require structure: transfer matrices for a Gibbs chain, sparse linear algebra, simulation, or mean-field approximation.

Computation in this monograph serves four roles.

1. It evaluates formulas whose sums are finite but tedious.
2. It tests limiting approximations against the exact finite-state chain.
3. It displays parameter dependence that is clearer as a figure than as a table of derivatives.
4. It checks reproducibility: every plotted quantity is regenerated by the accompanying Python program from declared parameters and a fixed random seed.

Numerical agreement is not used as a substitute for a proof. Theorems establish structural claims; computation reports the size of effects in specified examples.

4

The static choice problem

4.1 Geometry before psychology

Consider a row of stalls at walking distances

$$0 = x_1 < x_2 < \cdots < x_n$$

from the point at which the entrant can commit to a stall. A baseline cost can contain any fixed stall attributes:

$$g_i = ae^{-x_i/\lambda} + wx_i + z_i^{\top}\boldsymbol{\gamma}. \tag{4.1}$$

The parameters have direct interpretations:

- $a > 0$ is the cost of exposure at the entrance;
- $\lambda > 0$ is the distance over which that exposure decays;
- $w > 0$ is marginal walking cost;
- z_i records fixed features such as a side wall, larger floor area, a changing table, a damaged latch, or an accessibility designation.

The decomposition is intentionally physical. An abstract middle indicator can fit a middle effect, but it cannot predict what happens when the row is longer, the door moves, a partition is added, or the first stall is set back from the entrance.

4.1.1 The continuous compromise

Ignore the fixed covariates and write

$$g(x) = ae^{-x/\lambda} + wx, \qquad 0 \le x \le L.$$

Its derivatives are

$$g'(x) = -\frac{a}{\lambda}e^{-x/\lambda} + w, \qquad g''(x) = \frac{a}{\lambda^2}e^{-x/\lambda} > 0. \tag{4.2}$$

Thus g is strictly convex and has at most one unconstrained minimum. Write

$$x_0 = \lambda \log\!\left(\frac{a}{\lambda w}\right)$$

for its unconstrained stationary point.

Proposition 4.1 (Interior positional optimum)**.** *The minimizer of g on $[0, L]$ is*

$$x^\star = [x_0]_{[0,L]}\,, \tag{4.3}$$

where $[y]_{[0,L]} = \min\{L, \max\{0, y\}\}$. An interior minimum exists exactly when

$$\lambda w < a < \lambda w e^{L/\lambda}. \tag{4.4}$$

Proof. Strict convexity makes any stationary point the unique unconstrained minimizer. Solving $g'(x) = 0$ gives $x = x_0$. The stationary point lies in $(0, L)$ exactly under (4.4); otherwise convexity places the constrained minimum at the appropriate endpoint. □

The result supplies the central intuition. Near the entrance, moving farther away can sharply reduce exposure. Farther into the room, that benefit saturates while every additional step still costs w. A middle stall is the balance point whenever it lies inside the installed row.

4.1.2 The exact discrete optimum

Real stalls are discrete. For equal spacing $\Delta > 0$, let $x_i = (i-1)\Delta$. The forward difference is

$$g_{i+1} - g_i = w\Delta - a e^{-x_i/\lambda}\big(1 - e^{-\Delta/\lambda}\big). \tag{4.5}$$

The right side increases with i, so the sequence decreases and then increases. Define

$$x_{\text{cross}} = \lambda \log\left[\frac{a(1 - e^{-\Delta/\lambda})}{w\Delta}\right]. \tag{4.6}$$

The discrete minimizer is the first stall i for which $x_i \geq x_{\text{cross}}$, with the index clipped to the installed row; if a forward difference is exactly zero, the two adjacent stalls tie. As $\Delta/\lambda \to 0$,

$$\frac{1 - e^{-\Delta/\lambda}}{\Delta} \to \frac{1}{\lambda},$$

and (4.6) converges to the unconstrained point x_0. After clipping to the installed interval, the discrete minimizer converges to the constrained optimum $x^\star$ in (4.3).

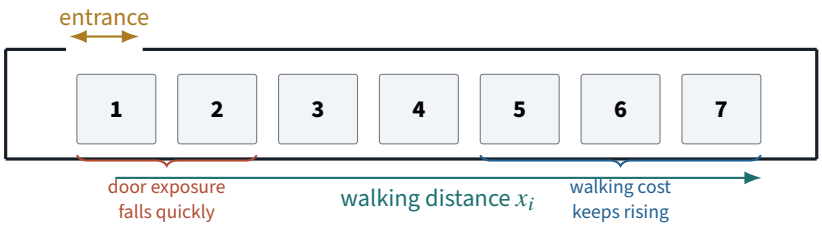


Figure 4.1: The positional compromise. Stall labels are indices, not assumed preferences. The cost minimum can lie in the interior because two physical gradients point in opposite directions.

4.2 Immediate privacy as a spatial interaction

Let $d_{ij} = |x_i - x_j|$. A symmetric privacy kernel $K_{ij} = K_{ji} \geq 0$ converts occupied positions into the cost of choosing i. Set $K_{ii} = 0$. For equally spaced stalls, one convenient specification normalized to unit adjacent

interaction is

$$K_{ij} = \begin{cases} \exp[-(|i-j|-1)/\ell], & i \neq j, \\ 0, & i = j, \end{cases} \tag{4.7}$$

where $\ell > 0$ is a screening length measured in stall spacings. The current privacy cost is

$$r_i(S) = J \sum_{j \in S} K_{ij}. \tag{4.8}$$

Adjacent occupancy contributes J; more distant occupancy contributes less. Physical partitions can be represented by multiplying K_{ij} by a screening factor in $(0, 1)$, and asymmetric sightlines can be represented by a directed kernel. Symmetry is retained in the main theory because it creates a potential and permits exact energy calculations.

4.2.1 Boundary geometry

Let $r = e^{-1/\ell} \in (0, 1)$. If every other stall were equally likely to be occupied, the total exposure of stall i would be proportional to its kernel row sum

$$\begin{aligned} R_i &= \sum_{j \neq i} K_{ij} \\ &= \sum_{d=1}^{i-1} r^{d-1} + \sum_{d=1}^{n-i} r^{d-1} \\ &= \frac{2 - r^{i-1} - r^{n-i}}{1-r}. \end{aligned} \tag{4.9}$$

At the left edge,

$$R_1 = \frac{1 - r^{n-1}}{1-r};$$

an interior stall receives positive contributions from both sides. The row sum is smallest at the edges and largest at the middle position or middle pair. This is the exact boundary advantage that a privacy-only theory produces when occupancy is diffuse.

Privacy does not select a first stall in an empty room

If $S = \varnothing$, then $r_i(S) = 0$ for every i. A rule based only on current privacy is indifferent among all free stalls. It cannot explain a systematic middle choice on first entry. A privacy explanation of first choice must be forward-looking: it must value where later users are expected to stand.

4.3 Conditional choice in a fixed state

Combining position and current privacy, the cost of a free stall is

$$C_i(S) = g_i + J \sum_{j \in S} K_{ij}.$$

The logit probability is

$$P_i(S) = \frac{\mathbf{1}\{i \notin S\} e^{-\beta C_i(S)}}{Z(S)}, \qquad Z(S) = \sum_{k \notin S} e^{-\beta C_k(S)}. \tag{4.10}$$

The normalizer $Z(S)$ is a choice-set partition function. It depends on the state and will later be responsible for nonreversibility.

4.3.1 Odds and marginal effects

For two free stalls i and j,

$$\log \frac{P_i(S)}{P_j(S)} = -\beta\{g_i - g_j\} - \beta J \sum_{k \in S} (K_{ik} - K_{jk}). \tag{4.11}$$

The log odds separate baseline position from the observed occupancy configuration. This is the key identification equation for event-level data.

Differentiating the softmax with respect to costs gives

$$\frac{\partial P_i}{\partial C_j} = -\beta P_i(\mathbf{1}\{i = j\} - P_j). \tag{4.12}$$

Increasing the cost of stall i lowers its own probability at rate $\beta P_i(1 - P_i)$ and raises every alternative's probability at rate $\beta P_i P_j$. The total derivative sums to zero because probabilities must still sum to one.

4.3.2 A five-stall example

Take five identical stalls, set $g_i = 0$, and suppose stall 3 is occupied. With the kernel (4.7), the free-stall costs are

$$(Jr,\ J,\ \text{occupied},\ J,\ Jr).$$

Therefore

$$P_1 = P_5 = \frac{e^{-\beta Jr}}{2e^{-\beta Jr} + 2e^{-\beta J}}, \tag{4.13}$$

$$P_2 = P_4 = \frac{e^{-\beta J}}{2e^{-\beta Jr} + 2e^{-\beta J}}. \tag{4.14}$$

Because $r < 1$, the outer stalls are favored. The ratio is

$$\frac{P_1}{P_2} = e^{\beta J(1-r)}.$$

The product βJ measures effective privacy sensitivity; the screening factor r determines how much farther distance helps.

4.4 Unavailable, unacceptable, and outside options

An occupied stall is unavailable and has probability zero by definition. A free stall can be unacceptable for a different reason. To model refusal, add an outside option 0 with cost C_0, representing waiting, leaving, or

using a urinal:

$$P_i^{\text{out}}(S) = \frac{e^{-\beta C_i(S)}}{e^{-\beta C_0} + \sum_{k \notin S} e^{-\beta C_k(S)}}, \qquad i \notin S, \qquad P_0(S) = \frac{e^{-\beta C_0}}{e^{-\beta C_0} + \sum_{k \notin S} e^{-\beta C_k(S)}}. \tag{4.15}$$

The accepted arrival rate becomes

$$\Lambda[1 - P_0(S)],$$

which is state dependent. This distinction separates two systems:

- **forced allocation**: every entrant takes a stall if any is free;
- **elastic demand**: some entrants wait or leave when all free choices are costly.

The first system has a simple busy-count law under common service rates. The second can jam below physical capacity and is the natural soft extension of random sequential exclusion.

4.5 Fixed heterogeneity and user types

The average entrant is not a literal person. Let θ index a user type, with type-specific walking cost w_θ, privacy sensitivity J_θ, hygiene sensitivity H_θ, choice precision β_θ, and valuation of stall attributes $\boldsymbol{\gamma}_\theta$. Conditional choice is

$$P_i(S, \boldsymbol{m}, \theta) \propto \exp\{-\beta_\theta [g_i(\theta) + J_\theta (K\boldsymbol{s})_i + H_\theta m_i]\}.$$

If types arrive with distribution F_Θ, the observed probability is the mixture

$$\overline{P}_i(S, \boldsymbol{m}) = \int P_i(S, \boldsymbol{m}, \theta) \, \mathrm{d}F_\Theta(\theta). \tag{4.16}$$

This mixed logit allows some users to value the nearest stall and others to value the far wall. It also prevents the accessible stall from being treated as a nominally identical alternative: users for whom its features are necessary have a distinct choice set or coefficient vector.

The men who choose a toilet stall rather than a urinal are themselves a selected subpopulation. Their duration and privacy coefficients need not match those estimated from urinal use. The mathematical framework transfers; the numerical parameters do not transfer without data.

4.6 Scope of the static model

For a declared state S, hygiene vector $\boldsymbol{m}$, geometry, and parameter vector, (4.10) gives a complete distribution of the next choice. It predicts how moving the door, adding a partition, occupying a neighbor, or changing a hygiene indicator changes the odds.

It does not yet give long-run usage. The state seen by the next entrant is generated by all previous choices and departures. A stall favored in the empty room becomes unavailable more often; a privacy-sensitive entrant changes the state faced by the next user. Closing that feedback loop is the subject of Chapter 5.

5

From choices to an occupancy process

5.1 Closing the feedback loop

The static rule gives the next choice conditional on the present state. Long-run use requires the distribution of the state itself. This chapter constructs that distribution and resolves a subtle modeling point: an exponential choice rule does not automatically imply a Gibbs stationary law. The arrival protocol determines whether detailed balance holds.

For now, hold hygiene fixed and write the energy increment associated with occupying free stall i in state S as

$$\Delta_i E(S) = g_i + J \sum_{j \in S} K_{ij}. \tag{5.1}$$

The corresponding state energy is

$$E(S) = \sum_{i \in S} g_i + J \sum_{\substack{i<j \\ i,j \in S}} K_{ij}, \tag{5.2}$$

so that $E(S \cup \{i\}) - E(S) = \Delta_i E(S)$.

5.2 Protocol N: normalize over the free stalls

Under the **normalized-entry protocol**, potential arrivals form a Poisson process of rate Λ. If at least one stall is free, the entrant always takes one. Conditional on state S,

$$\lambda_i^{\mathrm{N}}(S) = \Lambda \frac{e^{-\beta \Delta_i E(S)}}{Z_{\mathrm{N}}(S)}, \qquad Z_{\mathrm{N}}(S) = \sum_{k \notin S} e^{-\beta \Delta_k E(S)}. \tag{5.3}$$

If $S = \mathcal{I}$, the arrival is blocked. Assume initially that each occupant departs at rate μ, independently of stall and state:

$$Q_{S,S\setminus\{i\}} = \mu, \qquad i \in S. \tag{5.4}$$

The important identity is

$$\sum_{i \notin S} \lambda_i^{\mathrm{N}}(S) = \Lambda \quad \text{for every nonfull } S. \tag{5.5}$$

Choice rearranges the arrival rate across free stalls but does not change the total accepted rate.

5.2.1 The busy-count law

Theorem 5.1 (Erlang loss law under arbitrary forced choice)**.** *Suppose potential arrivals are Poisson with rate Λ, every arrival enters whenever at least one stall is free, and all holding times are independent exponentials*

with rate μ. The number of occupied stalls $K(t)$ is a birth–death chain with rates

$$k \to k+1 \text{ at rate } \Lambda \quad (k < n), \qquad k \to k-1 \text{ at rate } k\mu \quad (k > 0),$$

regardless of how an entrant chooses among the free stalls. Its stationary distribution is

$$\mathbb{P}\{K = k\} = \frac{\rho^k/k!}{\sum_{r=0}^{n} \rho^r/r!}, \qquad \rho = \frac{\Lambda}{\mu}. \tag{5.6}$$

Proof. From every state with $k < n$ occupants, the total rate of a transition to level $k+1$ is Λ by (5.5). From every state with $k > 0$, any of the k occupants can depart, so the total rate to level $k-1$ is $k\mu$. These rates depend only on k, which makes the partition by busy count strongly lumpable. Detailed balance for the one-dimensional birth–death chain gives

$$\pi_{k+1}(k+1)\mu = \pi_k \Lambda,$$

hence $\pi_k = \pi_0 \rho^k/k!$. Normalization yields (5.6). □

The blocking probability is the Erlang–B formula

$$B(n, \rho) = \mathbb{P}\{K = n\} = \frac{\rho^n/n!}{\sum_{r=0}^{n} \rho^r/r!}. \tag{5.7}$$

Poisson arrivals see time averages, so the fraction of arrivals that encounter a full system is also $B(n, \rho)$. The accepted throughput is

$$\Lambda_{\text{acc}} = \Lambda[1 - B(n, \rho)], \tag{5.8}$$

and rate conservation gives

$$\mathbb{E}K = \rho[1 - B(n, \rho)]. \tag{5.9}$$

These identities are the classical loss-system result and its modern loss-network setting (Erlang 1918; Kelly 1979, 1991).

What the Erlang loss model answers

The Erlang loss distribution determines how many stalls are occupied, and its full-state probability is the Erlang–B blocking formula. It does not determine which stalls are occupied. Positional and privacy parameters disappear from (5.6) because forced choice changes only the allocation within each busy-count level.

5.3 The full stationary distribution

Let $\pi^{\mathrm{N}}(S)$ denote the stationary probability of state S under Protocol N. It is the unique solution of

$$\pi^{\mathrm{N}} Q^{\mathrm{N}} = 0, \qquad \sum_{S \subseteq \mathcal{I}} \pi^{\mathrm{N}}(S) = 1. \tag{5.10}$$

The marginal occupancy and entry rates are

$$q_i^{\mathrm{N}} = \sum_{S \ni i} \pi^{\mathrm{N}}(S), \tag{5.11}$$

$$a_i^{\mathrm{N}} = \sum_{S : i \notin S} \pi^{\mathrm{N}}(S) \lambda_i^{\mathrm{N}}(S). \tag{5.12}$$

Flow into and out of each stall balances separately:

$$a_i^{\mathrm{N}} = \mu q_i^{\mathrm{N}}. \tag{5.13}$$

Thus the normalized entry share is

$$p_i^{\mathrm{N}} = \frac{a_i^{\mathrm{N}}}{\sum_j a_j^{\mathrm{N}}} = \frac{q_i^{\mathrm{N}}}{\mathbb{E}K}. \tag{5.14}$$

This equality uses common service rates. With rate μ_i in stall i, $a_i = \mu_i q_i$, so the entry and occupancy rankings can differ.

5.4 Why normalized logit is generally nonreversible

It is tempting to insert the energy (5.2) into a Gibbs formula and declare the problem solved. The normalization in (5.3) prevents that step.

Theorem 5.2 (Cycle obstruction for normalized entry)**.** *Assume a symmetric interaction kernel and common departure rate μ. For a state S and two distinct free stalls i, j, reversibility of the normalized chain requires*

$$Z_{\mathrm{N}}(S \cup \{i\}) = Z_{\mathrm{N}}(S \cup \{j\}). \tag{5.15}$$

If this equality fails for any such square, the chain is nonreversible and its transitions cannot be interpreted as a reversible Gibbs dynamics for the energy in (5.2).

Proof. Kolmogorov's cycle criterion on the square that adds i and j in opposite orders requires

$$\lambda_i^{\mathrm{N}}(S) \lambda_j^{\mathrm{N}}(S \cup \{i\}) = \lambda_j^{\mathrm{N}}(S) \lambda_i^{\mathrm{N}}(S \cup \{j\}).$$

The numerator along either path is the same:

$$\begin{aligned} & e^{-\beta \Delta_i E(S)} e^{-\beta \Delta_j E(S \cup \{i\})} \\ &\quad = e^{-\beta [E(S \cup \{i,j\}) - E(S)]} \\ &\quad = e^{-\beta \Delta_j E(S)} e^{-\beta \Delta_i E(S \cup \{j\})}. \end{aligned}$$

The common factor $\Lambda^2 / Z_{\mathrm{N}}(S)$ also cancels. What remains is (5.15). □

The equality is exceptional. A positional gradient makes the sets remaining after occupying i or j have different total attractiveness. Even with equal baselines, a boundary position and an interior position screen the remaining row differently. The normalized chain can therefore support stationary probability currents around state-space cycles.

The behavioral meaning is simple. An entrant who must choose one of the free stalls compares every free option. Removing one option changes the denominator and therefore changes every remaining birth rate. A local energy increment no longer determines the transition ratio.

5.5 Protocol G: local detailed balance

Now consider a different mechanism. Free stall i receives accepted occupancy attempts at rate

$$\lambda_i^{\mathrm{G}}(S) = \nu e^{-\beta \Delta_i E(S)}, \qquad i \notin S, \tag{5.16}$$

and occupied stalls empty at rate μ. The total accepted arrival rate is state dependent. One interpretation is a local proposal-and-accept process: opportunities directed toward a stall occur at a baseline rate, and the chance of settlement falls exponentially with the incremental cost. Unlike Protocol N, this process permits balking even when another stall is free. Product-form links between state-dependent loss systems and Markov random fields belong to the broader loss-network literature (Zachary and Ziedins 1999); the point here is to state the behavioral arrival protocol that makes such a law appropriate for this stall system.

Theorem 5.3 (Exact lattice-gas stationary law)**.** *Under* (5.16) *and common departure rate μ, the chain is reversible with stationary distribution*

$$\pi^{\mathrm{G}}(S) = \frac{1}{\mathcal{Z}} \zeta^{|S|} e^{-\beta E(S)}, \qquad \zeta = \frac{\nu}{\mu}, \tag{5.17}$$

where

$$\mathcal{Z} = \sum_{T \subseteq \mathcal{I}} \zeta^{|T|} e^{-\beta E(T)}. \tag{5.18}$$

Proof. For any $i \notin S$,

$$\begin{aligned} \frac{\pi^{\mathrm{G}}(S \cup \{i\})}{\pi^{\mathrm{G}}(S)} &= \zeta e^{-\beta [E(S \cup \{i\}) - E(S)]} \\ &= \frac{\nu}{\mu} e^{-\beta \Delta_i E(S)} = \frac{\lambda_i^{\mathrm{G}}(S)}{\mu}. \end{aligned}$$

This is detailed balance for every adjacent pair of states. □

Equation (5.17) is a one-dimensional lattice gas with an external field g_i, repulsive interaction JK_{ij}, and activity ζ. It permits transfer-matrix methods and exact derivatives of $\log \mathcal{Z}$. For example,

$$\mathbb{E}_{\mathrm{G}} |S| = \zeta \frac{\partial}{\partial \zeta} \log \mathcal{Z}. \tag{5.19}$$

The construction is the birth–death analogue of standard lattice-gas and Glauber dynamics (Glauber 1963; Lee and Yang 1952); the forward/reverse rate ratio, rather than the presence of an exponential alone, creates the Gibbs law.

5.6 The two protocols are different models

The protocols can be calibrated to have similar mean occupancy at one parameter setting. They need not have similar state correlations or response to a positional perturbation. The difference is empirically observable: under Protocol N, accepted throughput is exactly $\Lambda(1 - B)$ and depends on costs only through full-state occupancy; under Protocol G, a costly configuration can reduce accepted throughput before physical capacity is reached.

Table 5.1: Normalized forced entry and local detailed balance.

Property	Protocol N	Protocol G
Arrival behavior	always selects a free stall	acceptance rate depends on incremental cost
Total birth rate	Λ in every nonfull state	state dependent
Busy-count law	Erlang loss distribution under common μ; Erlang–B blocking	generally not the Erlang loss law
Reversibility	generally no	yes
Stationary law	solution of $\pi Q = 0$	explicit Gibbs lattice gas
Outside option	absent unless added	intrinsic through unaccepted proposals
Best use	traffic allocation with forced entry	local acceptance, balking, and statistical-mechanical analysis

5.7 Exact finite-state computation

For modest n, no approximation is necessary. Index the subsets of $\mathcal{I}$, construct the $2^n \times 2^n$ generator using the declared birth and death rates, and solve (5.10). Numerically, one row of Q^{T} is replaced by ones and the corresponding right-hand side by 1; the resulting linear system enforces both stationarity and normalization. Residuals

$$\|\pi Q\|_\infty, \qquad \left|\sum_S \pi(S) - 1\right|, \qquad \min_S \pi(S) \tag{5.20}$$

provide direct accuracy checks.

For Protocol N, a second check is that the aggregated probabilities

$$\widehat{\pi}_k = \sum_{S:|S|=k} \pi(S)$$

agree with (5.6). For Protocol G, direct stationarity must agree with (5.17). These independent identities catch generator orientation, normalization, and indexing errors.

5.8 Low- and high-load limits

5.8.1 Sparse traffic

Let $\rho = \Lambda/\mu$ under Protocol N. As $\rho \downarrow 0$, the empty state has probability $1 - O(\rho)$. Therefore the entry share is

$$p_i^{\mathrm{N}} = \frac{e^{-\beta g_i}}{\sum_k e^{-\beta g_k}} + O(\rho). \tag{5.21}$$

Current privacy appears only in the $O(\rho)$ correction because it requires another occupant to be present. Low-load data therefore identify baseline position especially cleanly.

5.8.2 Saturated traffic

As $\rho \uparrow \infty$, the system is full with probability tending to one and each stall's occupancy probability tends to one:

$$q_i^{\mathrm{N}} \to 1.$$

By stall-level flow balance,

$$a_i^{\mathrm{N}} = \mu q_i^{\mathrm{N}} \to \mu.$$

Consequently

$$p_i^{\mathrm{N}} \to \frac{1}{n}. \tag{5.22}$$

At very high forced load, every departure creates a vacancy that is soon filled, so every stall turns over at the same service-limited rate. Positional and privacy differences can create a finite-load center-to-edge crossover, but they cannot sustain unequal entry shares in the infinite-load limit when service rates are common.

Crossover is not a thermodynamic phase transition

For fixed finite n, the stationary probabilities of an irreducible finite-state chain are smooth functions of positive transition rates. A change from center-peaked to edge-peaked use is therefore a finite-system shape crossover, not a singular phase transition. One-dimensional finite-range Gibbs systems likewise have no ordinary finite-temperature bulk phase transition. A true bifurcation can occur in the mean-field equilibrium or in the delayed dynamic system; those claims are stated separately.

5.9 When the Erlang reduction fails

Theorem 5.1 relies on three structural conditions:

1. total accepted arrival rate is Λ in every nonfull state;
2. each of the k occupants departs at a common rate μ;
3. arrival and holding processes have the memoryless Markov form, or a more general insensitivity result is invoked for the count process.

An outside option violates the first condition. Stall-specific duration violates the second. Proximity-dependent duration makes the downward rate depend on the configuration, not just k. Time-varying traffic makes the distribution transient. These extensions are not defects; they are experimentally meaningful alternatives. Discrete phase variables can be added to a larger finite Markov state. A continuous hygiene stock instead produces a piecewise-deterministic Markov process, and a delayed signal requires a history state; neither retains the automatic 2^n finite enumeration.

6

Population equilibrium and theory of mind

6.1 Population-scale closure

The exact occupancy chain is the right object for a particular restroom with a small number of stalls. A population model is useful for different questions: How does a persistent distribution of beliefs reproduce itself? When is that distribution unique? How does privacy favor spatial patterns? How strongly does hygiene flatten a positional gradient?

Let $\boldsymbol{p} \in \Delta_n$ be the vector of long-run entry shares. A reduced mean-field cost takes the form

$$c_i(\boldsymbol{p}) = g_i + \alpha(K\boldsymbol{p})_i + hp_i. \tag{6.1}$$

The coefficient $\alpha \geq 0$ converts population entry share into expected concurrent privacy exposure. Under sparse traffic, common mean duration $1/\mu$, and negligible blocking, the occupancy probability satisfies $q_i \approx (\Lambda/\mu)p_i$, so $\alpha \approx J\Lambda/\mu$. At finite load, $\alpha \approx J\rho[1 - B(n, \rho)]$ is a more natural common-duration closure when occupancy and entry shares align. In either case, α is a reduced-form approximation to the exact map from entry shares to occupancy. The coefficient $h \geq 0$ is the hygiene own-load derived in Chapter 7.

6.2 The potential game

Because K is symmetric, the marginal costs (6.1) integrate to the potential

$$V(\boldsymbol{p}) = \boldsymbol{g}^\top \boldsymbol{p} + \frac{\alpha}{2}\boldsymbol{p}^\top K\boldsymbol{p} + \frac{h}{2}\|\boldsymbol{p}\|_2^2. \tag{6.2}$$

Indeed,

$$\frac{\partial V}{\partial p_i} = g_i + \alpha(K\boldsymbol{p})_i + hp_i = c_i(\boldsymbol{p}).$$

This is a nonatomic congestion or potential game (Monderer and Shapley 1996; Rosenthal 1973). Privacy creates cross-congestion: use of j raises the cost of nearby i. Hygiene creates own-congestion: use of i raises its own future cost.

Random utility adds an entropy term. Define the free-energy potential

$$\Phi_\beta(\boldsymbol{p}) = V(\boldsymbol{p}) + \frac{1}{\beta}\sum_{i=1}^{n} p_i \log p_i. \tag{6.3}$$

At $\beta = 0$ the expression is understood through the uniform-choice limit; for the analysis take $\beta > 0$.

Theorem 6.1 (Logit equilibrium as a variational problem)**.** *Every global minimizer of* (6.3) *on* Δ_n *is strictly*

positive and satisfies

$$p_i = \frac{\exp\{-\beta[g_i + \alpha(K\boldsymbol{p})_i + hp_i]\}}{\sum_{k=1}^{n} \exp\{-\beta[g_k + \alpha(K\boldsymbol{p})_k + hp_k]\}}. \tag{6.4}$$

Conversely, every solution of (6.4) *is a stationary point of* Φ_β *under the simplex constraint.*

Proof. Continuity on the compact simplex gives a minimizer. The derivative of $p \log p$ tends to $-\infty$ as $p \downarrow 0$, so moving a small amount of mass from a positive coordinate into a zero coordinate lowers the objective; a minimizer is interior. Introduce a multiplier λ for $\sum_i p_i = 1$. The first-order conditions are

$$g_i + \alpha(K\boldsymbol{p})_i + hp_i + \frac{1}{\beta}(1 + \log p_i) + \lambda = 0.$$

Solving for p_i and normalizing yields (6.4). Reversing the calculation proves the converse. □

Equation (6.4) is a logit quantal-response equilibrium: population shares equal noisy best responses to the costs generated by those same shares (Blume 1993; McKelvey and Palfrey 1995; Sandholm 2010). The theorem guarantees existence of at least one equilibrium. It does not declare every stationary point stable or unique.

6.3 When equilibrium is unique

The Hessian on the interior is

$$\nabla^2 \Phi_\beta(\boldsymbol{p}) = \alpha K + hI + \frac{1}{\beta} \operatorname{diag}(1/p_i).$$

Let the simplex tangent space be

$$\mathcal{T} = \{\boldsymbol{v} \in \mathbb{R}^n : \mathbf{1}^\top \boldsymbol{v} = 0\}.$$

Define

$$\lambda_{\min}^{\mathcal{T}}(K) = \min_{\substack{\boldsymbol{v} \in \mathcal{T} \\ \|\boldsymbol{v}\|_2 = 1}} \boldsymbol{v}^\top K \boldsymbol{v} \tag{6.5}$$

be the smallest privacy eigenvalue on the simplex tangent space.

Proposition 6.2 (A global uniqueness condition)**.** *If*

$$h + \frac{2}{\beta} + \alpha \lambda_{\min}^{\mathcal{T}}(K) > 0, \tag{6.6}$$

then Φ_β *is strictly convex, and* (6.4) *has a unique solution.*

Proof. For every nonzero tangent vector $\boldsymbol{v}$,

$$\sum_i \frac{v_i^2}{p_i} \geq \left(\sum_i |v_i| \right)^2 \geq 2\|\boldsymbol{v}\|_2^2,$$

where the first inequality is Cauchy–Schwarz and the second uses $\sum_i v_i = 0$. Adding the hygiene and privacy quadratic forms yields

$$\boldsymbol{v}^\top \nabla^2 \Phi_\beta \boldsymbol{v} \geq \left(h + \frac{2}{\beta} + \alpha \lambda_{\min}^{\mathcal{T}}(K) \right) \|\boldsymbol{v}\|_2^2 > 0.$$

Strict convexity on the simplex gives a unique minimizer and stationary point. □

The condition is sufficient, not necessary. Its interpretation is exact: entropy contributes diagonal curvature of at least $1/\beta$, and hygiene contributes h. The privacy term can contribute negative curvature because a repulsive off-diagonal matrix can have negative eigenvalues. If privacy overwhelms the two regularizers, alternating or otherwise anti-coordinated patterns can appear as multiple local minima. Hygiene does more than flatten shares; it stabilizes a unique population equilibrium.

6.4 The weak-heterogeneity expansion

Let

$$\boldsymbol{u} = \frac{1}{n}\mathbf{1}, \qquad P = I - \frac{1}{n}\mathbf{1}\mathbf{1}^{\mathsf{T}}$$

be the uniform vector and the projection onto zero-sum perturbations. Write

$$\boldsymbol{p} = \boldsymbol{u} + \boldsymbol{\epsilon}, \qquad \mathbf{1}^{\mathsf{T}}\boldsymbol{\epsilon} = 0.$$

For small $\boldsymbol{\epsilon}$,

$$\log(\boldsymbol{u} + \boldsymbol{\epsilon}) = -(\log n)\mathbf{1} + n\boldsymbol{\epsilon} + O(\|\boldsymbol{\epsilon}\|^2),$$

where the logarithm acts coordinatewise. Projecting the first-order conditions onto the tangent space gives

$$\left[\left(\frac{n}{\beta} + h\right) I + \alpha PKP\right] \boldsymbol{\epsilon} = -P(\boldsymbol{g} + \alpha K\boldsymbol{u}) + O(\|\boldsymbol{\epsilon}\|^2). \tag{6.7}$$

This formula contains the center-versus-edge argument in one line.

- $P\boldsymbol{g}$ is the centered positional gradient.
- $PK\boldsymbol{u}$ is the centered row-sum profile of the privacy kernel. On a finite line, its middle coordinates are positive and its edge coordinates negative. The leading privacy correction $-\alpha PK\boldsymbol{u}$ therefore raises edge shares and lowers middle shares.
- $n/\beta + h$ resists all nonuniformity.
- PKP changes the resistance by spatial mode; modes aligned with negative eigenvalues of K are easiest for repulsion to amplify.

When privacy is absent, (6.7) becomes

$$\boldsymbol{\epsilon} = -\frac{\beta}{n + \beta h} P\boldsymbol{g} + O(\|\boldsymbol{g}\|^2). \tag{6.8}$$

This is the precise scope of the inverse-dispersion law. In the weak-heterogeneity regime, every centered positional contrast is attenuated by the factor

$$\frac{n}{n + \beta h} \tag{6.9}$$

relative to the $h = 0$ logit response.

6.4.1 Privacy-only edge bias

Set $\boldsymbol{g} = \mathbf{0}$ and $h = 0$. If α is small enough for the linearization to apply, then

$$\boldsymbol{\epsilon} = -\frac{\alpha\beta}{n} PK\boldsymbol{u} + O(\alpha^2). \tag{6.10}$$

For the decreasing line kernel in (4.7), the row sums are smallest at the ends, so the end coordinates of $\boldsymbol{\epsilon}$ are positive. This proves a local statement: privacy alone moves use toward the boundaries at first order away from uniformity.

The local qualifier matters. At strong interaction, negative-eigenvalue modes can create alternating peaks and multiple equilibria. A universal theorem that privacy can *never* make a center stall locally popular would be false for arbitrary kernels, row lengths, or equilibrium branches. What is robust is the finite-line boundary mechanism and its weak-interaction sign.

6.5 Current privacy versus anticipated privacy

An empty room makes current privacy identical across stalls. A forward-looking entrant can still prefer an end because he anticipates later arrivals. To make that statement mathematical, fix an initial choice i. Let the initial user's departure time be exponential with rate μ, and consider the first competing event: either he departs or another person arrives at rate Λ. The probability that the next event is an arrival is

$$\frac{\Lambda}{\Lambda + \mu}. \tag{6.11}$$

If the later entrant uses the state-conditional rule $P_j(\{i\})$, a one-arrival approximation to the first user's expected cumulative privacy cost is

$$R_i^{(1)} = \frac{\Lambda}{\Lambda + \mu} \frac{J}{2\mu} \sum_{j \neq i} K_{ij} P_j(\{i\}). \tag{6.12}$$

After the later arrival, both users depart independently at rate μ, so their remaining joint-occupancy time has mean $1/(2\mu)$. With stall-specific departure rates, this factor becomes $1/(\mu_i + \mu_j)$. A user with foresight conversion rate θ chooses according to

$$P_i^{\text{forward}} \propto \exp\{-\beta[g_i + \theta R_i^{(1)}]\}. \tag{6.13}$$

Because $R_i^{(1)}$ is a time integral of privacy cost, it has units cost–time; θ has units 1/time so that $\theta R_i^{(1)}$ is a cost. Equivalently, one may first divide exposure by a declared planning horizon and use a dimensionless weight.

With equal baselines and approximately uniform later choice, the sum in (6.12) is proportional to the kernel row sum. An edge stall has one inward flank and a smaller expected exposure. A sufficiently forward-looking first user therefore favors an edge even though current privacy is tied.

6.6 Exact expected future exposure

The one-arrival approximation ignores additional arrivals and departures. Finite-state Markov reward theory gives the exact quantity for a declared occupancy protocol.

Fix an initial user in stall i. Consider the transient states that contain i and kill the process when that user departs. Let A_i be the resulting subgenerator, including all other arrivals and departures and the killing rate in its diagonal. Define the instantaneous privacy-cost vector

$$r_i(S) = J \sum_{j \in S \setminus \{i\}} K_{ij}. \tag{6.14}$$

With discount rate $\delta \geq 0$, the vector of expected cumulative future costs is

$$\boldsymbol{V}_i = (\delta I - A_i)^{-1} \boldsymbol{r}_i. \tag{6.15}$$

The component $V_i(\{i\})$ is the expected cumulative privacy exposure of an initial user who enters an otherwise empty room. Equation (6.15) follows from conditioning on the first infinitesimal transition, or equivalently from the resolvent identity

$$\boldsymbol{V}_i = \int_0^\infty e^{-\delta t} e^{A_i t} \boldsymbol{r}_i \, \mathrm{d}t.$$

It incorporates the entire arrival process, vacancies created by later departures, blocking, and the initial user's random remaining time. The vector $\boldsymbol{V}_i$ has units cost–time and enters a contemporaneous choice cost only after multiplication by the same conversion rate θ.

6.7 Level-k reasoning is a protocol, not a theorem

Cognitive-hierarchy models begin with a specified level-0 rule and define higher levels as best responses to lower levels (Camerer et al. 2004). In a restroom example one might set

level 0: choose from baseline and current occupancy,

level 1: choose while forecasting a level-0 successor,

level 2: forecast a level-1 successor,

and continue.

For a symmetric stylized row, some specifications alternate between an interior choice and an edge choice. That alternation is not general. It changes with the level-0 rule, the positional gradient, the horizon, the probability of a later arrival, and whether higher levels believe the population is homogeneous. Likewise, a quantal-response equilibrium is not automatically the limit of level-k iteration. It is a fixed point in which choice probabilities respond to costs generated by the same probability vector.

The mathematically clean treatment is therefore:

1. declare a cognitive hierarchy when the scientific question concerns bounded reasoning;
2. use (6.15) when the question concerns optimal anticipation of future occupancy;
3. use (6.4) when the question concerns a self-consistent population distribution.

These are complementary models of theory of mind, not interchangeable labels.

6.8 Dynamics toward equilibrium

Potential structure gives disciplined adjustment dynamics. Under asynchronous better response in a finite potential game, every strict improvement reduces the potential and the process cannot cycle indefinitely (Monderer and Shapley 1996). For positive initial shares, an entropy-compatible population adjustment is the mirror-gradient flow

$$\dot{p}_i = -\gamma p_i \left[\partial_i \Phi_\beta(\boldsymbol{p}) - \sum_k p_k \partial_k \Phi_\beta(\boldsymbol{p}) \right], \qquad \gamma > 0,$$

where γ has units $1/(\text{cost} \cdot \text{time})$. Positive shares remain positive and sum to one. Along the flow,

$$\frac{\mathrm{d}}{\mathrm{d}t}\Phi_\beta(\boldsymbol{p}) = -\gamma \left\{ \sum_i p_i (\partial_i \Phi_\beta)^2 - \left(\sum_i p_i \partial_i \Phi_\beta \right)^2 \right\} \leq 0.$$

The entropy derivative is finite throughout this interior trajectory. Simultaneous discrete best responses need not share that convergence property.

Hygiene introduces a state that evolves on its own clock. If users respond to current hygiene, the joint system can still settle. If they respond to delayed hygiene, no static potential in $\boldsymbol{p}$ alone governs the trajectory. The next two chapters develop that temporal layer.

Part III

Hygiene remembers

7

Hygiene as memory and own-load congestion

7.1 From a verbal preference to a state variable

The statement "I want the less frequently used stall" is a preference over history. It cannot be represented by current occupancy alone. Introduce a nonnegative hygiene-load state $m_i(t)$ for stall i. The state can represent a declared physical assay, visible disorder, odor, or perceived recent use. It is not automatically an infection probability.

A flexible continuous-time model is

$$\mathrm{d}m_i(t) = -\kappa_0 m_i(t)\ \mathrm{d}t + \delta_m\ \mathrm{d}N_i^{\mathrm{in}}(t) + \sigma s_i(t)\ \mathrm{d}t - \chi m_i(t^-)\ \mathrm{d}N_i^{\mathrm{cl}}(t). \tag{7.1}$$

The four terms mean:

- natural decay or passive removal at rate κ_0;
- an instantaneous deposit δ_m when a visit begins;
- continuous accumulation σ per occupied unit time;
- cleaning events counted by $N_i^{\mathrm{cl}}(t)$, each removing fraction $\chi \in (0, 1]$ of the pre-cleaning load.

The parameters can be stall specific, visit-type specific, or random. Their common form is sufficient to derive the population mechanism.

7.1.1 The convolution representation

With no discrete cleaning events and an initial condition far in the past, (7.1) has the solution

$$m_i(t) = \int_{-\infty}^{t} e^{-\kappa_0 (t-u)} \{\delta_m\ \mathrm{d}N_i^{\mathrm{in}}(u) + \sigma s_i(u)\ \mathrm{d}u\}. \tag{7.2}$$

Every use contributes to the future with exponentially decaying weight. Privacy and hygiene can now be written as two projections of the occupancy history:

$$\text{privacy at } (i,t) = J \sum_j K_{ij} s_j(t), \tag{7.3}$$

$$\text{hygiene at } (i,t) = H \int_{-\infty}^{t} e^{-\kappa_0 (t-u)} \{\delta_m\ \mathrm{d}N_i^{\mathrm{in}}(u) + \sigma s_i(u)\ \mathrm{d}u\}. \tag{7.4}$$

Privacy is temporally local and spatially nonlocal. The minimal hygiene component modeled here is spatially local and temporally nonlocal. Shared air, contact surfaces, or cleaning tools can couple stalls and would add a spatial hygiene kernel.

7.2 Mean hygiene load under cleaning

Let a_i be the long-run entry rate and q_i the long-run occupancy probability, equivalently the occupied-time fraction. The mean input rate to hygiene load is

$$u_i = \delta_m a_i + \sigma q_i. \tag{7.5}$$

7.2.1 Continuous decay

If cleaning is represented by continuous exponential removal at total rate $\kappa > 0$, then

$$\frac{\mathrm{d}}{\mathrm{d}t}\mathbb{E}m_i(t) = u_i - \kappa \mathbb{E}m_i(t),$$

and the stationary mean is

$$\overline{m}_i = \frac{u_i}{\kappa}. \tag{7.6}$$

7.2.2 Poisson reset cleaning

Suppose physical decay occurs at rate κ_0 and cleaning events form a Poisson process with rate c, independently of m_i. Each event removes fraction χ. Taking expectations in (7.1) gives

$$\frac{\mathrm{d}}{\mathrm{d}t}\mathbb{E}m_i(t) = u_i - (\kappa_0 + c\chi)\mathbb{E}m_i(t), \tag{7.7}$$

so

$$\overline{m}_i = \frac{u_i}{\kappa_0 + c\chi}. \tag{7.8}$$

Random resets act like an additional decay rate for the first moment.

7.2.3 Periodic perfect cleaning

Suppose input is constant at u_i, natural decay is κ_0, and perfect cleaning resets the load to zero every T time units. At age $t \in [0, T)$ since cleaning,

$$m_i(t) = \frac{u_i}{\kappa_0}(1 - e^{-\kappa_0 t}).$$

Averaging uniformly over the cleaning cycle yields

$$\overline{m}_i^{\mathrm{per}} = \frac{u_i}{\kappa_0}\left[1 - \frac{1 - e^{-\kappa_0 T}}{\kappa_0 T}\right]. \tag{7.9}$$

When $\kappa_0 \downarrow 0$, a Taylor expansion gives the sawtooth limit

$$\overline{m}_i^{\mathrm{per}} \longrightarrow \frac{u_i T}{2}. \tag{7.10}$$

Periodic and exponential cleaning therefore have the same mean only after their parameters are matched; they are not automatically equivalent. Even with matched means, their variances and age profiles differ.

7.3 The steady-state own-load

Under common service rate μ, stall-level flow balance gives $q_i = a_i/\mu$. If the total accepted entry rate is $a_\bullet = \sum_j a_j$ and $p_i = a_i/a_\bullet$, then

$$u_i = a_\bullet \left(\delta_m + \frac{\sigma}{\mu}\right) p_i. \tag{7.11}$$

With continuous decay or Poisson cleaning, substitute an effective removal rate κ into (7.6):

$$H\overline{m}_i = \underbrace{\frac{Ha_\bullet}{\kappa}\left(\delta_m + \frac{\sigma}{\mu}\right)}_{h} p_i. \tag{7.12}$$

The complicated physical history reduces at steady state to a marginal own-load cost hp_i.

This reduction supplies the temporal counterpart to current-state privacy. Privacy charges stall i for *others' nearby current use*; hygiene charges stall i for *its own accumulated past use*.

7.4 Why the potential is quadratic

If the marginal hygiene cost at share p_i is hp_i, its integral from zero to p_i is

$$\int_0^{p_i} hx \, \mathrm{d}x = \frac{h}{2} p_i^2.$$

Summing across stalls gives

$$V_{\mathrm{hyg}}(\boldsymbol{p}) = \frac{h}{2} \sum_{i=1}^{n} p_i^2. \tag{7.13}$$

The concentration index $\sum_i p_i^2$ is the Herfindahl–Hirschman index in economics and the complement of a common Simpson diversity measure in ecology. It lies between $1/n$ and 1:

$$\frac{1}{n} \le \sum_i p_i^2 \le 1. \tag{7.14}$$

The lower bound occurs at uniform use; the upper bound occurs when every visit uses one stall. Hygiene is therefore an explicit penalty on concentration.

Entropy also favors dispersion, but the functions are different:

$$\text{hygiene regularizer:} \quad \frac{h}{2} \sum_i p_i^2, \tag{7.15}$$

$$\text{choice-noise regularizer:} \quad \frac{1}{\beta} \sum_i p_i \log p_i. \tag{7.16}$$

The quadratic has finite slope at $p_i = 0$. The entropy derivative tends to $-\infty$. As a result, deterministic hygiene can abandon a sufficiently unattractive stall, while finite logit noise assigns every finite-cost stall positive probability.

7.5 The deterministic water-filling solution

Set privacy aside and take the deterministic limit $\beta \to \infty$. The equilibrium minimizes

$$\min_{\boldsymbol{p} \in \Delta_n} \left\{ \boldsymbol{g}^\top \boldsymbol{p} + \frac{h}{2} \|\boldsymbol{p}\|_2^2 \right\}. \tag{7.17}$$

Theorem 7.1 (Hygiene water filling)**.** *For $h > 0$, the unique solution of (7.17) is*

$$p_i^\star = \frac{(\theta_{\mathrm{w}} - g_i)_+}{h}, \qquad (x)_+ = \max\{x, 0\}, \tag{7.18}$$

where the water level θ_{w} is the unique value satisfying

$$\sum_{i=1}^{n} (\theta_{\mathrm{w}} - g_i)_+ = h. \tag{7.19}$$

Proof. The Karush–Kuhn–Tucker conditions for multiplier λ on $\sum_i p_i = 1$ and nonnegative multipliers ν_i are

$$g_i + h p_i + \lambda - \nu_i = 0, \quad p_i \geq 0, \quad \nu_i \geq 0, \quad \nu_i p_i = 0.$$

For $p_i > 0$, $p_i = (-\lambda - g_i)/h$. For $p_i = 0$, stationarity requires $-\lambda - g_i \leq 0$. Set $\theta_{\mathrm{w}} = -\lambda$ to obtain (7.18). The left side of (7.19) is continuous and strictly increasing once positive, so the normalizing threshold is unique. □

The water metaphor is exact. Imagine bars of height g_i inside vessels of equal cross-section. Add total volume h until the common surface reaches θ_{w}. Stalls whose baseline bars remain above the surface receive zero use. As h grows, the surface rises and more stalls enter the active set.

7.6 Finite-noise equilibrium and the Lambert-W function

With finite β, no privacy, and $h > 0$, the first-order condition is

$$g_i + h p_i + \frac{1}{\beta}(1 + \log p_i) + \lambda = 0. \tag{7.20}$$

Rearranging gives

$$p_i e^{\beta h p_i} = A e^{-\beta g_i}, \qquad A = e^{-1-\beta\lambda}.$$

The Lambert function W is defined by $W(x)e^{W(x)} = x$. Therefore

$$p_i = \frac{1}{\beta h} W\left(\beta h A e^{-\beta g_i}\right), \tag{7.21}$$

with $A > 0$ chosen so that $\sum_i p_i = 1$. This is a one-dimensional normalization problem and an exact closed form up to the scalar A. When $h = 0$, the model reduces continuously to ordinary logit, $p_i \propto e^{-\beta g_i}$.

Proposition 7.2 (Baseline ranking is preserved)**.** *In the hygiene-plus-entropy model without privacy, if $g_i < g_j$, then the unique equilibrium satisfies $p_i > p_j$.*

Proof. Subtract the first-order conditions:

$$g_i - g_j + h(p_i - p_j) + \frac{1}{\beta}\log\frac{p_i}{p_j} = 0.$$

The function $x \mapsto hx + \beta^{-1}\log x$ is strictly increasing on $(0,\infty)$. If $p_i \le p_j$, the last two terms are nonpositive, which cannot cancel $g_i - g_j < 0$. □

Hygiene flattens the positional ranking; it does not arbitrarily reverse it in the static homogeneous equilibrium. Reversal can occur transiently under delayed information or across heterogeneous user groups.

7.7 How fast persistent inequality shrinks

For weak baseline heterogeneity, (6.8) gives

$$\boldsymbol{p} - \frac{1}{n}\mathbf{1} \approx -\frac{\beta}{n+\beta h}P\boldsymbol{g}.$$

Any homogeneous measure of small dispersion therefore scales with

$$\frac{1}{n+\beta h}. \tag{7.22}$$

For example,

$$\mathrm{sd}(\boldsymbol{p}) \approx \frac{\beta}{n+\beta h}\,\mathrm{sd}(\boldsymbol{g}). \tag{7.23}$$

The formula is an asymptotic result, not an all-parameter identity. Privacy adds the matrix correction in (6.7); large positional differences activate the nonlinear Lambert-W response.

7.8 The self-erasing recommendation

Suppose a fraction $f \in [0,1]$ of entrants responds to historical use and the remainder ignores it. Near uniformity and with common β, non-hygiene users have first-order perturbation

$$\boldsymbol{\epsilon}_0 = -\frac{\beta}{n}P\boldsymbol{g} + O(\|P\boldsymbol{g}\|^2),$$

while hygiene-aware users respond to the aggregate share $\boldsymbol{p} = \boldsymbol{u} + \boldsymbol{\epsilon}$:

$$\boldsymbol{\epsilon}_H = -\frac{\beta}{n}(P\boldsymbol{g} + h\boldsymbol{\epsilon}) + O(\|P\boldsymbol{g}\|^2).$$

The aggregate perturbation is

$$\boldsymbol{\epsilon} = (1-f)\boldsymbol{\epsilon}_0 + f\boldsymbol{\epsilon}_H.$$

Solving gives

$$\boldsymbol{\epsilon} = -\frac{\beta}{n+f\beta h}P\boldsymbol{g} + O(\|P\boldsymbol{g}\|^2). \tag{7.24}$$

Equation (7.24) formalizes the self-defeating advice effect. A piece of information has the greatest value when few people act on it. As f grows, aggregate use moves against the original positional inequality, so the less-used stall's advantage shrinks. At $f = 1$ the formula reduces to (6.8); at $f = 0$ there is no feedback.

This has the structure of the El Farol and minority-game family: an action is attractive when fewer others take it, and common prediction changes the outcome being predicted (Arthur 1994; Challet and Zhang 1997). The restroom model is a congestion game with physical memory, not the canonical binary minority game, but the informational externality is the same.

7.9 Partial feedback and learning

Most entrants do not observe the full vector $\boldsymbol{m}(t)$. They see the stall they enter, perhaps glance at a few open stalls, and remember previous visits. This creates censored feedback:

- choosing i yields information about i;
- not choosing j usually yields no new information about j;
- choosing i also changes the future state of i;
- cleaning can improve an option while it is not chosen.

The decision problem is related to rotting bandits, in which pulling an arm reduces its future reward (Levine et al. 2017), and to restless bandits, in which unchosen arms also evolve. It is not a textbook rotting bandit because cleaning can restore quality and because many users jointly change the arms. The appropriate label is an *endogenous restless bandit with congestion*. The resting recovery component connects it to recovering-reward formulations (Simchi-Levi et al. 2024; Whittle 1988).

A belief state $\widehat{m}_i(t)$ can be modeled by

$$\dot{\widehat{m}}_i(t) = -\gamma_i\{\widehat{m}_i(t) - m_i(t-\tau_i)\} + \text{observation innovations}. \tag{7.25}$$

The response rate γ_i and delay τ_i turn static hygiene avoidance into a dynamic feedback system. The next chapter analyzes the simplest exact version of that system and derives its stability boundary.

7.10 Cleaning has two distinct effects

Increasing effective cleaning rate κ lowers mean physical load and, from (7.12), lowers the static feedback coefficient h. Lower h means that baseline positional inequality is less strongly counteracted: frequent equal cleaning can make the use histogram look *more* positional even while every stall is cleaner.

Cleaning also increases the rate at which a hygiene perturbation disappears. In the delayed system, that faster decay stabilizes the uniform state. These effects are compatible:

1. static cross-stall hygiene differences become smaller;
2. dynamic deviations die out faster;
3. the remaining entry-share differences more closely reflect fixed geometry.

Thus a flatter use histogram is not, by itself, evidence of better cleaning. It can indicate stronger hygiene avoidance under worse cleaning.

8

Delayed hygiene feedback and oscillation

8.1 Why negative feedback can fail to settle

Immediate hygiene avoidance is stabilizing. If stall i becomes more used, its hygiene cost rises, its next-choice probability falls, and use shifts elsewhere. Delay changes the timing:

1. a stall becomes popular;
2. its load rises;
3. users continue acting on an older, cleaner estimate;
4. the delayed signal eventually makes them avoid it;
5. avoidance continues after the stall has already recovered.

This is the standard mechanism of delayed negative-feedback oscillation. The restroom model is useful because the feedback gain can be derived from traffic, choice sensitivity, cleaning, and the curvature of the population equilibrium.

8.2 The primitive delayed model

Let q be the total admitted visit-start rate. Use an event-driven hygiene stock in which each visit contributes dose δ_m, with removal rate κ:

$$\dot{\boldsymbol{m}}(t) = \delta_m q\, \boldsymbol{p}(t) - \kappa \boldsymbol{m}(t). \tag{8.1}$$

Users respond to hygiene information delayed by $\tau \geq 0$. Let $\mathcal{S}(\boldsymbol{m})$ be a locally unique minimizer over the simplex of

$$\Psi(\boldsymbol{p}; \boldsymbol{m}) = \boldsymbol{g}^\top \boldsymbol{p} + \frac{\alpha}{2} \boldsymbol{p}^\top K \boldsymbol{p} + \frac{h_0}{2} \|\boldsymbol{p}\|_2^2 + \frac{1}{\beta} \sum_i p_i \log p_i + \eta \boldsymbol{m}^\top \boldsymbol{p}. \tag{8.2}$$

The dynamic closure is

$$\boldsymbol{p}(t) = \mathcal{S}(\boldsymbol{m}(t - \tau)). \tag{8.3}$$

The coefficient η is cost per unit of the primitive hygiene stock. The coefficient h_0 is an optional *instantaneous* own-load not already carried by $\boldsymbol{m}$.

This notation prevents double counting. In the static event-dose reduction, $h = \eta \delta_m q / \kappa$. The dynamic feedback gain contains $q\eta\delta_m$, not qh.

8.3 Equilibrium and its linear response

At an equilibrium $(\boldsymbol{p}^\star, \boldsymbol{m}^\star)$,

$$\boldsymbol{m}^\star = \frac{\delta_m q}{\kappa} \boldsymbol{p}^\star, \qquad \boldsymbol{p}^\star = \mathcal{S}(\boldsymbol{m}^\star). \tag{8.4}$$

Let

$$P = I - \frac{1}{n}\mathbf{1}\mathbf{1}^{\mathsf{T}}$$

project onto zero-sum directions. The tangent Hessian of the base choice problem is

$$\mathcal{H} = P\left[\alpha K + h_0 I + \frac{1}{\beta}\operatorname{diag}(1/p_i^\star)\right]P\Big|_{\mathcal{T}}, \qquad \mathcal{T} = \{\boldsymbol{x} : \mathbf{1}^{\mathsf{T}}\boldsymbol{x} = 0\}. \tag{8.5}$$

Assume $\mathcal{H}$ is positive definite, so the local choice minimizer is isolated.

Proposition 8.1 (Sensitivity of equilibrium choice to hygiene)**.** *For a small hygiene perturbation $\delta\boldsymbol{m}$, the induced choice perturbation satisfies*

$$\delta\boldsymbol{p} = -\eta\mathcal{H}^{-1}P\,\delta\boldsymbol{m} + O(\|\delta\boldsymbol{m}\|^2). \tag{8.6}$$

Proof. Project the constrained first-order condition for (8.2) onto $\mathcal{T}$ and differentiate at the equilibrium:

$$\mathcal{H}\,\delta\boldsymbol{p} + \eta P\,\delta\boldsymbol{m} = 0.$$

Invertibility of $\mathcal{H}$ gives the result. □

Let $\boldsymbol{y}(t) = \boldsymbol{m}(t) - \boldsymbol{m}^\star$. Substituting (8.6) into (8.1) gives the linearized nonuniform dynamics

$$\dot{\boldsymbol{y}}(t) = -\kappa\boldsymbol{y}(t) - \delta_m q\eta\mathcal{H}^{-1}P\boldsymbol{y}(t-\tau). \tag{8.7}$$

The uniform hygiene mode does not change choice, because adding the same cost to every stall cancels in the softmax. It simply decays at rate κ.

8.4 Decomposition into spatial modes

Let $d_r > 0$, $r = 1, \dots, n-1$, be the eigenvalues of $\mathcal{H}$ on the tangent space, with orthonormal eigenvectors $\boldsymbol{v}_r$. Expanding

$$\boldsymbol{y}(t) = \sum_{r=1}^{n-1} y_r(t)\boldsymbol{v}_r$$

decouples (8.7):

$$\dot{y}_r(t) = -\kappa y_r(t) - A_r y_r(t-\tau), \qquad A_r = \frac{\delta_m q\eta}{d_r}. \tag{8.8}$$

Every spatial mode has its own feedback gain. A soft direction of the choice potential has small d_r; a small hygiene contrast then produces a large choice response, so A_r is large.

At a uniform equilibrium on a geometry whose privacy kernel has constant row sum,

$$d_r = h_0 + \frac{n}{\beta} + \alpha\lambda_r(K), \qquad A_r = \frac{\delta_m q\eta}{h_0 + n/\beta + \alpha\lambda_r(K)}. \tag{8.9}$$

A negative privacy eigenvalue lowers curvature and makes that spatial mode more vulnerable to delayed instability. With no privacy and no instantaneous own-load, all nonuniform modes have the common gain

$$A = \frac{\delta_m q\beta\eta}{n}. \tag{8.10}$$

8.5 The exact scalar stability boundary

Trying $y_r(t) = e^{zt}$ in (8.8) gives

$$z + \kappa + A_r e^{-z\tau} = 0. \tag{8.11}$$

Theorem 8.2 (Delay stability for one mode)**.** *Let $A, \kappa > 0$.*

(i) If $A \le \kappa$, every root of

$$z + \kappa + Ae^{-z\tau} = 0$$

has negative real part for every $\tau \ge 0$.

(ii) If $A > \kappa$, the equilibrium is linearly stable precisely for $0 \le \tau < \tau_c$, where

$$\tau_c(A, \kappa) = \frac{\arccos(-\kappa/A)}{\sqrt{A^2 - \kappa^2}}. \tag{8.12}$$

The first crossing frequency and period are

$$\omega_c = \sqrt{A^2 - \kappa^2}, \qquad T_c = \frac{2\pi}{\sqrt{A^2 - \kappa^2}}. \tag{8.13}$$

Further imaginary-axis crossings occur at

$$\tau_k = \frac{\arccos(-\kappa/A) + 2\pi k}{\sqrt{A^2 - \kappa^2}}, \qquad k = 0, 1, 2, \dots. \tag{8.14}$$

At the first crossing, the conjugate roots move from the stable to the unstable half-plane as τ increases.

The scalar root criterion is classical in the theory of difference–differential equations (Hale and Verduyn Lunel 1993; Hayes 1950). Its use here is to map the abstract gain A to restroom traffic, cleaning, and choice curvature.

Proof. An imaginary-axis root has the form $z = i\omega$. Separating real and imaginary parts gives

$$\kappa + A\cos(\omega\tau) = 0, \qquad \omega - A\sin(\omega\tau) = 0. \tag{8.15}$$

Squaring and adding gives

$$\omega^2 = A^2 - \kappa^2.$$

A nonzero real frequency exists only when $A > \kappa$. The smallest positive phase with positive sine is

$$\omega\tau = \arccos(-\kappa/A),$$

which proves (8.12); adding full rotations gives (8.14). At $\tau = 0$, the only relevant root is $z = -(A + \kappa) < 0$. Standard root-continuity for the scalar delay equation, together with the absence of an imaginary root when $A \le \kappa$, establishes all-delay stability in the first case and stability until the first crossing in the second.

For the crossing direction, implicit differentiation of (8.11) gives

$$\frac{\mathrm{d}z}{\mathrm{d}\tau} = \frac{Aze^{-z\tau}}{1 - A\tau e^{-z\tau}}.$$

At $z = i\omega_c$,

$$\Re \frac{\mathrm{d}z}{\mathrm{d}\tau} = \frac{\omega_c^2}{(1+\kappa\tau)^2 + \omega_c^2\tau^2} > 0. \tag{8.16}$$

□

Corollary 8.3 (First unstable spatial pattern)**.** *The full linearized system is stable for every delay when* $\max_r A_r \le \kappa$*. Otherwise its first loss of stability occurs at*

$$\tau_{\text{first}} = \min_{r:A_r>\kappa} \tau_c(A_r, \kappa). \tag{8.17}$$

If the minimizing mode is unique, its eigenvector $\mathbf{v}_r$ *gives the spatial shape of the incipient oscillation. If several modes attain the same first threshold, the critical spatial pattern lies in their eigenspace and the linear criterion alone does not select a unique combination.*

8.6 Comparative statics of instability

The simple gain (8.10) makes the stability criterion

$$\frac{\delta_m q \beta \eta}{n} \le \kappa. \tag{8.18}$$

The direction of every parameter is transparent:

- larger admitted traffic q makes stale information more dangerous;
- larger dose per visit δ_m strengthens the historical feedback;
- larger hygiene sensitivity η makes the response stronger;
- larger choice precision β makes users move together more sharply;
- more stalls n dilutes a given perturbation;
- faster removal or cleaning κ stabilizes the system.

If the reduced static coefficient is

$$h = \frac{\eta \delta_m q}{\kappa},$$

then

$$A = \frac{\kappa \beta h}{n}, \tag{8.19}$$

not $q\beta h/n$. The latter expression would count traffic twice.

For A only slightly above κ, the denominator of (8.12) is small and a long delay is required. When $A \gg \kappa$,

$$\tau_c \sim \frac{\pi}{2A}, \qquad T_c \sim \frac{2\pi}{A}.$$

Strong, synchronized responsiveness therefore destabilizes even relatively fresh information.

8.7 Numerical scale of the threshold

For the illustrative values $A = 4.29$ and $\kappa = 1$,

$$\omega_c = \sqrt{4.29^2 - 1^2} \approx 4.17, \tag{8.20}$$

$$\tau_c = \frac{\arccos(-1/4.29)}{4.17} \approx 0.433, \tag{8.21}$$

$$T_c = \frac{2\pi}{4.17} \approx 1.51. \tag{8.22}$$

These values give the scalar linearization a concrete time scale. A simulated period near six time units at $\tau = 2$ would be a far-from-threshold nonlinear period, not the linear crossing period (8.13).

8.8 Scope of the linear stability result

Theorem 8.2 proves loss of linear stability through an oscillatory pair. A classical nonlinear Hopf theorem additionally requires a simple critical eigenpair and a nonzero first Lyapunov coefficient. In the fully symmetric no-privacy model, all $n - 1$ nonuniform modes have the same gain, so the crossing is degenerate. Spatial asymmetry or privacy often splits the modes and restores a simple first crossing (Hassard et al. 1981).

The threshold alone does not determine the nonlinear attractor. Depending on the full choice map and symmetry, the post-threshold system can exhibit:

- a stable standing oscillation in one spatial mode;
- competition among several modes;
- switches between metastable popularity patterns;
- a stable periodic orbit;
- more complicated aperiodic behavior at large delay.

A standing center-versus-edge mode alternates which group is cleaner; it does not march the identity of the cleanest stall through positions $1, 2, \ldots, n$. A traveling sequence requires an appropriate pair of phase-shifted modes or asymmetry. It is a numerical prediction of a specified nonlinear model, not a consequence of (8.12).

8.9 Delay, filtering, and distributed memory

A pure delay is the sharpest analytical model. Human information is often filtered. For example,

$$\dot{\widehat{\boldsymbol{m}}} = \gamma[\boldsymbol{m}(t) - \widehat{\boldsymbol{m}}(t)], \qquad \boldsymbol{p}(t) = \mathcal{S}(\widehat{\boldsymbol{m}}(t)) \tag{8.23}$$

creates an exponentially weighted perception rather than a fixed lag. The combined system is an ordinary differential equation in $(\boldsymbol{m}, \widehat{\boldsymbol{m}})$ and has a different characteristic polynomial. A moving average creates a distributed delay. Chosen-stall-only learning creates a stochastic, user-specific filter.

The empirical task is to estimate the response kernel, not to assume that every entrant carries the same τ. The pure-delay theorem remains valuable because it gives an exact boundary and shows which combination of gain, cleaning, and information age governs oscillation.

8.10 Stationary Gibbs laws and delayed dynamics

For a finite reversible Markov chain, the generator is similar to a symmetric matrix. Its eigenvalues are real and nonpositive, and quantities such as relative entropy decrease toward stationarity. Individual observables need not be monotone, but the probability law has no attracting periodic orbit.

A finite nonreversible Markov chain can have complex eigenvalues and damped oscillatory transients; irreducibility still drives it to a stationary distribution. Nonreversibility alone does not create sustained cycles.

The present oscillation mechanism is different. Delayed feedback makes the state infinite dimensional because the future depends on a history segment. The nonlinear choice map feeds that delayed state back into deposition. That combination, not merely departure from a Gibbs formula, permits an attracting cycle.

8.11 Cleaning schedules as forcing

Periodic cleaning can create a periodic hygiene signal even when behavioral feedback is stable. If all stalls are cleaned simultaneously every T units, their common load drops together; this affects average hygiene but not relative choice when the drop is exactly equal. If stalls are cleaned sequentially, the schedule creates a moving exogenous advantage that can look like endogenous popularity migration.

A field study must therefore record cleaning at stall resolution. The diagnostic distinction is:

- exogenous cycles are phase-locked to the cleaning schedule;
- endogenous delay cycles change period with traffic, information age, and behavioral sensitivity according to the stability mechanism above.

This distinction turns cleaning from a nuisance variable into an experimental control.

Part IV

What the model says

9

Exact computation and numerical results

9.1 Numerical questions

The finite-row model contains no hidden asymptotics: for seven stalls it has $2^7 = 128$ occupancy states. Exact enumeration can therefore answer the stationary question directly. Simulation is reserved for historical hygiene and delay, where the state is continuous or depends on a past interval.

The numerical study asks six questions.

1. Does the normalized-entry chain reproduce the Erlang loss law after aggregation?
2. How different are Protocol N and the Gibbs Protocol G when their mean occupancies are comparable?
3. How do positional and privacy forces change the entry histogram with offered load?
4. How strongly does hygiene flatten baseline inequality?
5. How far apart can entry, occupancy, and depletion shares move under duration and paper-use heterogeneity?
6. Does the delayed nonlinear system agree with the analytical stability threshold, and what spatial pattern appears beyond it?

9.2 Computational definitions

For Protocol N, the stationary vector is obtained from the generator described in Appendix C. Reported visit-start shares are conditional on admission:

$$p_i^{\mathrm{N}} = \frac{\sum_{S: i \notin S} \pi^{\mathrm{N}}(S) \lambda_i^{\mathrm{N}}(S)}{\Lambda[1 - B(n, \rho)]}. \tag{9.1}$$

Occupancy probabilities are

$$q_i^{\mathrm{N}} = \sum_{S \ni i} \pi^{\mathrm{N}}(S).$$

For Protocol G, state probabilities are computed both from the stationary linear system and the direct Gibbs weights (5.17).

Mean-field equilibria minimize (6.3) on the simplex. Every reported solution satisfies the simplex constraint, the projected first-order equations, and the fixed-point equation (6.4). Multiple initial points are used when the Hessian is not globally positive.

The delayed system uses a method-of-steps integration of (8.1)–(8.3). The history over $[-\tau, 0]$ is declared, probabilities are recomputed from the nonlinear softmax at every step, and the integration interval contains many cleaning times $1/\kappa$. Reported oscillation is diagnosed after discarding the initial transient.

9.3 A scale-free reading of the Christenfeld aggregate

The direct field aggregate can be translated into one log-odds contrast without choosing a cost unit (Christenfeld 1995, p. 52). Suppose, only for this calculation, that the two middle stalls have a common choice probability and the two ends have a common choice probability, that every observation represents a comparable free set, and that depletion is proportional to visits. The per-stall middle-to-end odds ratio is

$$\frac{52/2}{34/2} = \frac{52}{34} \approx 1.529.$$

The implied dimensionless cost contrast is

$$\beta(g_{\text{end}} - g_{\text{middle}}) = \log \frac{52}{34} \approx 0.425. \tag{9.2}$$

At $\beta = 3$, this would be a cost difference of about 0.142 in the chosen cost units.

Equation (9.2) is not a fit of the dynamic model. It is the only symmetric empty-room logit contrast carried by the headline aggregate. Occupancy, asymmetric end positions, duration, and consumption can all change the mapping.

9.4 Validation identities

Every exact run is required to satisfy:

$$\max_S |(\pi Q)_S| < 10^{-10}, \tag{9.3}$$

$$\left| \sum_S \pi(S) - 1 \right| < 10^{-12}, \tag{9.4}$$

$$\max_i |a_i - \mu q_i| < 10^{-10}, \tag{9.5}$$

$$\max_k \left| \sum_{|S|=k} \pi^{\mathrm{N}}(S) - \frac{\rho^k/k!}{\sum_{r=0}^{n} \rho^r/r!} \right| < 10^{-10}. \tag{9.6}$$

Protocol G is additionally checked against direct normalized Gibbs weights. Random sequential adsorption is checked against the finite recurrence (G.1). Delay thresholds are calculated independently from (8.12).

9.5 Position can favor the near end, an interior stall, or the far end

Figure 9.1 begins with the mechanism that exists before another person enters the room. Three parameter sets in

$$g(x) = ae^{-x/\lambda} + wx$$

produce the three possible constrained regimes. The nearest-door example has an unconstrained stationary point at $x_0 = -0.482$ stall spacings, so the first physical stall minimizes cost. The interior example has $x_0 = 2.911$, making stall 4 the exact discrete minimizer. The far-end example has $x_0 = 13.540$, beyond an eleven-stall row, so stall 11 minimizes cost. Panel B evaluates the discrete minimizer over 39,600 parameter pairs. The staircase boundaries arise when the convex continuous profile is evaluated on a finite grid and its exact adjacent-stall comparisons change sign; they are not numerical noise or a rule that blindly rounds x_0.

This calculation gives a mechanistic interpretation to a middle-peaked empty room profile. It does not

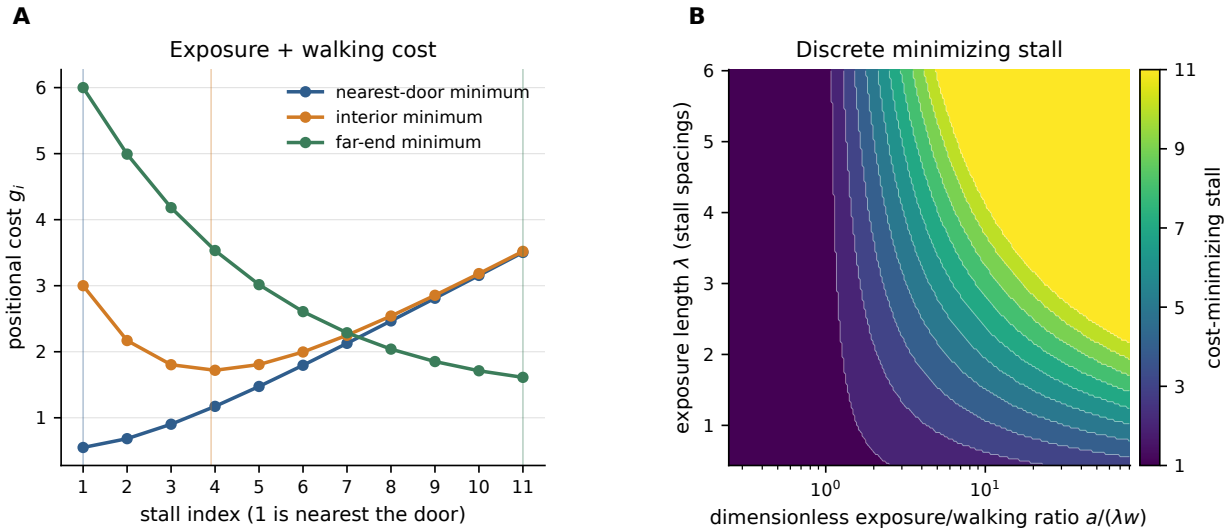


Figure 9.1: Positional cost regimes for eleven stalls. Panel A uses $(a, \lambda, w) = (3, 2, 0.35)$ for the interior case, together with near- and far-boundary examples. The faint vertical lines mark the continuous minimizers after boundary clipping. Panel B holds $w = 1$ without loss of generality and varies the dimensionless ratio $a/(\lambda w)$ and decay length λ. Color gives the exact discrete cost-minimizing stall.

require an intrinsic attraction to the center. Conversely, a far-end or near-door preference is not evidence against the model; it locates the facility in another part of the same parameter plane.

9.6 The arrival protocol changes the stationary physics

The next computation makes the distinction between Protocol N and Protocol G quantitative. It uses five stalls, $\beta = 1.7$, privacy strength $J = 0.8$, screening length $\ell = 1$, common service rate $\mu = 1$, and a positional field generated by $(a, \lambda, w) = (1.8, 1.4, 0.28)$ after subtracting its minimum. Protocol G has site activity $z = 0.85$. The Protocol N offered load is chosen as $\rho = 1.201828$ so that both systems have the same mean occupancy, $\mathbb{E}K = 1.194267$.

Protocol G and the direct Gibbs calculation agree with maximum absolute error 4.75×10^{-17}. The Protocol N law is separated from the Gibbs law by total-variation distance

$$\frac{1}{2} \sum_{S} \left| \pi^{\mathrm{N}}(S) - \pi^{\mathrm{G}}(S) \right| = 0.21916. \tag{9.7}$$

Thus two models can agree on the expected number busy while assigning substantially different probabilities to the configurations that produce that mean.

Figure 9.3 diagnoses why. For every elementary state-space square under Protocol G, the clockwise and counterclockwise rate products agree; the largest computed log-affinity is 7.77×10^{-16}. Protocol N reaches absolute affinity 1.7860 and maximum stationary edge current 0.012986. These are not sampling estimates: they are algebraic properties of the exact generator and stationary vector.

The operational lesson is exact. “Choose one of the currently free stalls” and “attempt a locally weighted stall and otherwise balk” are different behavioral rules. Exponentiating the same apparent cost does not make their stationary laws interchangeable.

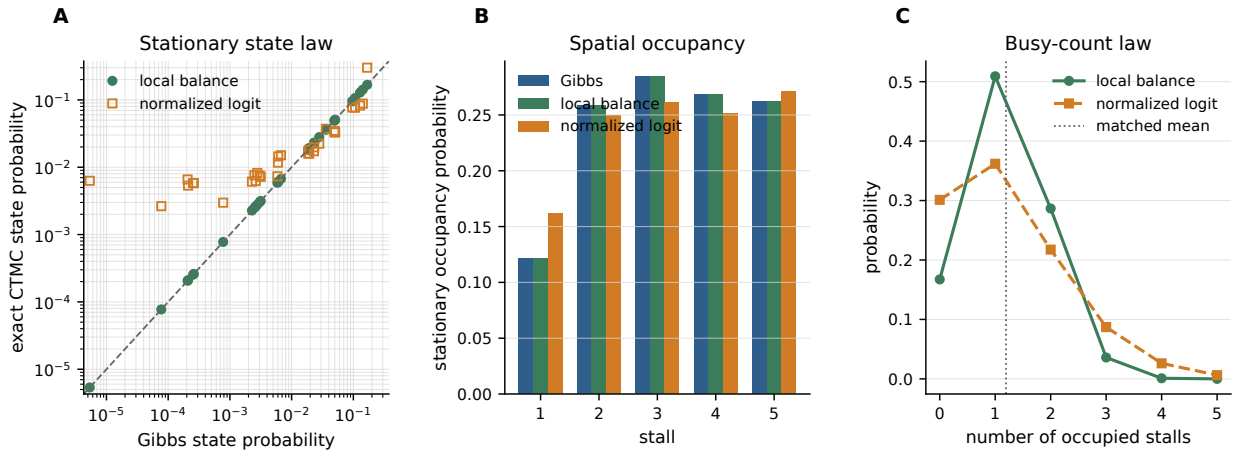


Figure 9.2: Exact comparison of the two arrival protocols by enumeration of all $2^5 = 32$ configurations. Panel A compares each exact CTMC state probability with the Gibbs probability. Protocol G lies on the identity line to floating-point precision; Protocol N does not. Panel B compares stallwise occupancy probabilities. Panel C shows that matching mean occupancy does not match the busy-count distribution.

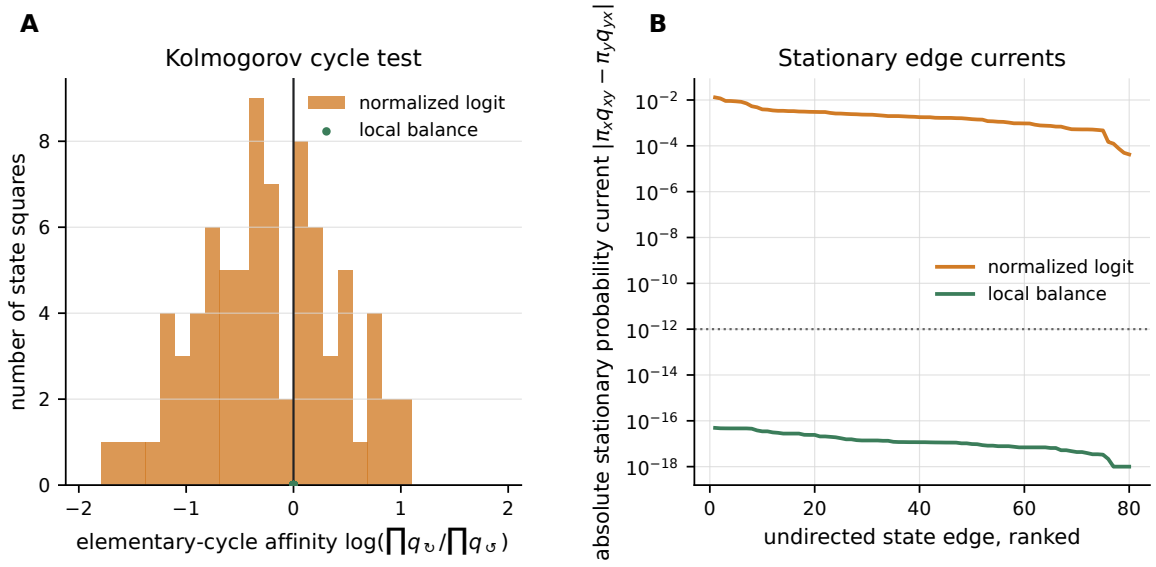


Figure 9.3: Cycle affinities and stationary currents for the same five-stall calculation as Figure 9.2. Panel A applies Kolmogorov's cycle test to elementary squares of the occupancy hypercube. Panel B ranks absolute stationary currents on its undirected edges. The gray line is a numerical reference at 10^{-12}, not a statistical significance threshold.

9.7 Offered load transforms a positional preference

For the principal seven-stall calculation, set

$$\beta = 3, \quad J = 1, \quad \ell = 1, \quad (a, \lambda, w) = (1.6, 1.35, 0.22), \quad \mu = 1.$$

The empty-room choice profile peaks at stall 3 with probability 0.2647; stall 1 has probability 0.0243. That histogram answers only the isolated choice question. The exact 128-state Protocol N chain gives the admitted-entry and time-occupancy observables in Figure 9.4.

At $\rho = 0.04$, blocking is negligible and admitted-entry shares are

$$(2.57,\ 15.65,\ 26.02,\ 23.67,\ 16.37,\ 10.00,\ 5.72)\%.$$

At $\rho \approx 2.95$, privacy and changing availability have reorganized them to

$$(12.91,\ 15.30,\ 14.92,\ 14.25,\ 13.31,\ 12.52,\ 16.79)\%.$$

Stall 7, not the empty-room favorite, now has the largest admitted-entry share. At $\rho = 15$, all seven shares lie between 14.11 % and 14.58 %; the blocking probability is 0.5761 and the mean number busy is 6.359. This is the finite-load approach to the uniform high-load limit proved in Equation (5.22).

The aggregate busy-count law remained the Erlang loss distribution at all 52 evaluated loads, including Erlang–B full-state blocking. The largest difference between exact aggregated state probabilities and the Erlang formula was 3.33×10^{-16}. That identity coexists with the large spatial changes in Panel B because the count law forgets which stalls are busy.

9.8 Privacy alone is an intermediate-load boundary force

To isolate privacy, Figure 9.5 sets $\boldsymbol{g} = \mathbf{0}$ while retaining $n = 7$, $\beta = 3$, $J = 1$, $\ell = 1$, and common service rate. The edge statistic is

$$R_{\text{edge}}(\rho) = \frac{[p_1(\rho) + p_7(\rho)]/2}{[p_2(\rho) + \cdots + p_6(\rho)]/5} - 1. \tag{9.8}$$

At $\rho = 0.02$, $R_{\text{edge}} = 0.0122$: the row is nearly uniform because another occupant is rarely present. It rises to 0.5867 near $\rho = 2$, then falls to 0.00787 by $\rho = 30$ because vacancies are filled at nearly equal service-limited rates. The calculation therefore sharpens the phrase "privacy favors the ends." Privacy favors the ends locally for the distance-decaying open-row kernel, but the observable excess is concentrated at intermediate traffic.

9.9 Static hygiene awareness flattens persistent inequality

Return to the seven-stall positional field and remove privacy so that the hygiene mechanism is visible by itself. The equilibrium minimizes

$$\boldsymbol{g}^{\top}\boldsymbol{p} + \frac{h}{2}\|\boldsymbol{p}\|_2^2 + \frac{1}{\beta}\sum_i p_i \log p_i$$

with $\beta = 3$. Figure 9.6 solves this problem over 80 values of h.

With $h = 0$, the largest-to-smallest probability ratio is 10.90 and the Herfindahl index is 0.1924. At $h \approx 5.18$, the ratio is 2.26 and the index is 0.1510. At $h = 100$, the seven shares range only from 13.79 % to 14.57 %; the ratio is 1.056 and the Herfindahl index is 0.142905, close to its uniform lower bound

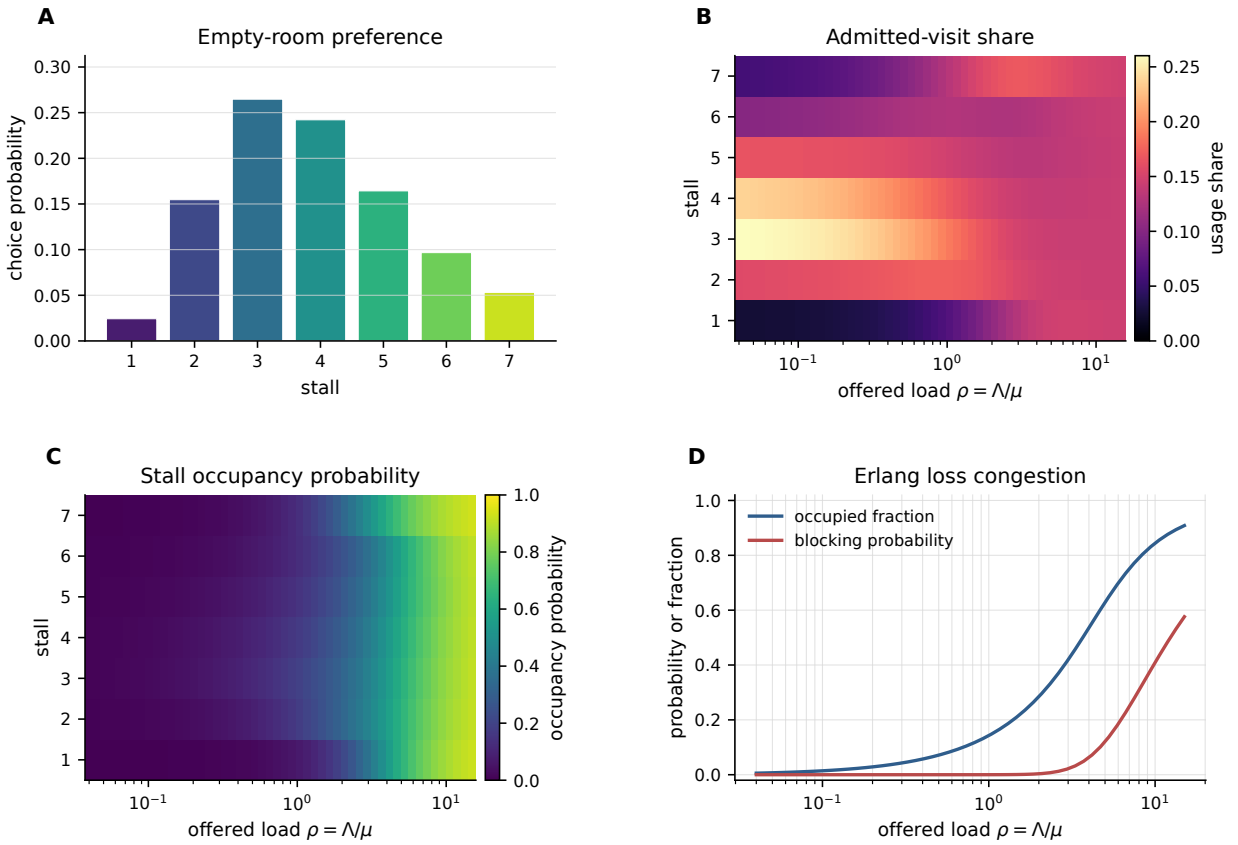


Figure 9.4: Exact load-dependent choice and occupancy for the seven-stall Protocol N chain. Panel A is the empty-room logit profile. Panel B gives each stall's share of admitted visit starts; Panel C gives its stationary probability of being occupied. Panel D gives the fraction of capacity occupied and the blocking probability. The horizontal axis in Panels B–D is offered load $\rho = \Lambda/\mu$, not admitted throughput.

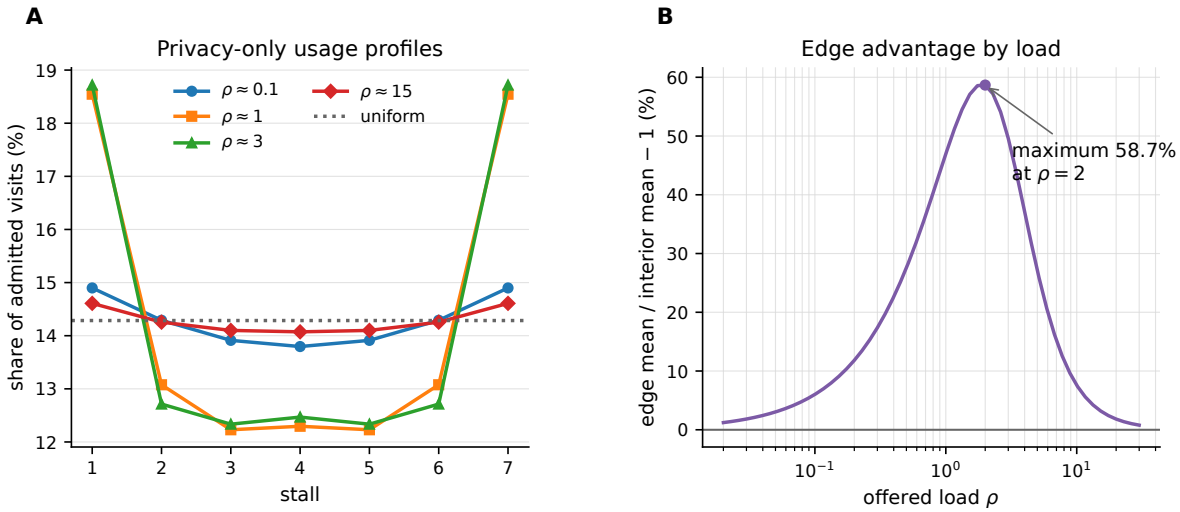


Figure 9.5: Privacy-only edge effect in the exact seven-stall Protocol N chain. Panel A shows admitted-entry profiles at four loads; the dotted line is $1/7$. Panel B shows the edge statistic (9.8). The maximum is 58.7 % at $\rho \approx 2.00$ for this parameter set.

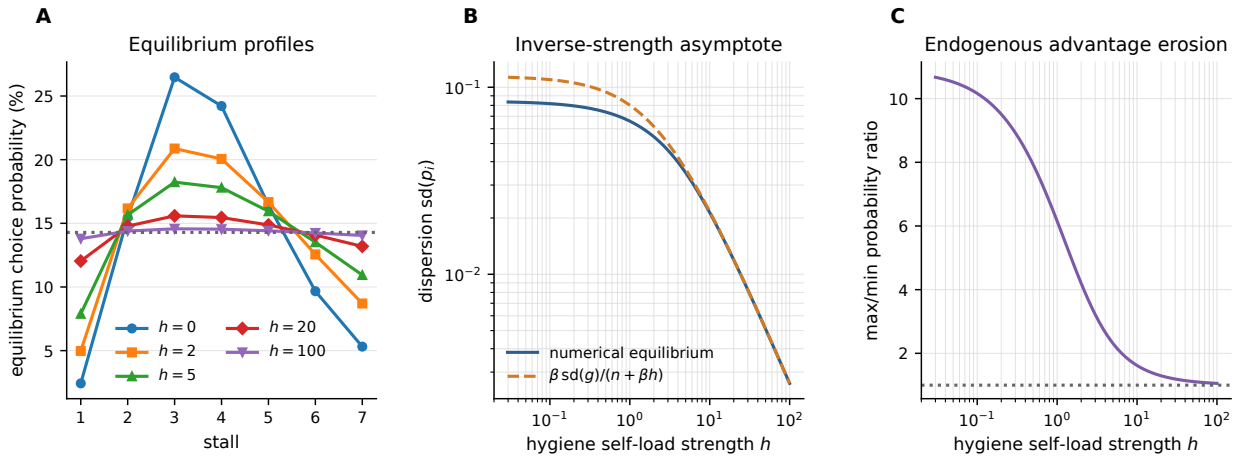


Figure 9.6: Static hygiene-driven flattening. Panel A shows equilibrium profiles at selected own-load strengths. Panel B compares the exact standard deviation of the shares with the large-h prediction $\beta\,\mathrm{sd}(g)/(n+\beta h)$. Panel C shows the ratio of the largest to smallest equilibrium share. The positional field and β are the same as in Figure 9.4; privacy is zero.

$1/7 \approx 0.142857$. For $h \geq 15$, the computed dispersion differs from the linear inverse-strength approximation by at most 0.61 %.

This is the mathematical answer to the proposed "reverse" motive. A preference for less accumulated use acts as negative own-load feedback. It does not select one permanently clean stall. It transfers demand toward the underused stalls until the original positional demand advantage has been consumed.

9.10 Hygiene changes where privacy can reverse the positional ranking

Figure 9.7 combines the positional field, privacy, and static hygiene in the seven-stall mean-field equilibrium. Its order parameter is

$$M(\rho, J, h) = 7\left[\frac{p_3 + p_4 + p_5}{3} - \frac{p_1 + p_7}{2}\right]. \tag{9.9}$$

Red means center-heavy and blue means edge-heavy. The reduced closure uses

$$\alpha(\rho, J) = J\overline{N}(\rho) = J\rho[1 - B_7(\rho)],$$

where $\overline{N}(\rho)$ is the exact Erlang carried load for seven stalls. Thus $\alpha K \boldsymbol{p}$ matches the expected privacy field $JK\,\mathbb{E}\boldsymbol{s}$ under the closure $\mathbb{E}\boldsymbol{s} \approx \overline{N}\boldsymbol{p}$. This is a mean-field reduction of the finite-state forced-entry chain, not an exact replacement for it.

With $h = 0$, the declared positional advantage remains center-heavy over the displayed rectangle. Hygiene first weakens that baseline inequality. Privacy can then move the selected equilibrium across the center/edge sign boundary at combinations of load and interaction strength that were insufficient without hygiene. This is not a contradiction: own-load feedback and spatial cross-load act in different directions. Hygiene erodes the privileged stall's inherited advantage; privacy then rewards the lower-interaction geometry of the boundaries.

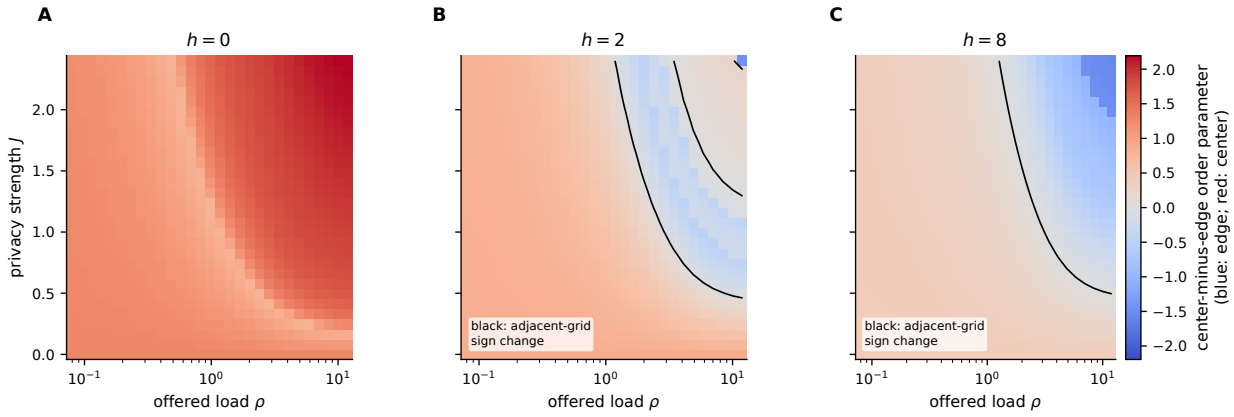


Figure 9.7: Finite-row center-to-edge crossover maps for hygiene strengths $h = 0, 2, 8$. The positional field has $(a, \lambda, w) = (1.6, 1.35, 0.22)$, $\beta = 3$, and the privacy kernel has $\ell = 1$. The black curve is the numerical center/edge sign-change boundary. Each pixel is solved directly by deterministic multistart constrained minimization at $\alpha = J\rho[1 - B_7(\rho)]$; no interpolation is made across equilibrium branches, and the lowest converged potential value is retained. Dense simplex-lattice and seeded-random audits at the two strongest-interaction points reproduce the selected minima. The finite row can exchange lowest-potential branches, but this is not a thermodynamic-limit phase-transition claim.

9.11 Visit starts, occupied time, and depletion can disagree

The computation in Figure 9.8 keeps the exact seven-stall choice process but assigns mean durations

$$(0.72,\ 0.85,\ 0.95,\ 1.08,\ 1.25,\ 1.48,\ 1.78)$$

and paper-use marks $0.65 + 0.55\,\mathbb{E}L_i$ at potential arrival rate 2.6. Stallwise flow balance is verified to 6.11×10^{-16}:

$$q_i = a_i \mathbb{E}L_i,$$

where a_i is the admitted visit-start rate.

Stall 2 receives 18.51 % of admitted visits, the largest entry share, but only 14.22 % of occupied time. Stall 7 receives 12.34 % of admitted visits yet accounts for 19.86 % of occupied time because its mean visit is longest. Its depletion share is 15.98 %, between the other two. A roll-count histogram, an occupancy snapshot, and a door-event histogram can therefore rank the same stalls differently even when all three are measured without error.

9.12 Delayed hygiene response has a computable stability boundary

The delayed experiment uses the primitive system

$$\dot{\boldsymbol{m}} = \delta_m q\,\boldsymbol{p} - \kappa \boldsymbol{m}, \qquad p_i(t) = \frac{e^{-\beta\eta m_i(t-\tau)}}{\sum_j e^{-\beta\eta m_j(t-\tau)}},$$

with $n = 7$, admitted visit-start rate $q = 10$, $\beta = 3$, $\eta = 1$, $\delta_m = 1$ in the chosen hygiene-load units, and $\kappa = 1$. Its nonuniform gain is $A = \delta_m q\beta\eta/n = 30/7$, giving the exact linear threshold

$$\tau_c = 0.433434.$$

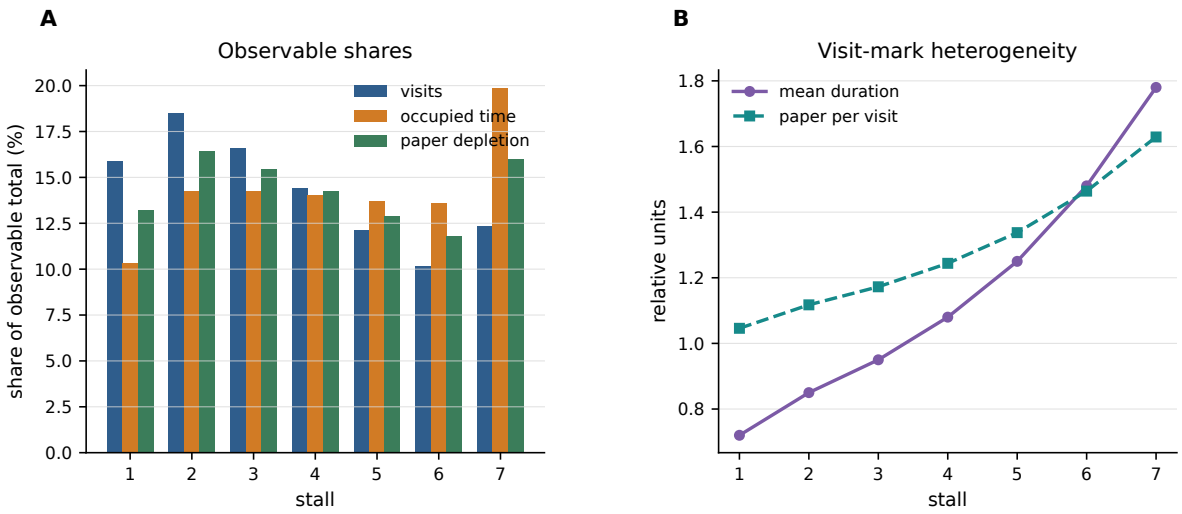


Figure 9.8: Three observables under heterogeneous visit marks. Panel A normalizes admitted visit starts, occupied time, and expected paper depletion separately to sum to 100 %. Panel B gives the imposed mean durations and paper use per visit. The divergence is generated by the declared mark model, not fitted to the historical roll counts.

At $\tau = 0.2$, the late root-mean-square deviation from uniform choice is 3.10×10^{-15}. At $\tau = 0.8$, the nonlinear softmax bounds the growing linear mode and the integration remains in a bounded oscillatory regime through $T = 200$, with RMS amplitude 0.149116. Across 54 peak intervals on $t \in [50, 200]$, the mean period is 2.720074 and its coefficient of variation is 4.94×10^{-4}; a one-period recurrence comparison has RMS discrepancy 0.00881 relative to trajectory RMS. Repeating the calculation with half the time step changes the measured amplitude by only 0.060,2 %. These diagnostics establish the stated long-horizon numerical trajectory for the declared history and parameters. The threshold theorem itself establishes loss of linear stability, not the global form of every post-threshold attractor.

Figure 9.10 varies the removal rate. For fixed delay $\tau = 0.8$, the characteristic-root boundary is $\kappa_c = 3.089775$. Below it, the largest real part is positive and the nonlinear run retains a large late amplitude. Above it, the real part is negative and the late amplitude falls rapidly toward zero.

Continuous stock removal has consequently acquired a stability interpretation in addition to its direct hygienic effect. Raising effective removal beyond every modal feedback gain prevents stale-information amplification for every delay; at a specified delay, the exact characteristic boundary gives the smaller rate needed for stability. A periodic janitorial reset is a distinct hybrid system: it must be mapped to an effective removal law or analyzed directly before this boundary is applied to a cleaning schedule.

9.13 A symmetric post-threshold trajectory switches its argmax label

The same $\tau = 0.8$ run shows that the identity of the stall currently most attractive under the delayed signal need not be fixed. In Figure 9.11, the maximum choice probability changes identity 91 times over $t \in [165, 200]$, and every one of the seven labels becomes the maximum at least once.

The figure supplies a finite-window numerical example inside the specified nonlinear model: over $t \in [165, 200]$, no fixed label remains the maximizer. It does not convert the linear threshold into a traveling-wave theorem. Geometry, asymmetry, current privacy, filtering, and the history select which post-threshold pattern is realized.

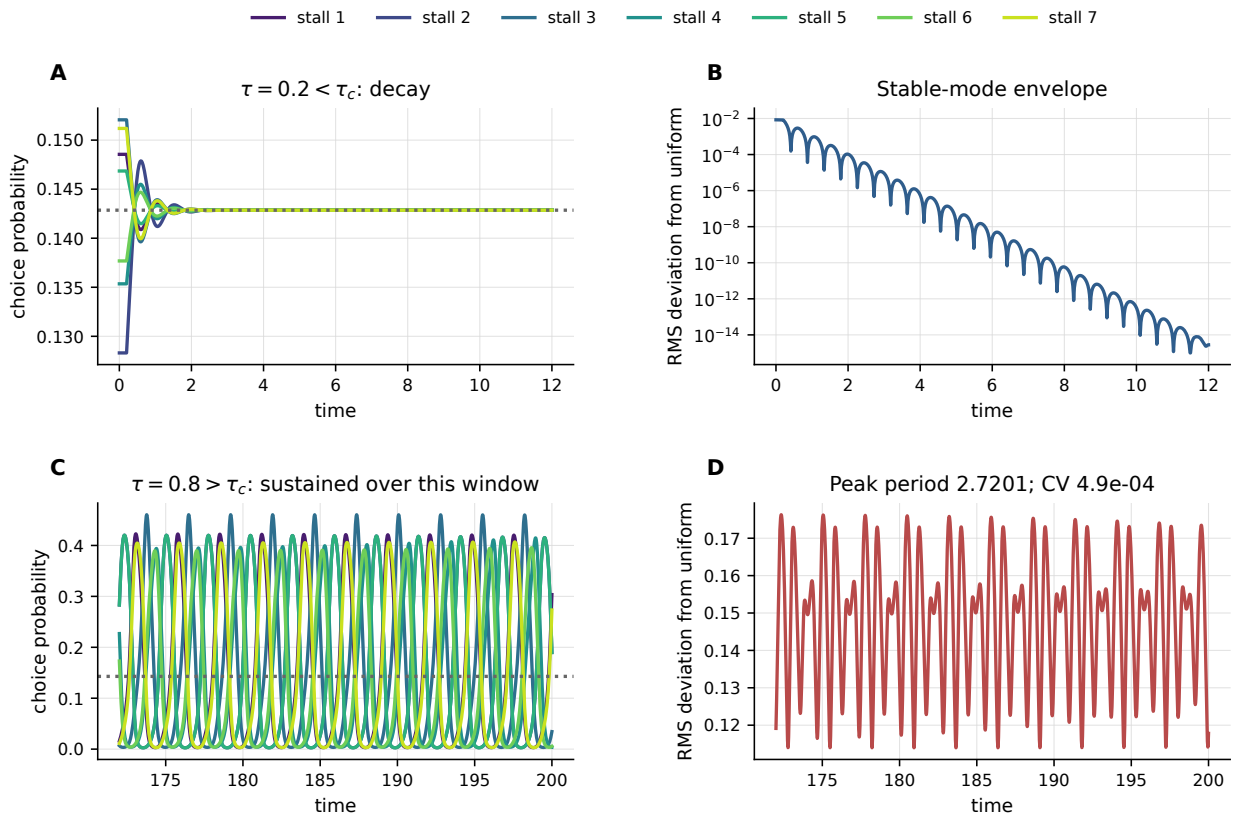


Figure 9.9: Delayed hygiene dynamics below and above the linear oscillatory-instability threshold. Panels A–B use $\tau = 0.2 < \tau_c$ and show convergence to uniform choice. Panels C–D use $\tau = 0.8 > \tau_c$ and show a bounded oscillation through $T = 200$ from the declared seeded history. The late RMS amplitude is 0.1491 and the mean peak-to-peak period is 2.720.

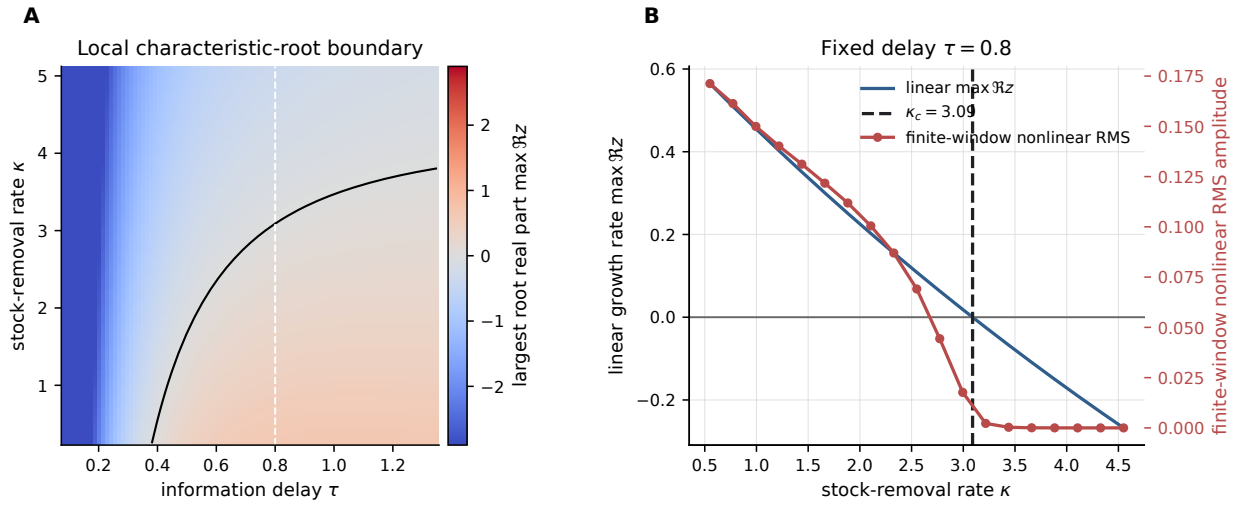


Figure 9.10: Removal rate and delayed-feedback stability for gain $A = 30/7$. Panel A evaluates the largest real part of the scalar characteristic roots over delay and removal rate; black is the zero contour. Panel B takes the slice $\tau = 0.8$, comparing the exact linear growth rate with late RMS amplitudes from nonlinear integrations. The vertical line is the computed critical removal rate $\kappa_c = 3.089775$.

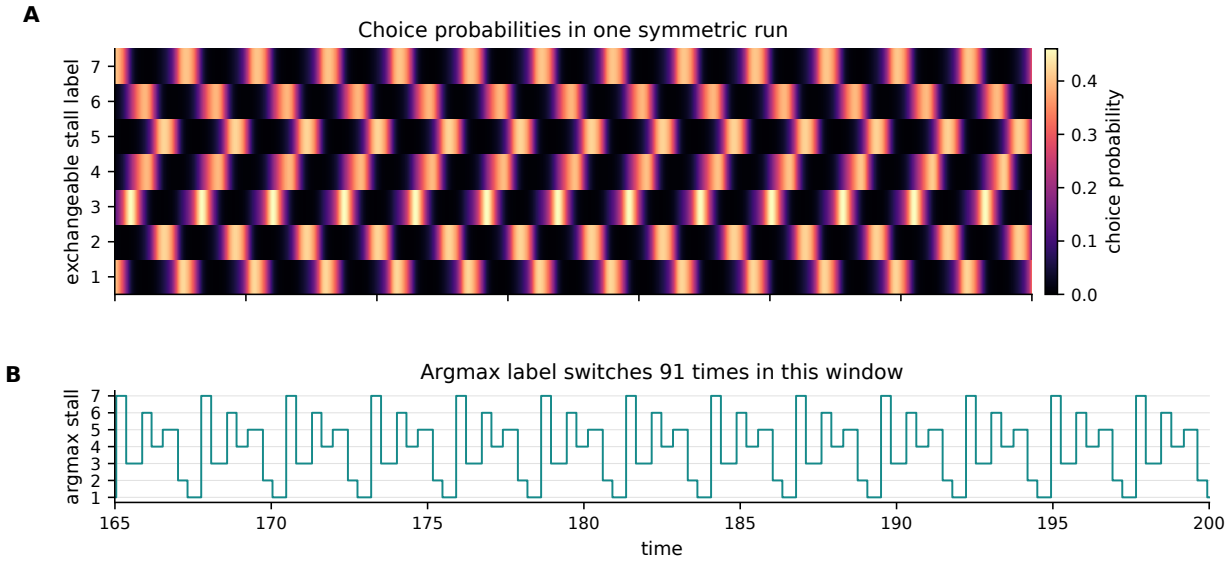


Figure 9.11: Argmax-label switching in the particular post-threshold trajectory from Figure 9.9. Panel A is a time–label heat map of choice probability. Panel B plots the instantaneous maximizing label. Because this experiment has $\boldsymbol{g} = \mathbf{0}$ and no spatial privacy kernel, its seven labels are exchangeable. The seeded history selects the displayed ordering; the figure is not a spatial traveling wave or a claim about the empirically least-used stall.

9.14 The historical aggregate does not identify the mechanism

Finally, Figure 9.12 asks how much can be learned from a historical four-stall aggregate in which the middle two stalls accounted for $52/86 = 0.604651$ of observed toilet-paper depletion (Christenfeld 1995, p. 52). To expose the identification problem in its most favorable form, the calculation temporarily treats that proxy as an empty-room choice share. One hundred thousand parameter vectors were sampled from

$$a \in [0.25, 4.5] \text{ and } w \in [0.018, 0.85] \quad \text{log-uniformly},$$
$$\lambda \in [0.45, 4.5], \quad \beta \in [0.5, 5.5], \quad J \in [0, 2.5],$$

and 999 produced a middle share within 0.004 of $52/86$, namely in $[0.60066, 0.60864]$.

Among accepted vectors, the dimensionless ratio $a/(\lambda w)$ has a 5 % to 95 % interval from 1.40 to 93.36; the choice precision has a corresponding interval from 0.82 to 5.25. These parameter sets agree on the calibration statistic yet extrapolate to widely different seven-stall profiles.

The local Jacobian makes the information loss explicit. A single middle-share aggregate has rank one. The full empty-room profile has rank three, the maximum allowed by four probabilities that sum to one, and its derivative with respect to J is exactly zero because no current occupant generates privacy cost. Three state-averaged load profiles raise the numerical rank to four. The remaining scale invariance reflects the fact that logit choices identify dimensionless products such as βg and βJ, not an absolute cost unit. This calculation explains why state variation, timing, and experimental perturbations are needed in Chapter 10.

9.15 Computational scope and reproducibility

The source program writes machine-readable tables for every numerical experiment, a declared parameter ledger, a figure manifest, and a validation record for the numerical assertions. The final run records 31

successful checks. The most diagnostic residuals are summarized in Table 9.1.

These checks certify the computation relative to its declared models. The exact CTMC results concern the specified finite chain; the mean-field regime maps concern the reduced equilibrium closure; and the delayed trajectories concern the stated history and numerical integration. Keeping those objects separate is what permits all three to be strong results rather than one ambiguous simulation.

Table 9.1: Selected numerical validation checks.

Identity or diagnostic	observed error or signal	required bound
Protocol G stationary law versus Gibbs weights	4.75×10^{-17}	$\leq 2 \times 10^{-12}$
Protocol N stationary residual	9.71×10^{-17}	$\leq 2 \times 10^{-12}$
Protocol N maximum absolute cycle affinity	1.786	$\geq 10^{-3}$
Erlang busy-count identity over 52 loads	3.33×10^{-16}	$\leq 5 \times 10^{-12}$
Heterogeneous-duration stallwise flow balance	6.11×10^{-16}	$\leq 5 \times 10^{-12}$
Subcritical late delayed amplitude	3.10×10^{-15}	$\leq 2 \times 10^{-7}$
Supercritical late delayed amplitude	0.149116	≥ 0.02
Nonlinear amplitude step-size discrepancy	6.02×10^{-4}	≤ 0.025
Characteristic abscissa at the stability boundary	1.02×10^{-14}	$\leq 2 \times 10^{-8}$

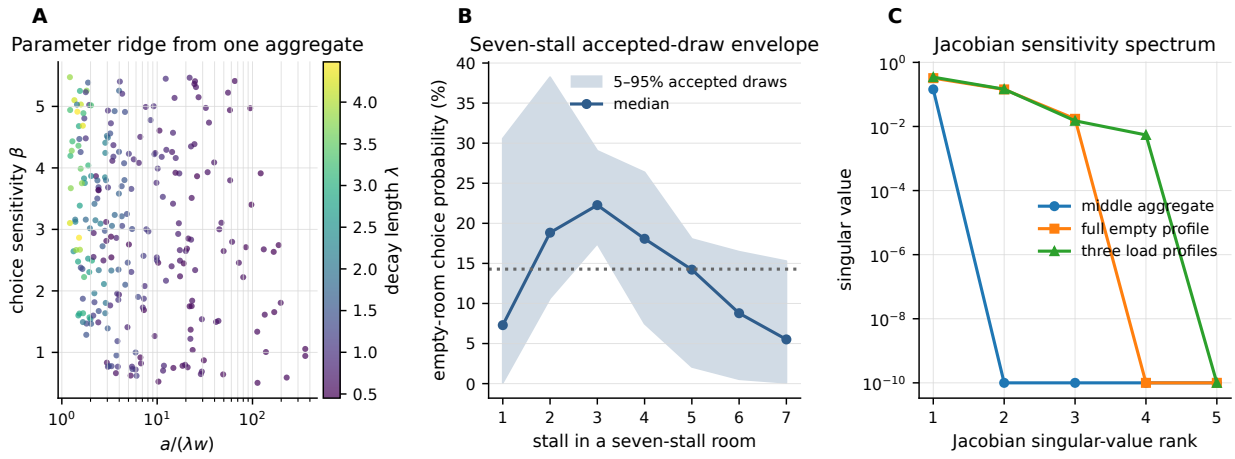


Figure 9.12: Sensitivity and identifiability. Panel A shows part of the broad parameter ridge consistent with one four-stall middle-depletion aggregate when it is treated as a choice share; color is λ. Panel B extrapolates 900 accepted positional calibrations to seven stalls and shows the proposal-and-tolerance-conditioned 5 % to 95 % pointwise envelope, not a confidence interval. Panel C gives singular values of local observation Jacobians for one aggregate, the full four-stall empty-room profile, and full profiles at offered loads 0.25, 1.5, and 5.

10

Identification, inference, and a decisive field study

10.1 The inferential target

The model is useful only if its terms can be separated in data. The primitive behavioral parameter vector can include

$$\vartheta = (a, \lambda, w, \boldsymbol{\gamma}, J, \ell, H, \beta),$$

along with arrival, service, deposition, decay, and cleaning parameters. The central estimands are not all of these numbers in isolation. They are:

1. the empty-room positional probability vector;
2. the change in log odds caused by a current occupant at each distance;
3. the response to own-stall history after conditioning on position and current occupancy;
4. per-stall visit-start and holding-time distributions;
5. the hygiene decay and information-response timescales;
6. the resulting stationary or time-varying distribution of entry, occupancy, and depletion.

Only products such as βa, βw, βJ, and βH enter the conditional logit. A cost scale must therefore be normalized. One can set $\beta = 1$ and interpret every coefficient in log-odds units, or fix one behavioral coefficient using an external calibration and estimate β.

Hygiene stock has a second, distinct scale invariance. For any $b > 0$, the transformation

$$m_i' = b m_i, \qquad \delta_m' = b\delta_m, \qquad \sigma' = b\sigma, \qquad H' = H/b, \qquad \eta' = \eta/b$$

leaves every choice probability unchanged. An assay with fixed physical units, an externally calibrated deposition coefficient, or one explicit stock normalization is therefore required to identify the absolute hygiene-load scale. Decay rates and dimensionless cleaning fractions are unaffected by this rescaling.

10.2 Event-level conditional likelihood

For arrival r, let F_r be the free set, I_r the chosen stall, $\boldsymbol{s}_r$ the pre-choice occupancy vector, and $\widehat{\boldsymbol{m}}_r$ the hygiene information available to the user. Define

$$C_{ir}(\vartheta) = g_i(\vartheta) + J(K(\ell)\boldsymbol{s}_r)_i + H\widehat{m}_{ir}. \tag{10.1}$$

The conditional log-likelihood is

$$\ell_C(\vartheta) = \sum_{r=1}^{R} \left[-\beta C_{I_r r}(\vartheta) - \log \sum_{k \in F_r} e^{-\beta C_{kr}(\vartheta)} \right]. \tag{10.2}$$

Conditioning on F_r removes the need to model the arrival rate in order to estimate choice. It also prevents availability bias: occupied stalls are not treated as rejected alternatives. Empty-room events identify g_i without a current privacy term. Events in which the same focal alternatives i and j are both free, while the occupancy of other stalls varies, identify J and ℓ. Events with comparable current states but different own histories identify H.

10.2.1 A direct odds interpretation

For free stalls i and j in event r,

$$\log \frac{P_{ir}}{P_{jr}} = -\beta(g_i - g_j) - \beta J[(K\boldsymbol{s}_r)_i - (K\boldsymbol{s}_r)_j] - \beta H(\widehat{m}_{ir} - \widehat{m}_{jr}). \tag{10.3}$$

An adjacent occupied neighbor changes a stall's odds by a multiplicative factor $e^{-\beta J}$ when the comparison stall's exposure is unchanged and adjacent kernel weight is one. A one-unit hygiene difference changes the odds by $e^{-\beta H}$. These are observable contrasts, not latent narrative labels.

10.3 The full continuous-time likelihood

If arrival and departure times are observed, the state process has a standard point-process likelihood. Let event times be $0 < t_1 < \cdots < t_M < T$, let $r_{e_m}(X(t_m^-); \theta)$ be the rate of the observed transition type, and let

$$R(X(t); \theta) = \sum_e r_e(X(t); \theta)$$

be the total event rate. The log-likelihood is

$$\ell_{\mathrm{CT}}(\theta) = \sum_{m=1}^{M} \log r_{e_m}(X(t_m^-); \theta) - \int_0^T R(X(t); \theta)\, \mathrm{d}t. \tag{10.4}$$

For Protocol N with common Λ and μ, this likelihood separates into arrival-time, conditional-choice, and holding-time parts. With balking, state-dependent arrival rates, or hygiene-dependent duration, the integral term contains scientifically meaningful parameters and should be retained.

10.4 Holding times and visit classes

Let L_r be the duration of visit r. A proportional-hazards model can test whether privacy affects departure:

$$\mu_r(t) = \mu_0(t; Z_r) \exp\{\theta_{\mathrm{stall},I_r} + \theta_{\mathrm{near}}(K\boldsymbol{s}(t))_{I_r}\}, \tag{10.5}$$

where Z_r contains anonymous contextual covariates such as time block and whether a urinal was available. A mixture model can instead represent short and long visit classes. Duration modeling matters for two reasons:

1. it determines the mapping from visit starts to time occupancy;
2. it tests whether privacy changes the service process, one implication of the lavatory proximity literature.

Per-stall entry shares and occupancy shares should be reported separately even when their rankings coincide.

10.5 Reconstructing hygiene history

Given entry, occupancy, and cleaning records, the stock (7.1) can be reconstructed for any proposed $(\delta_m, \sigma, \kappa_0, \chi)$ once its initial value is specified. A documented complete reset sets $\boldsymbol{m}(0) = \mathbf{0}$; otherwise the initial stocks are nuisance parameters, or an explicit burn-in period must make their effect negligible. If direct hygiene samples Y_{ik} are available at times t_{ik}, an observation model such as

$$\log \frac{Y_{ik} + c_Y}{Y_0} = \alpha_i + \zeta \log \frac{m_i(t_{ik}) + c_m}{m_0} + \epsilon_{ik} \tag{10.6}$$

links the latent load to the declared assay. Here α_i is a stall-specific calibration intercept, ζ is the assay response to a log-unit change in modeled load, and ϵ_{ik} is a centered measurement error. The positive offsets c_Y and c_m have the units of the assay and modeled stock, respectively; Y_0 and m_0 are fixed positive reference scales with the same respective units. The offsets, reference scales, and error law are chosen before analysis. This observation model fixes the absolute stock scale only when its calibration parameters or reference standards are externally anchored.

If users act on displayed recent-use counts rather than physical measurements, $\widehat{m}_i$ is known by design. That creates the cleanest test of behavioral hygiene avoidance because it separates perception from physical load. A second-stage study can then connect physical assays to perceived cleanliness.

10.6 Identification as a rank condition

Let $\boldsymbol{x}_{ir}$ contain the stall-event covariates in (10.1). Conditional logit identifies coefficient contrasts when the within-choice-set design has full rank after removing event-specific constants. In practical terms:

- entrance distance must vary independently of permanent stall defects;
- neighbor exposure must vary while both comparison stalls remain free;
- own-stall history must vary independently of current occupancy;
- cleaning must not always occur at the same time and in the same order;
- accessible-stall use must be modeled by eligibility or user type, not absorbed into an unexplained fixed effect.

Different temporal kernels create an opportunity for identification, not an automatic guarantee. Current privacy and historical hygiene are separable when the columns

$$\{(K\boldsymbol{s}_r)_i - (K\boldsymbol{s}_r)_j\}_r \quad \text{and} \quad \{\widehat{m}_{ir} - \widehat{m}_{jr}\}_r$$

contain independent variation. Randomized cleaning and information timing create that variation deliberately.

10.7 Why aggregates cannot estimate the full model

An aggregate histogram $\widehat{\boldsymbol{p}}$ supplies $n - 1$ independent numbers. A flexible model with $n - 1$ baseline contrasts, privacy, screening length, hygiene response, and service heterogeneity has more unknown contrasts than the histogram contains. Moreover, averaging discards the state that makes privacy observable.

The problem is structural, not a shortage of numerical optimization. For any candidate J, baseline costs can often be adjusted so that the stationary chain reproduces the same aggregate shares. For any candidate visit-start vector, stall-specific paper-use means can reproduce the same depletion vector. Additional weeks reduce sampling error around the aggregate but do not restore the missing conditioning information.

10.8 A privacy-preserving sensor study

The complete event log can be collected without imaging, audio, device identifiers, or persistent user identifiers. A suitable system contains:

- a beam or pressure counter at the restroom entrance;
- magnetic door and anonymous occupancy sensors for each stall;
- a random one-use visit token during instrumented research blocks, used only to associate one entrance with one stall entry and destroyed at exit;
- timestamps recorded at a resolution sufficient for event ordering and coarsened only in the analysis release;
- a cleaning interface that records which stalls were serviced and when;
- a one-time geometry survey;
- optional surface samples identified only by stall and time.

The raw event stream should be transformed immediately into anonymous state transitions. The one-use token is a data-association device, not an identity: it has no stable device field, is never reused, and cannot link visits across time. Where a token is unavailable, the choice likelihood includes only transitions with a unique entrance-to-stall association; concurrent or otherwise ambiguous transitions are flagged rather than guessed. The scientific unit is an arrival event, not an individual identity.

10.8.1 Data association and sensor error

Let $Y_i(t)$ be the recorded occupancy state and $s_i(t)$ the latent true state. A minimal independently calibrated sensor model is

$$\mathbb{P}\{Y_i = 1 \mid s_i = 1\} = 1 - \nu_i, \qquad \mathbb{P}\{Y_i = 1 \mid s_i = 0\} = \phi_i,$$

where ν_i is the false-negative probability and ϕ_i the false-positive probability. Staged occupied and unoccupied validation intervals estimate both rates for every stall. The event likelihood then either marginalizes over the latent occupancy sequence with these observation probabilities or restricts the primary analysis to manually verified intervals and uses the calibrated model as a sensitivity analysis. Clock drift, simultaneous timestamps, and missed entrance events receive analogous predeclared status flags. This layer is necessary because an incorrectly reconstructed free set changes the denominator of the conditional choice probability, not merely its variance.

10.8.2 Site sampling

A multi-site design should deliberately span:

1. rows of three through at least eight stalls;
2. entrances at an end and entrances offset from the row;
3. end walls of different construction;

4. high- and low-traffic facilities;
5. frequent and infrequent cleaning schedules;
6. uniform rows and rows containing clearly nonidentical stalls.

The parametric position model earns scientific value by predicting across these geometries. A separate fixed effect for every stall can fit each site but cannot predict a redesigned site.

10.9 Three randomized perturbations

10.9.1 Availability

Brief rotating maintenance closures alter the free set. Because the closure order is randomized across comparable traffic blocks, the displacement of choice identifies substitution patterns and tests the logit's independence structure. Closures must never be used to reduce legally or practically required accessible capacity.

10.9.2 Cleaning

Within an operationally acceptable range, randomize whether otherwise identical stalls are cleaned early or late in a block. Direct samples validate the resulting load contrast. The post-clean choice response estimates H; the recovery curve estimates κ.

10.9.3 Information age

Display an accurate recent-cleaning or recent-use signal in some blocks and no signal in others. When operationally and ethically approved, randomize the signal's averaging window among accurate windows. This changes information age without falsifying the underlying measurement. The effect of window length on autocorrelation and switching estimates the response kernel.

10.10 Power calculations from first principles

10.10.1 A middle-versus-end share

Suppose the outcome is a binary classification: middle pair versus end pair in a four-stall row. To test $p_0 = 0.50$ against $p_1 = 0.60$ with two-sided type-I error 0.05 and power 0.80, a normal approximation gives

$$N \approx \frac{\left[z_{0.975}\sqrt{p_0(1-p_0)} + z_{0.80}\sqrt{p_1(1-p_1)}\right]^2}{(p_1 - p_0)^2}. \tag{10.7}$$

Using $z_{0.975} = 1.96$ and $z_{0.80} = 0.84$ gives

$$N \approx 194$$

independent events. Correlation within days, cleaning blocks, or roll replacement intervals increases the requirement by the design effect $1 + (m - 1)\rho_c$, where m is average cluster size and ρ_c an intracluster correlation.

10.10.2 A log-odds contrast

For a balanced binary comparison with covariate contrast one, the Fisher information per event for a log-odds coefficient is at most $1/4$. Detecting coefficient magnitude Δ with the same nominal error and power therefore

requires approximately

$$N \gtrsim \frac{4(z_{0.975} + z_{0.80})^2}{\Delta^2} = \frac{31.4}{\Delta^2}. \tag{10.8}$$

An odds ratio of 0.75 has $|\Delta| = |\log 0.75| \approx 0.288$ and gives about 379 informative balanced comparisons. Only events in which the relevant stalls are simultaneously available and the neighbor contrast varies are informative for J, so total traffic must be larger.

10.11 Model checking

A fitted model should pass checks at every scale.

1. **Choice calibration.** Within bins of predicted probability, the empirical choice frequency should agree with the prediction.
2. **State calibration.** Exact or simulated stationary configuration frequencies should match observed patterns, not only per-stall marginals.
3. **Busy-count calibration.** Under Protocol N with common service, the observed count distribution should match the Erlang loss distribution, including its full-state probability given by Erlang–B; systematic failure indicates time variation, balking, or service heterogeneity.
4. **Duration calibration.** Survival curves should be checked by visit class, stall, and neighbor state.
5. **Temporal calibration.** Residual per-stall choice and hygiene series should have no unexplained autocorrelation at the cleaning or predicted delay timescales.
6. **Out-of-site prediction.** Geometry-based parameters should predict a held-out restroom, the test that distinguishes mechanism from stall-specific curve fitting.

10.12 Predeclared tests of the theory

These tests divide the integrated theory into components. A failure of one does not make the remaining mechanisms unmeasurable; it identifies which link in the generative chain needs replacement.

Table 10.1: Falsifiable model predictions and their decisive measurements.

Test	Prediction	Measurement
P1	At sparse traffic, entry shares approach the empty-room positional logit in (5.21).	Empty-state arrivals across traffic strata.
P2	A current neighbor lowers conditional odds according to (10.3).	Contrasts with the same focal alternatives available and randomized or naturally varying other-stall occupancy.
P3	With flat baselines and weak privacy, the first mean-field correction favors boundaries.	Comparable rows or temporary geometry balancing.
P4	Under forced entry and common service, entry shares approach $1/n$ at very high offered load.	High-load state and duration logs.
P5	Hygiene response attenuates small positional contrasts by $n/(n+\beta h)$ when privacy is negligible.	Randomized accurate recent-use information.
P6	Event, occupancy, and depletion shares diverge in proportion to duration and mark heterogeneity.	Door sensors plus supply or paper measurements.
P7	Delayed nonuniform modes are stable when $\max_r A_r \le \kappa$ and lose stability at (8.17).	Traffic, cleaning, signal-age, and spectral estimates.
P8	The first unstable spatial pattern aligns with the softest Hessian mode.	Cross-stall phase and eigenvector comparison.
P9	Cleaning-phase-locked cycles remain locked when traffic changes; endogenous delay cycles shift with q.	Joint intervention on traffic strata and cleaning phase.

11

Facility design and regime-dependent conclusions

11.1 The facility-design objective

An entrant minimizes personal cost. A facility designer balances privacy, hygiene, access, capacity, waiting, and cleaning effort across everyone. The two objectives are related but not identical. A general planning objective is

$$\mathcal{W} = \underbrace{\sum_i a_i g_i}_{\text{position and amenities}} + \underbrace{J \sum_i a_i \mathbb{E}_i^{\text{ent}}\left[\sum_j K_{ij} s_j\right]}_{\text{privacy cost at entry}} + \underbrace{H \sum_i a_i \mathbb{E}_i^{\text{ent}}[m_i]}_{\text{hygiene exposure at entry}} + \underbrace{c_B \Lambda B}_{\text{blocking or waiting}} + \underbrace{C(\boldsymbol{c})}_{\text{cleaning effort}}, \quad (11.1)$$

subject to accessibility, building, and operating constraints. The vector $\boldsymbol{c}$ denotes stall-specific cleaning intensity, and $\mathbb{E}_i^{\text{ent}}$ is the Palm expectation seen at an accepted entry into stall i. Thus every displayed term is a cost rate rather than a mixture of arrival-weighted and snapshot quantities.

The equation is not proposed as a universal monetary valuation. It is a map of the tradeoffs. Every term corresponds to a separately measurable outcome, and the design variables enter through specific mechanisms.

11.2 Geometry and privacy

11.2.1 Partitions dominate behavioral spacing

Relying on users to leave adjacent stalls empty sacrifices effective capacity. In the hard-exclusion benchmark, random irreversible filling on a long row has limiting density $(1 - e^{-2})/2 \approx 0.4323$ occupants per installed position, compared with the optimal alternating density $1/2$. With ongoing departures the exact number changes, but the operational lesson survives: privacy supplied by construction does not consume a vacancy.

Longer partitions, better acoustic isolation, and screened door sightlines reduce J or shorten the kernel K_{ij}. That simultaneously lowers experienced privacy cost and reduces strategic interaction between choices.

11.2.2 Entrance placement changes the baseline

The unconstrained stationary point

$$x_0 = \lambda \log\left(\frac{a}{\lambda w}\right)$$

is measured from the commitment point; the physical optimum is $x^\star = [x_0]_{[0,L]}$. Moving or screening the entrance changes the exposure profile; shortening the path changes walking cost. Consequently, a row can be designed so that no stall carries a large fixed penalty. A turn, vestibule, or visual screen near the entrance can

reduce a without requiring users to walk farther.

11.2.3 End walls are amenities

An end wall lowers expected exposure because it removes one flank. Treating it as a covariate in g_i rather than only through the interaction kernel allows for a wall benefit even in an empty room. If use becomes excessively edge-concentrated, improving interior partitions addresses the cause more directly than attempting to nudge people away from the walls.

11.3 Cleaning allocation

Suppose hygiene input is event driven at rate $\delta_m a_i$, and stall i is cleaned through an effective continuous rate c_i. Its mean load is

$$\overline{m}_i = \frac{\delta_m a_i}{c_i}.$$

The following rules solve a conditional one-step allocation problem: the visit-start rates a_i are held fixed at their measured or forecast values while the cleaning budget is assigned. Two planning objectives then give transparent exact solutions.

11.3.1 Minimizing the sum of stall loads

With a cleaning-rate budget $\sum_i c_i = C$, minimizing

$$\sum_i \overline{m}_i = \delta_m \sum_i \frac{a_i}{c_i}$$

gives, by Lagrange multipliers,

$$c_i = C \frac{\sqrt{a_i}}{\sum_j \sqrt{a_j}}. \tag{11.2}$$

Cleaning intensity is proportional to the square root of visit rate.

11.3.2 Minimizing load encountered by visits

Add the closure that starts into stall i sample its stationary stock with mean $\overline{m}_i$. This is exact for an exogenous Poisson start stream independent of m_i by PASTA, and it is a declared mean-field approximation when choice depends on hygiene. Under that closure, weighting each stall's load by how many entrants encounter it gives

$$\sum_i a_i \overline{m}_i = \delta_m \sum_i \frac{a_i^2}{c_i},$$

the optimum is

$$c_i = C \frac{a_i}{\sum_j a_j}. \tag{11.3}$$

Cleaning in proportion to use equalizes $\overline{m}_i$ across stalls:

$$\overline{m}_i = \frac{\delta_m \sum_j a_j}{C}.$$

This rule directly removes the evidence base for a "cleanest stall" based on visit count.

If users observe the new cleaning policy and change stalls in response, the rates a_i are endogenous and the coupled allocation–choice fixed points must be computed. Candidate fixed points can be sought by

damped alternation of the allocation rule and occupancy-choice model, followed by a residual check, multiple initializations, and comparison of the planner objective; convergence or uniqueness does not follow from the conditional formulas. Equations (11.2) and (11.3) remain the exact cleaning step conditional on each iterate's demand rates.

Operational cleaning rule

Conditional on forecast visit-start rates, if the aim is to equalize the mean event-driven load encountered by users, clean each stall in proportion to those rates. If the aim is instead to minimize the unweighted sum of stall loads, allocate cleaning in proportion to the square root of visit rate. A uniform schedule is optimal only when use is uniform or when other constraints dominate.

11.4 Information design

Publicly announcing that one stall is least used changes its use. A static ranking therefore has three defects:

1. it is based on an observable that may not equal hygiene;
2. it becomes stale after traffic and cleaning;
3. common adoption erases the advantage it reports.

Accurate last-cleaned information is more interpretable than a popularity ranking. If recent-use information is displayed, smoothing and update frequency should be chosen with the stability gain in mind. Highly synchronized users responding to a delayed ranking can create avoidable oscillation. Randomized or individualized routing would reduce synchronization, but a public restroom should preserve free choice and collect no personal identity. Better physical privacy and responsive cleaning solve the underlying problem without coercive assignment.

11.5 Accessibility is a constraint, not a nuisance parameter

An accessible stall is not interchangeable with a standard stall. Its larger space, fixtures, transfer clearances, and location are essential to some users and optional amenities to others. The correct model gives eligible users a different utility or choice set and imposes accessibility requirements on any closure or routing policy.

Operational attempts to "balance" usage must not make the accessible stall unavailable to the people who need it. Data should therefore be interpreted at the level of anonymous choice contexts, not used to discourage lawful access or infer disability.

11.6 An individual decision rule

For a particular entrant at a particular moment, the model gives an exact decision rule:

$$i^\star(t) \in \arg\min_{i \in F(t)} \left\{ g_i + J \sum_j K_{ij} s_j(t) + H\widehat{m}_i(t) + \theta V_i(S(t) \cup \{i\}) \right\}, \tag{11.4}$$

where V_i is optional anticipated future privacy and $\widehat{m}_i$ is the information actually available. The conversion rate θ has units $1/\text{time}$ because V_i is cumulative cost–time. At finite choice precision, replace the argmin by the logit probabilities.

Without direct hygiene information, position is only a prior. In the four-stall setting reported by Christenfeld (1995), the end pair had lower paper depletion than the middle pair. That makes an end stall the lower-depletion

prior for a comparable setting. It does not establish that the first stall is physically cleaner, that the same ordering holds in a different geometry, or that the ordering survives when many users know the result.

Visible condition and recent verified cleaning are more informative than a population positional prior. Current adjacency informs privacy, not past hygiene. The practical hierarchy is therefore:

1. respect accessibility and actual availability;
2. use direct visible and cleaning information;
3. for present privacy, prefer greater screened distance from occupants;
4. use positional depletion evidence only as a weak prior when the room is otherwise symmetric.

11.7 The distribution by traffic regime

Table 11.1: Dominant mechanisms across regimes under the normalized forced-entry model with common service rates.

Regime	Leading distributional force	Consequence
Sparse traffic	baseline positional cost g_i	entry shares approach the empty-room logit; current privacy is a first-order-in-load correction
Intermediate traffic	availability and spatial privacy	finite-line boundaries can gain; center-to-edge or alternating crossovers depend on parameters
Saturated forced traffic	common service capacity	all $q_i \to 1$ and entry shares return to $1/n$
Strong current hygiene response	own-use congestion hp_i	persistent positional contrasts flatten; deterministic support follows water filling
Strong delayed hygiene response	gain A_r versus cleaning κ	the first mode loses stability at its exact delay threshold and may generate oscillatory popularity

11.8 Testable social predictions

The integrated model yields predictions that are sharper than "people prefer the middle" or "people avoid neighbors."

1. **Traffic crossover.** With an interior baseline advantage and privacy repulsion, the middle-versus-edge contrast can change sign at intermediate load and must return toward zero under extreme forced load.
2. **Boundary response.** Holding baseline position fixed, an interior free stall loses more probability than a boundary stall when diffuse occupancy rises.
3. **Hygiene flattening.** Accurate evidence of accumulated use reduces persistent cross-stall dispersion; the local attenuation is $n/(n+\beta h)$ without privacy.
4. **Adoption attenuation.** The benefit of a least-used-stall recommendation declines with the fraction of users acting on it, as in (7.24).
5. **Cleaning stabilization.** Raising effective removal above every delayed mode gain prevents delay-driven instability at any information age.

6. **Mode selection.** When privacy softens an alternating spatial direction, that mode can destabilize before a broad center-versus-edge mode.
7. **Observable divergence.** Long-duration sorting creates larger occupancy contrasts than entry contrasts; paper-use sorting creates depletion contrasts not present in either.

11.9 Contribution of the synthesis

The component mathematics has distinguished lineages: random utility, Erlang loss systems, lattice gases, potential games, random sequential adsorption, and delay equations. The scientific contribution here is their integration around a measurement-defined question and the separation of models that superficially share the same exponential form.

The central synthesis consists of:

- a single cost with a fixed positional field, instantaneous spatial cross-load, and historical own-load;
- an exact theorem showing why normalized forced choice and a Gibbs lattice gas are generally incompatible microscopic descriptions;
- the coexistence of an exact Erlang busy-count law with a nonreversible configuration process;
- a quadratic hygiene potential derived from deposition and cleaning;
- a full matrix response showing how privacy changes both boundary forcing and stability by spatial mode;
- a mode-dependent delayed-feedback threshold expressed in estimable primitives;
- a measurement theory that separates visits, occupied time, depletion, and hygiene load.

11.10 Regime-dependent allocation

The public-stall allocation problem, resolved

The long-run use of public toilet stalls is the stationary or time-varying output of a state-dependent choice system.

Position determines the empty-room prior. Door exposure falling with distance and walking cost rising with distance can create an interior optimum. Current privacy adds repulsion from occupied stalls and gives boundaries a geometric advantage because they have one inward flank. Poisson arrivals and exponential holding times turn those choices into an occupancy process: if every nonfull arrival takes a stall and service rates are common, the number busy has the Erlang loss distribution, with Erlang–B blocking, even though the allocation among stalls is generally nonreversible.

Hygiene is the temporal reverse of privacy in the precise kernel sense. The minimal component modeled here is own-stall history filtered by decay and cleaning; shared air, contact surfaces, or cleaning tools require a cross-stall hygiene kernel. At steady state, linear deposition makes its marginal cost proportional to own use, and its potential is quadratic in the usage shares. Hygiene awareness therefore flattens persistent popularity. A recommendation to use the least-used stall erases its own advantage as adoption grows.

When hygiene information is delayed, the same negative feedback can oscillate. Each spatial mode has

gain $A_r = \delta_m q\eta/d_r$ and is stable for every delay if $A_r \le \kappa$. If $A_r > \kappa$, it first becomes unstable at

$$\tau_c = \frac{\arccos(-\kappa/A_r)}{\sqrt{A_r^2 - \kappa^2}}.$$

The identity of the stall most preferred under the delayed cleanliness signal can then be a dynamic quantity rather than a fixed position.

There is consequently no context-free best stall. There is a self-contained conditional model, an exact finite-state occupancy solution for fixed hygiene in finite rows, and a sensor study designed to identify each declared mechanism up to the explicit cost and hygiene-stock scale normalizations.

11.11 Empirical research program

The next empirical advance should be one multi-site, event-level study rather than another aggregate popularity poll. Anonymous state sensors, cleaning logs, and randomized information windows can estimate the positional field, privacy kernel, service process, hygiene response, and delay in the same data. The model predicts both the average histogram and the sequence of choices conditional on every state.

That study would answer the practical question for real facilities and test the general scientific claim: collective use is produced by a spatially coupled system whose present choices write the history that future choices read.

Part V

Technical appendices

A

Dimensions, conventions, and model map

A.1 Dimensional analysis

Dimensional consistency prevents several attractive but incorrect formulas. Choose abstract units of time T, cost C, and hygiene load M. The principal quantities have dimensions shown in Table A.1.

Table A.1: Dimensions of the model primitives and reductions.

Symbol	Meaning	Dimension	Note
g_i	baseline cost	C	only differences affect choice
J	adjacent privacy cost	C	K_{ij} is dimensionless
H, η	hygiene cost per load	C/M	static and primitive dynamic notation, respectively
β	choice precision	1/C	makes βC dimensionless
Λ, q	potential and admitted start rates	1/T	$q \le \Lambda$
μ	departure rate	1/T	mean duration $1/\mu$
κ	load removal rate	1/T	decay or effective cleaning
δ_m	load per visit start	M	event-dose model
σ	load per occupied time	M/T	occupancy-dose model
$\varphi(v)$	hygiene-memory load rate	M/T	occupancy-time convolution
χ	cleaning removal fraction	1	fraction removed at each cleaning
m_i	hygiene stock	M	interpretation must be declared
p_i, q_i	entry share, occupancy probability	1	both dimensionless
h	steady own-load coefficient	C	marginal cost is hp_i
A_r	delayed-feedback gain	1/T	compared with κ
τ	information delay	T	$A_r\tau$ is dimensionless

For the event-dose stock

$$\dot{m}_i = \delta_m q p_i - \kappa m_i$$

the stock has units M. If cost is ηm_i, then

$$h = \frac{\eta \delta_m q}{\kappa}$$

has units of cost and

$$A = \frac{\delta_m q \beta \eta}{n} = \frac{\kappa \beta h}{n}$$

has units $1/\mathsf{T}$. The expression $q\beta h/n$ is also dimensionally an inverse time, so dimensions alone cannot expose its error. It is wrong because h is already the steady coefficient $\eta\delta_m q/\kappa$; multiplying by q again counts traffic twice.

A.2 From primitives to reduced models

Table A.2: How the exact, equilibrium, and delayed models are related.

Level	State and equation	Reduction
Microscopic choice	$P_i(S, \boldsymbol{m}) \propto e^{-\beta[g_i + J(K\boldsymbol{s})_i + H m_i]}$ over free stalls	describes one arrival conditional on the observed state
Occupancy CTMC	births $\Lambda P_i(S, \boldsymbol{m})$, deaths μ_i	stationary $\pi(S)$ gives exact finite-row use for fixed $\boldsymbol{m}$; continuous hygiene creates a hybrid process
Hygiene stock	$\dot{m}_i = \delta_m a_i + \sigma q_i - \kappa m_i$ in mean	$H\overline{m}_i = hp_i$ under common duration and constant shares
Population equilibrium	$p_i \propto e^{-\beta[g_i + \alpha(Kp)_i + hp_i]}$	mean-field closure for persistent shares
Delayed population	$\dot{m} = \delta_m q p - \kappa m$, $p(t) = \mathcal{S}(m(t-\tau))$	linear modes have gains $A_r = \delta_m q \eta / d_r$

No arrow in Table A.2 is an identity without its stated assumptions. In particular, the population equilibrium is not automatically the stationary marginal of the microscopic CTMC.

B

Exact positional calculations

B.1 Unequal stall spacing

For unequal distances $x_1 < \cdots < x_n$, the baseline sequence

$$g_i = ae^{-x_i/\lambda} + wx_i$$

need not have equal finite differences, but strict convexity of $g(x)$ still implies a unimodal sampled sequence. The exact discrete minimizer is any index i satisfying

$$g_i \le g_{i-1}, \qquad g_i \le g_{i+1}, \tag{B.1}$$

with one-sided conditions at the ends. Because a strictly convex continuous function crosses every horizontal slope once, there are at most two adjacent sampled minimizers.

The difference between two stalls has the direct log-odds interpretation

$$\log \frac{P_i(\varnothing)}{P_j(\varnothing)} = -\beta a(e^{-x_i/\lambda} - e^{-x_j/\lambda}) - \beta w(x_i - x_j) - \beta(z_i - z_j)^{\mathsf{T}}\boldsymbol{\gamma}. \tag{B.2}$$

Multiple restroom geometries identify the decay and walking terms because the same parameters predict different contrasts as the distances change.

B.2 Exact equal-spacing index

For $x_i = (i-1)\Delta$, define

$$\Delta_i = g_{i+1} - g_i = w\Delta - ae^{-(i-1)\Delta/\lambda}(1 - e^{-\Delta/\lambda}).$$

The increments satisfy

$$\Delta_{i+1} - \Delta_i = ae^{-(i-1)\Delta/\lambda}(1 - e^{-\Delta/\lambda})^2 > 0.$$

Let

$$r_\Delta = 1 + \frac{\lambda}{\Delta}\log\left[\frac{a(1 - e^{-\Delta/\lambda})}{w\Delta}\right]. \tag{B.3}$$

If $1 < r_\Delta < n$ and r_Δ is not an integer, the unique minimizer is $\lceil r_\Delta \rceil$. If r_Δ is an integer with $1 \le r_\Delta \le n-1$, stalls r_Δ and $r_\Delta + 1$ tie. For $r_\Delta < 1$ the unique minimizer is stall 1; for $r_\Delta \ge n$ it is stall n, apart from the already stated tie at $r_\Delta = 1$.

The discrete answer is not obtained by blindly rounding the continuous $x^\star$. The two agree asymptotically as $\Delta/\lambda \downarrow 0$ because

$$\frac{1 - e^{-\Delta/\lambda}}{\Delta} \to \frac{1}{\lambda}.$$

B.3 Putting the minimum at the geometric center

An interior optimum is not necessarily a geometric-center optimum. For odd n, the center index is $i_c = (n+1)/2$ and distance $x_c = (n-1)\Delta/2$. The continuous optimum equals x_c exactly when

$$a = \lambda w e^{x_c/\lambda}. \tag{B.4}$$

For even n, a tie between the two middle stalls requires the finite difference at the left middle index to vanish. If $i_L = n/2$,

$$a = \frac{w\Delta e^{(i_L-1)\Delta/\lambda}}{1-e^{-\Delta/\lambda}}. \tag{B.5}$$

These equations emphasize that center location is a parameter implication, not an assumption hidden in the functional form.

C

Markov-chain derivations and exact computation

C.1 Generator construction

Represent a subset S by the integer

$$b(S) = \sum_{i \in S} 2^{i-1}.$$

This gives a one-to-one map from occupancy configurations to $0, \dots, 2^n - 1$. The generator has at most $n2^n$ off-diagonal entries: each state connects only to configurations obtained by flipping one bit.

For Protocol N,

$$Q_{S,S\cup\{i\}} = \Lambda \frac{e^{-\beta\Delta_i E(S)}}{\sum_{k\notin S} e^{-\beta\Delta_k E(S)}}, \qquad i \notin S,\ |S| < n, \tag{C.1}$$

$$Q_{S,S\setminus\{i\}} = \mu_i(S), \qquad i \in S, \tag{C.2}$$

$$Q_{S,S} = -\sum_{T\neq S} Q_{S,T}. \tag{C.3}$$

Every row sums to zero. With the row-vector convention, $\dot{\pi}_t = \pi_t Q$ and stationarity is $\pi Q = 0$.

The linear system is singular because $Q\mathbf{1} = 0$. Replacing one row of $Q^\top$ by $\mathbf{1}^\top$ and the corresponding entry of the right-hand side by 1 produces a nonsingular system for an irreducible chain. A computed solution is accepted only when the three residuals in (5.20) meet the declared tolerance.

C.2 The full normalized-logit reversibility criterion

Theorem C.1 (Inclusive-value criterion)**.** *For the symmetric energy (5.2), common departure rate μ, and normalized-logit birth rates, the configuration chain is reversible if and only if*

$$Z_{\mathrm{N}}(S \cup \{i\}) = Z_{\mathrm{N}}(S \cup \{j\}) \tag{C.4}$$

for every S and every pair of distinct free stalls i, j. Equivalently, $Z_{\mathrm{N}}(S)$ depends only on $|S|$ for nonfull states.

When this condition holds, write $Z_{\mathrm{N}}(S) = z_{|S|}$. The stationary law is

$$\pi(S) \propto \rho^{|S|} e^{-\beta E(S)} \left[\prod_{r=0}^{|S|-1} z_r\right]^{-1}. \tag{C.5}$$

Proof. Necessity is the square-cycle calculation in Theorem 5.2. For a fixed cardinality k, the graph whose vertices are k-subsets and whose edges exchange one included element is connected. The square condition therefore makes the inclusive value constant on each cardinality level.

Conversely, take the ratio of (C.5) between $S \cup \{i\}$ and S:

$$\frac{\pi(S \cup \{i\})}{\pi(S)} = \frac{\rho e^{-\beta \Delta_i E(S)}}{z_{|S|}} = \frac{\lambda_i^{\mathrm{N}}(S)}{\mu}.$$

This is detailed balance. □

Even in the exceptional reversible case, the cardinality product in (C.5) is a many-body correction. It is absorbed into an ordinary fugacity $\zeta^{|S|}$ only if z_r is constant in r.

C.2.1 Pure random choice

When $\beta = 0$, every free stall is equally likely and $Z_{\mathrm{N}}(S) = n - |S|$. The reversible state weight is

$$\pi(S) = \pi(\varnothing)\rho^{|S|}\frac{(n - |S|)!}{n!}. \tag{C.6}$$

Conditional on $K = k$, all $\binom{n}{k}$ configurations are equally likely. Multiplying (C.6) by $\binom{n}{k}$ yields a quantity proportional to $\rho^k/k!$, recovering the Erlang busy-count law.

C.3 A stable Erlang–B recursion

Direct evaluation of factorials and powers is numerically unstable at large n or ρ. Let $B_m(\rho)$ be the blocking probability with m stalls. Starting from $B_0(\rho) = 1$, the recursion

$$B_m(\rho) = \frac{\rho B_{m-1}(\rho)}{m + \rho B_{m-1}(\rho)} \tag{C.7}$$

computes Erlang–B without large intermediate numbers.

To derive it, let

$$G_m = \sum_{k=0}^{m} \frac{\rho^k}{k!}.$$

Then

$$B_m = \frac{\rho^m/m!}{G_m} = \frac{(\rho/m)(\rho^{m-1}/(m-1)!)}{G_{m-1} + (\rho/m)(\rho^{m-1}/(m-1)!)}.$$

Divide numerator and denominator by G_{m-1} to obtain (C.7).

C.4 Transfer matrix for nearest-neighbor Gibbs interaction

For an exact Gibbs model with nearest-neighbor interaction,

$$E(s) = \sum_{i=1}^{n} g_i s_i + J \sum_{i=1}^{n-1} s_i s_{i+1}.$$

Set

$$z_i = \zeta e^{-\beta g_i}, \qquad c = e^{-\beta J}.$$

Let $v_i(0)$ and $v_i(1)$ be total weights of configurations on the first i sites ending in state 0 or 1. Then

$$\begin{pmatrix} v_i(0) \\ v_i(1) \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ z_i & z_i c \end{pmatrix} \begin{pmatrix} v_{i-1}(0) \\ v_{i-1}(1) \end{pmatrix}, \qquad \begin{pmatrix} v_0(0) \\ v_0(1) \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}. \tag{C.8}$$

The partition function is

$$\mathcal{Z}_n = v_n(0) + v_n(1).$$

This calculation is linear in n, unlike enumeration of 2^n configurations. Marginal occupancy follows by differentiating:

$$q_i = z_i \frac{\partial}{\partial z_i} \log \mathcal{Z}_n. \tag{C.9}$$

Forward and backward transfer products evaluate all n derivatives in linear time.

D

Potential games, convexity, and multiplicity

D.1 The finite simultaneous-placement game

Fix $N \le n$ labeled users assigned to distinct stalls. For an action profile $a = (a_1, \dots, a_N)$, give user r the cost

$$c_r(a) = g_{a_r} + J \sum_{s \neq r} K_{a_r a_s}. \tag{D.1}$$

Proposition D.1 (Exact finite potential)**.** *Assume K is symmetric. Then the function*

$$\Psi(a) = \sum_{r=1}^{N} g_{a_r} + J \sum_{1 \le r < s \le N} K_{a_r a_s} \tag{D.2}$$

is an exact potential. The finite game has a pure Nash equilibrium.

Proof. When player r moves unilaterally, terms not involving r cancel. Every remaining positional and pairwise term appears exactly once, so the change in Ψ equals the change in c_r. A global minimizer of Ψ permits no cost-reducing unilateral move. □

This is a contemporaneous placement game. Forward-looking choice is a dynamic game with an arrival horizon and transition law; its payoff is given by a Markov-reward calculation such as (6.15).

D.2 A sharper global curvature bound

For any tangent vector $\boldsymbol{x}$ and interior probability vector $\boldsymbol{p}$, Cauchy–Schwarz gives

$$\sum_i \frac{x_i^2}{p_i} \ge \left(\sum_i |x_i| \right)^2.$$

If $\sum_i x_i = 0$, the positive and negative parts have equal ℓ_1 mass, which implies

$$\|\boldsymbol{x}\|_1^2 \ge 2\|\boldsymbol{x}\|_2^2.$$

Let

$$\lambda_{\min}^{\mathcal{T}}(K) = \min_{\substack{\boldsymbol{x} \in \mathcal{T} \\ \|\boldsymbol{x}\|_2 = 1}} \boldsymbol{x}^{\mathsf{T}} K \boldsymbol{x}.$$

Then

$$\boldsymbol{x}^{\mathsf{T}} \nabla^2 \Phi_\beta(\boldsymbol{p}) \boldsymbol{x} \ge \left[h + \alpha \lambda_{\min}^{\mathcal{T}}(K) + \frac{2}{\beta} \right] \|\boldsymbol{x}\|_2^2. \tag{D.3}$$

Thus

$$h + \alpha\lambda_{\min}^{\mathcal{T}}(K) + \frac{2}{\beta} > 0 \tag{D.4}$$

is a sharper global sufficient condition for uniqueness than a bound using $1/\beta$.

D.3 An explicit symmetry-breaking equilibrium

Take $n = 2$, equal baselines, and

$$K = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

Write

$$p_1 = \frac{1+m}{2}, \qquad p_2 = \frac{1-m}{2}, \qquad -1 < m < 1.$$

The cost difference is

$$c_1 - c_2 = (h - \alpha)m.$$

Taking log odds in the equilibrium equation gives

$$2\,\mathrm{artanh}(m) = \beta(\alpha - h)m. \tag{D.5}$$

The even function

$$f(m) = \frac{2\,\mathrm{artanh}(m)}{m}$$

increases from 2 at $m = 0$ to infinity as $m \uparrow 1$. Therefore:

- if $\beta(\alpha - h) \le 2$, uniform use $m = 0$ is the only equilibrium;
- if $\beta(\alpha - h) > 2$, two asymmetric equilibria $\pm m^\star$ appear in addition to the symmetric stationary point.

The symmetric point changes from a local minimum to a local maximum along the tangent direction at the same threshold. Privacy favors anti-coordination; hygiene and entropy restore symmetry.

D.4 A contraction condition for fixed-point iteration

Define

$$T(\boldsymbol{p}) = \mathrm{softmax}\{-\beta[\boldsymbol{g} + (\alpha K + hI)\boldsymbol{p}]\}.$$

For any probability vector $\boldsymbol{r}$,

$$Q(\boldsymbol{r}) = \mathrm{diag}(\boldsymbol{r}) - \boldsymbol{r}\boldsymbol{r}^{\mathsf{T}}$$

is a categorical covariance matrix with spectral norm at most $1/2$. Since

$$DT(\boldsymbol{p}) = -\beta Q(T(\boldsymbol{p}))(\alpha K + hI),$$

the condition

$$\frac{\beta}{2}\|\alpha K + hI\|_2 < 1 \tag{D.6}$$

makes T a Euclidean contraction. The logit fixed point is then unique, and the simultaneous iteration $\boldsymbol{p}^{(k+1)} = T(\boldsymbol{p}^{(k)})$ converges from every starting distribution. Potential structure alone does not guarantee this particular simultaneous iteration converges when (D.6) fails.

D.5 Fourier symbol of the privacy kernel

On a long translation-invariant line, take

$$K_d = r^{d-1}, \qquad d = 1, 2, \ldots, \quad 0 < r < 1.$$

The Fourier symbol is

$$\begin{aligned} \lambda(\theta) &= 2 \sum_{d=1}^{\infty} r^{d-1} \cos(d\theta) \\ &= 2\Re \left[\frac{e^{i\theta}}{1 - r e^{i\theta}} \right] \\ &= \frac{2(\cos\theta - r)}{1 - 2r\cos\theta + r^2}. \end{aligned} \tag{D.7}$$

It is negative when $\cos\theta < r$. These high-frequency directions lower the privacy quadratic form and are candidates for alternating popularity patterns.

E

Hygiene and cleaning calculations

E.1 Event-dose versus occupancy-dose stocks

For visit-start counting process $N_i^{\text{in}}(t)$,

$$\mathrm{d}m_i = -\kappa m_i \, \mathrm{d}t + \delta_m \, \mathrm{d}N_i^{\text{in}}(t) \tag{E.1}$$

has stationary mean

$$\mathbb{E}m_i = \frac{\delta_m a_i}{\kappa}.$$

For continuous deposition during occupancy,

$$\dot{m}_i = \sigma s_i - \kappa m_i \tag{E.2}$$

has stationary mean

$$\mathbb{E}m_i = \frac{\sigma q_i}{\kappa}.$$

The two coincide after setting $\delta_m = \sigma/\mu$ only under common mean duration $1/\mu$ and the flow identity $q_i = a_i/\mu$.

E.2 Periodic resetting with natural decay

With constant input u, natural decay κ_0, and perfect resets every T,

$$m(t) = \frac{u}{\kappa_0}(1 - e^{-\kappa_0 t}), \qquad 0 \le t < T.$$

The cycle mean is (7.9). The second moment is

$$\begin{aligned} \overline{m^2} &= \frac{1}{T}\int_0^T \frac{u^2}{\kappa_0^2}(1 - 2e^{-\kappa_0 t} + e^{-2\kappa_0 t}) \, \mathrm{d}t \\ &= \frac{u^2}{\kappa_0^2}\left[1 - \frac{2(1 - e^{-\kappa_0 T})}{\kappa_0 T} + \frac{1 - e^{-2\kappa_0 T}}{2\kappa_0 T}\right]. \end{aligned} \tag{E.3}$$

The variance is $\overline{m^2} - (\overline{m})^2$. As $\kappa_0 \downarrow 0$, the load becomes uniformly distributed over the linear sawtooth ages and

$$\overline{m} \to \frac{uT}{2}, \qquad \mathrm{Var}(m) \to \frac{u^2T^2}{12}.$$

E.3 Poisson perfect resets

With linear input u, no natural decay, and Poisson resets at rate c, the stationary age since cleaning is exponential:

$$f_A(a) = ce^{-ca}, \qquad a \geq 0.$$

Since $m = uA$,

$$f_m(x) = \frac{c}{u}e^{-cx/u}, \qquad \mathbb{E}m = \frac{u}{c}, \qquad \mathrm{Var}(m) = \frac{u^2}{c^2}. \tag{E.4}$$

Periodic resets with $T = 2/c$ match the mean u/c, but their variance is $u^2/(3c^2)$, only one third as large. Matching a mean does not match the distribution seen by arrivals.

E.4 Derivation of the Lambert-W equilibrium

With no privacy, the finite-noise first-order condition is

$$g_i + hp_i + \frac{1}{\beta}(1 + \log p_i) + \lambda = 0.$$

Multiplication by β and exponentiation yield

$$p_i e^{\beta h p_i} = e^{-1-\beta\lambda} e^{-\beta g_i}.$$

Set $z = e^{-1-\beta\lambda}$ and multiply by βh:

$$(\beta h p_i) e^{\beta h p_i} = \beta h z e^{-\beta g_i}.$$

Applying the defining inverse gives

$$p_i = \frac{1}{\beta h} W(\beta h z e^{-\beta g_i}).$$

The left side of the normalization equation

$$\sum_i W(\beta h z e^{-\beta g_i}) = \beta h$$

is continuous and strictly increasing in $z > 0$, so a unique normalizer exists.

F

Delay analysis beyond the scalar formula

F.1 The uniform and nonuniform subspaces

Adding a common scalar to every stall cost does not change choice. Therefore the uniform hygiene mode $\overline{m}\mathbf{1}$ is invisible to the softmax. Under (8.1),

$$\frac{\mathrm{d}}{\mathrm{d}t}\overline{m} = \frac{\delta_m q}{n} - \kappa\overline{m},$$

which converges to $\delta_m q/(n\kappa)$ without delay.

Every distributional instability lies in the tangent space. This separation also explains why the feedback gain has the factor $1/n$ at uniformity: the softmax derivative on a tangent direction is

$$D\,\mathrm{softmax}(\mathbf{0})\,\boldsymbol{x} = \frac{1}{n}\boldsymbol{x}.$$

Applied to cost $-\beta\eta\boldsymbol{m}$, it gives $\delta p = -(\beta\eta/n)P\delta m$.

F.2 Root crossing in detail

For

$$F(z,\tau) = z + \kappa + Ae^{-z\tau} = 0,$$

implicit differentiation gives

$$F_z = 1 - A\tau e^{-z\tau}, \qquad F_\tau = -Aze^{-z\tau},$$

and hence

$$\frac{\mathrm{d}z}{\mathrm{d}\tau} = \frac{Aze^{-z\tau}}{1 - A\tau e^{-z\tau}}.$$

At the crossing, (8.15) implies

$$Ae^{-i\omega\tau} = -\kappa - i\omega.$$

Substitution and multiplication by the complex conjugate of the denominator give

$$\Re\frac{\mathrm{d}z}{\mathrm{d}\tau} = \frac{\omega^2}{(1+\kappa\tau)^2 + \omega^2\tau^2} > 0.$$

Thus increasing delay moves the critical pair into the unstable half-plane.

F.3 Distributed-delay generalization

Let users respond to a normalized memory kernel $w(v) \geq 0$ with $\int_0^\infty w(v)\ \mathrm{d}v = 1$. A scalar mode becomes

$$\dot{y}(t) = -\kappa y(t) - A \int_0^\infty w(v) y(t - v)\ \mathrm{d}v. \tag{F.1}$$

The characteristic equation is

$$z + \kappa + A\widehat{w}(z) = 0, \qquad \widehat{w}(z) = \int_0^\infty e^{-zv} w(v)\ \mathrm{d}v. \tag{F.2}$$

A point delay has $\widehat{w}(z) = e^{-z\tau}$. An exponential filter with $w(v) = \gamma e^{-\gamma v}$ has

$$\widehat{w}(z) = \frac{\gamma}{z + \gamma}$$

and yields the quadratic

$$(z + \kappa)(z + \gamma) + A\gamma = 0.$$

Its coefficients are positive and its roots have negative real part for all $A, \kappa, \gamma > 0$. A single first-order exponential perception filter does not generate the same delay instability as a sufficiently long point delay; additional dynamic stages or a sharper memory distribution are required.

F.4 Degeneracy and spatial form

When the base Hessian is a scalar multiple of the identity on $\mathcal{T}$, all $n - 1$ nonuniform modes cross together. The critical eigenspace is then high-dimensional, and a scalar Hopf theorem does not determine the nonlinear branch. Fixed positional heterogeneity, boundary effects, or a nontrivial privacy Hessian splits the eigenvalues d_r and typically selects a first mode.

If the selected eigenvector is real, the linear oscillation is

$$\boldsymbol{y}(t) \approx A_0 \cos(\omega t + \varphi)\boldsymbol{v}_r.$$

Its spatial pattern stands and changes sign. A traveling wave requires two independent spatial modes with a quarter-period phase difference, a possibility associated with symmetry and mode degeneracy rather than with the scalar threshold alone.

G

Random sequential exclusion

G.1 The irreversible random-greedy recurrence

Let E_n be the expected number of occupied positions when a path of n sites is filled by repeatedly choosing a uniformly random feasible site and permanently blocking its nearest neighbors. Set $E_0 = E_{-1} = 0$.

Condition on the first chosen site j. It contributes one occupied position and leaves independent subpaths of lengths $j-2$ and $n-j-1$. Hence

$$\begin{aligned} E_n &= 1 + \frac{1}{n}\sum_{j=1}^{n}(E_{j-2} + E_{n-j-1}) \\ &= 1 + \frac{2}{n}\sum_{k=0}^{n-2} E_k. \end{aligned} \tag{G.1}$$

G.2 Generating function

Let

$$\mathcal{E}(z) = \sum_{n\geq 0} E_n z^n.$$

Multiplying (G.1) by nz^n and summing over $n \geq 1$ gives

$$z\mathcal{E}'(z) = \frac{z}{(1-z)^2} + \frac{2z^2}{1-z}\mathcal{E}(z). \tag{G.2}$$

Solving the first-order equation with $\mathcal{E}(0) = 0$ yields

$$\mathcal{E}(z) = \frac{1-e^{-2z}}{2(1-z)^2}. \tag{G.3}$$

The coefficient formula is

$$E_n = \frac{1}{2}\left[(n+1) - \sum_{k=0}^{n}\frac{(-2)^k}{k!}(n-k+1)\right]. \tag{G.4}$$

The singular coefficient at $z = 1$ gives

$$E_n = \frac{1-e^{-2}}{2}(n+1) - e^{-2} + o(1), \tag{G.5}$$

and hence jamming density

$$\theta_{\text{jam}} = \frac{1-e^{-2}}{2}.$$

G.3 Capacity comparison

The largest independent set of a path has size $\lceil n/2 \rceil$, so its asymptotic density is $1/2$. Random greedy filling loses

$$\frac{\frac{1}{2} - \theta_{\text{jam}}}{1/2} = e^{-2} \approx 0.1353$$

relative to the hard-exclusion optimum. The absolute density gap is $e^{-2}/2 \approx 0.0677$.

This number is not the unused fraction of an ordinary soft-privacy restroom. It belongs to irreversible, permanent, nearest-neighbor exclusion with uniform selection among currently feasible positions.

G.4 Dynamic hard-core equilibrium

Allow departures. Give every feasible site birth rate ν and every occupied site death rate μ, while forbidding adjacent occupancy. The reversible stationary law is

$$\pi(S) \propto z^{|S|}, \qquad z = \nu/\mu,$$

over independent sets. For constant z, the path partition function obeys

$$Z_n = Z_{n-1} + zZ_{n-2}, \qquad Z_0 = 1, \quad Z_1 = 1 + z. \tag{G.6}$$

The dominant transfer eigenvalue is

$$\lambda_+(z) = \frac{1 + \sqrt{1 + 4z}}{2}.$$

The infinite-path occupied density is

$$\begin{aligned} \theta(z) &= z \frac{\mathrm{d}}{\mathrm{d}z} \log \lambda_+(z) \\ &= \frac{1}{2}\left(1 - \frac{1}{\sqrt{1 + 4z}}\right). \end{aligned} \tag{G.7}$$

As $z \to \infty$, $\theta(z) \to 1/2$. Departures allow the dynamic system to repair gaps that permanently jam the irreversible process.

H

Data dictionary and reproducibility specification

H.1 Minimal event table

Table H.1: Fields for a privacy-preserving event-level study.

Field	Type	Definition
`site_id`	categorical	pseudonymous restroom identifier
`visit_token`	random local token	one-use association key linking one entrance, stall entry, stall exit, and restroom exit; destroyed after the visit and never reused
`event_time`	timestamp	acquisition time sufficient to order state transitions; coarsened only in the analysis release
`event_type`	categorical	entrance, stall entry, stall exit, restroom exit, cleaning start, cleaning end, or sensor status
`stall_id`	categorical	local stall identifier; null for restroom-level events
`free_mask`	bit vector	free stalls immediately before an entry
`occupied_mask`	bit vector	occupied stalls immediately before an entry
`cleaning_block`	categorical	randomized or operational cleaning condition
`info_block`	categorical	information display and averaging-window condition
`traffic_block`	categorical	predeclared time-of-day or event stratum
`sensor_quality`	categorical	valid, ambiguous, outage, or manually verified aggregate
`association_status`	categorical	token-linked, uniquely inferred, ambiguous, or missing; only the first two enter the primary choice likelihood

No field identifies a person or links the same person across visits. The one-use token expires with the visit and is absent from public analysis files. Geometry, cleaning, and optional hygiene samples live in separate site- and time-indexed tables.

H.2 Geometry table

For every stall record walking distance from the commitment point, side-wall indicators, partition height and depth, door orientation, stall dimensions, accessibility features, fixtures, and any permanent defect during the study. Record geometry changes as versioned intervals rather than overwriting the site record.

H.3 Empirical input ledger

The source bundle contains `source_data.csv`. Each extracted numerical input has a stable data-set identifier, bibliography key, source title, page or table locator, DOI, raw numerator and denominator, derived value, measurement type, and extraction note. The figure-generation program reads the Christenfeld middle-depletion aggregate from this ledger, checks the stored ratio, and records the source metadata in the parameter table. This separates published observations from illustrative model parameters.

H.4 Computational reproducibility

The accompanying Python program regenerates every numerical figure and table from explicit parameter dictionaries and the empirical input ledger. Exact finite-state results use enumeration and a stationary linear solve; mean-field equilibria use constrained minimization and fixed-point residual checks; delayed trajectories use a declared integration step and stored history; stochastic experiments use fixed seeds.

Every result is checked by at least one independent identity:

- Protocol N busy-count probabilities against the Erlang loss (truncated-Poisson) law;
- Protocol G stationary probabilities against direct Gibbs weights;
- stall entry rates against departure flow;
- equilibrium probabilities against both simplex constraints and first-order residuals;
- delayed thresholds against the closed form (8.12);
- random sequential adsorption against (G.1).

Vector figures are saved as PDF for inclusion in the manuscript and as PNG for rapid inspection. Tabulated values are exported in machine-readable CSV or JSON form in the source bundle.

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